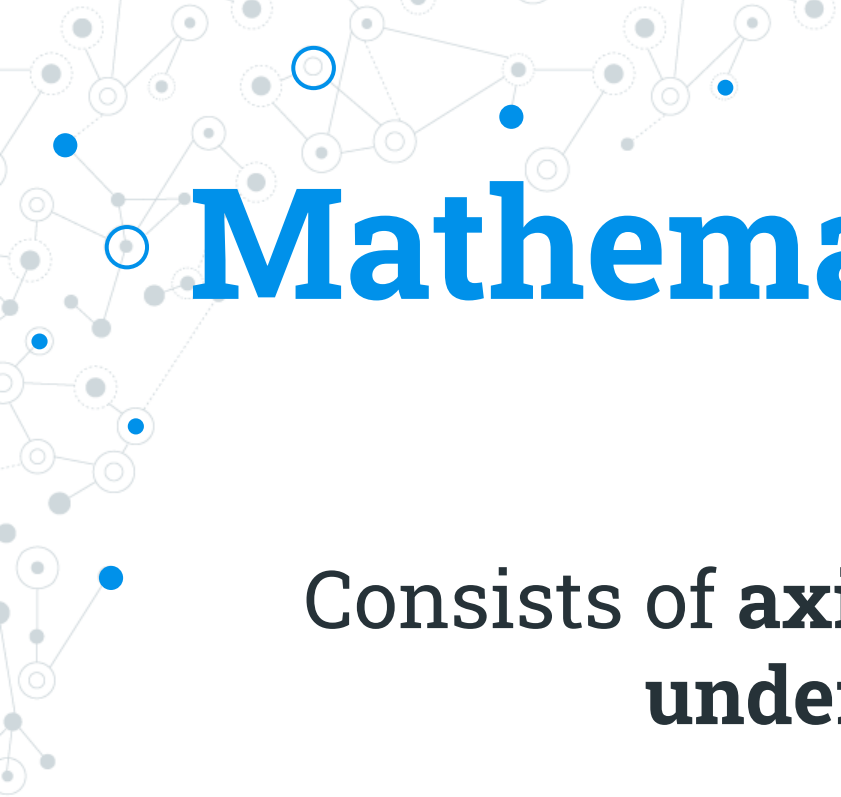




CSC 102– Discrete Structures I

Proofs





Mathematical system

Consists of **axioms, definitions and undefined terms.**



A decorative network diagram in the top-left corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall structure is dense and organic, resembling a molecular or biological network.

Axiom

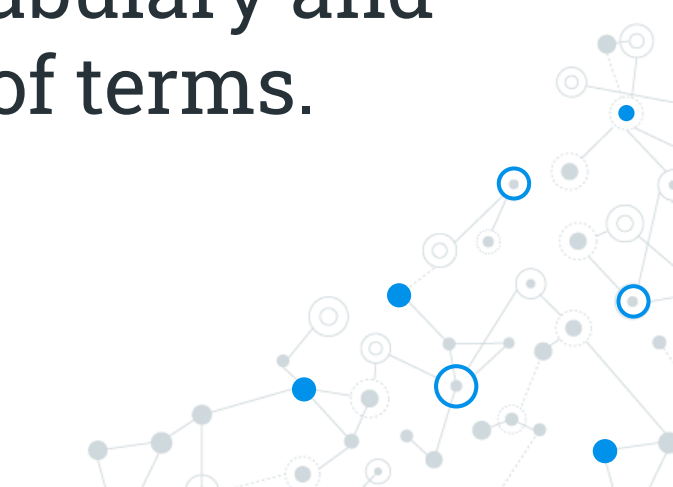
Assumed to be true.

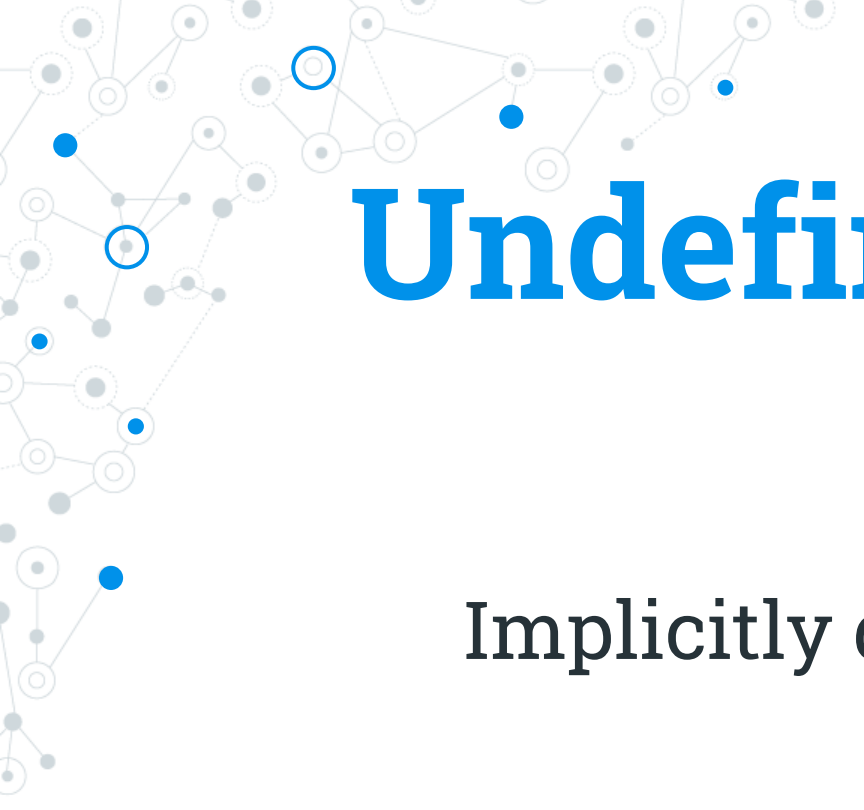
A decorative network diagram in the bottom-right corner of the slide. It features a complex web of interconnected nodes and edges. The nodes are represented by small circles, some of which are solid blue, some are solid grey, and some are hollow with a blue outline. The edges are thin grey lines connecting the nodes. The overall structure is dense and organic, resembling a molecular or biological network.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with solid blue dots. The lines are thin and grey, creating a mesh-like structure.

Definitions

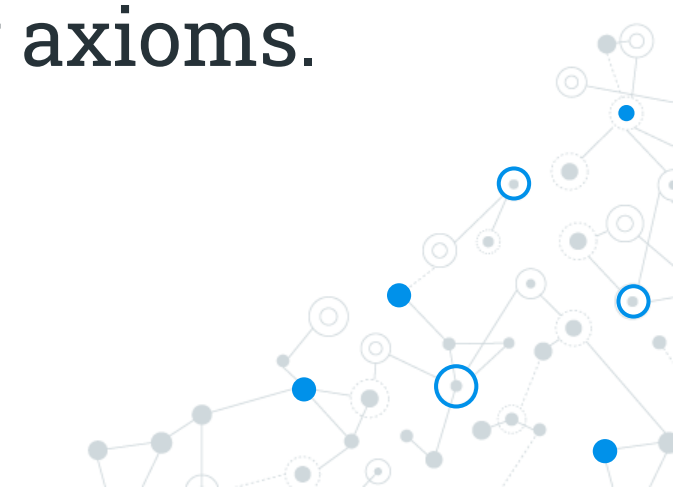
Provides necessary vocabulary and clarify the meaning of terms.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes connected by lines, with several nodes highlighted in blue.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and lines. Some nodes are highlighted with blue circles, and others with blue dots. The lines are thin and grey, creating a mesh-like structure.

Undefined Terms

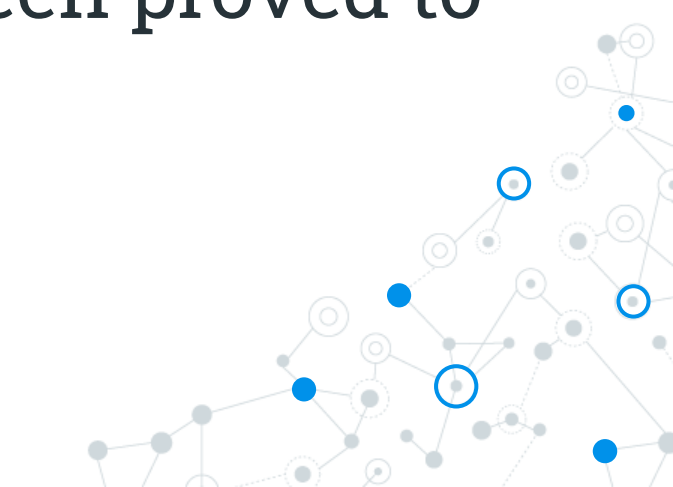
Implicitly defined by axioms.

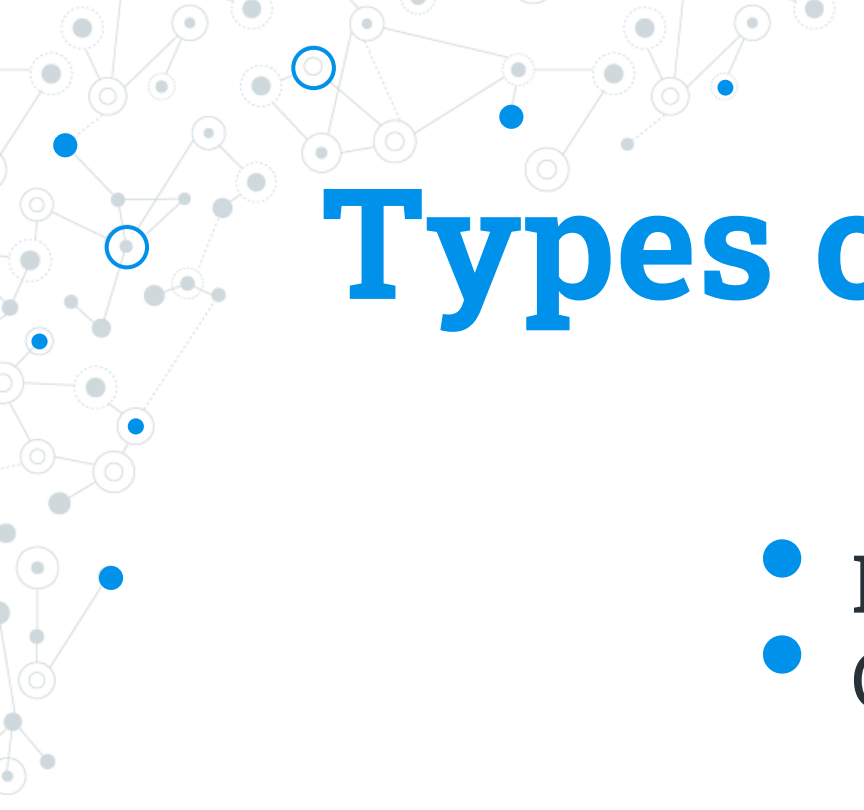
A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes connected by lines, with several nodes highlighted in blue.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles or dots, while others are grey. The edges are thin grey lines.

Theorem

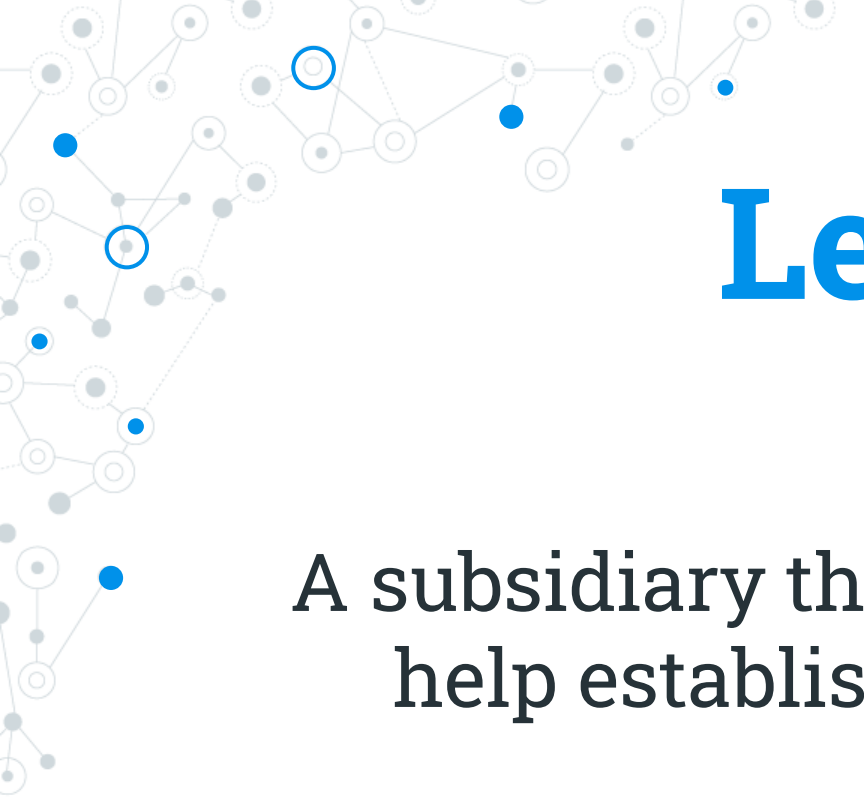
A proposition that has been proved to be true.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, showing a web of interconnected nodes and edges with some blue highlights.



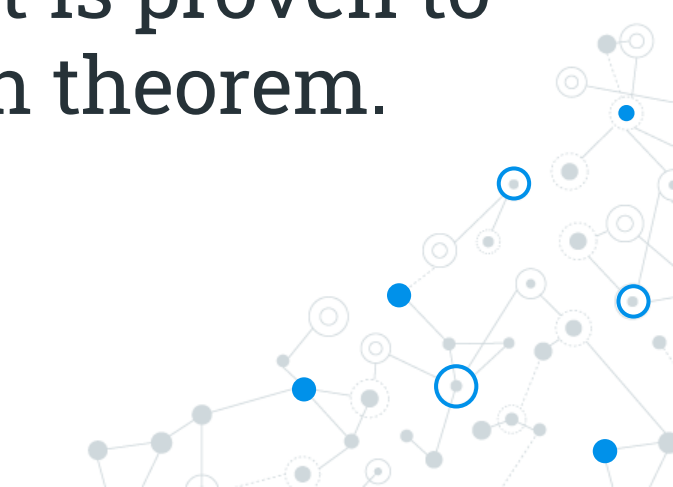
Types of Theorem

- Lemma
 - Corollary
- 

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with solid blue dots. The lines connecting them are thin and grey.

Lemma

A subsidiary theorem that is proven to help establish the main theorem.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, showing a web of nodes and edges with some nodes highlighted in blue.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and some edges are solid blue lines, while others are gray.

Corollary

A theorem that follows easily from
another theorem.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left, showing a web of nodes and edges with some blue highlights.

A decorative network graph pattern in the top-left corner, consisting of various sized grey and blue nodes connected by thin grey lines. Some nodes are solid blue circles, while others are grey circles with concentric outlines. The pattern is more dense in the top-left and fades towards the center.

○ Lemma:

Any multiple of 2 is an even number.

A decorative network graph pattern in the bottom-right corner, similar to the one in the top-left, featuring grey and blue nodes connected by lines, with some nodes highlighted with concentric circles.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are hollow blue circles. The nodes are connected by thin, light gray lines, creating a mesh-like structure that extends towards the center of the slide.

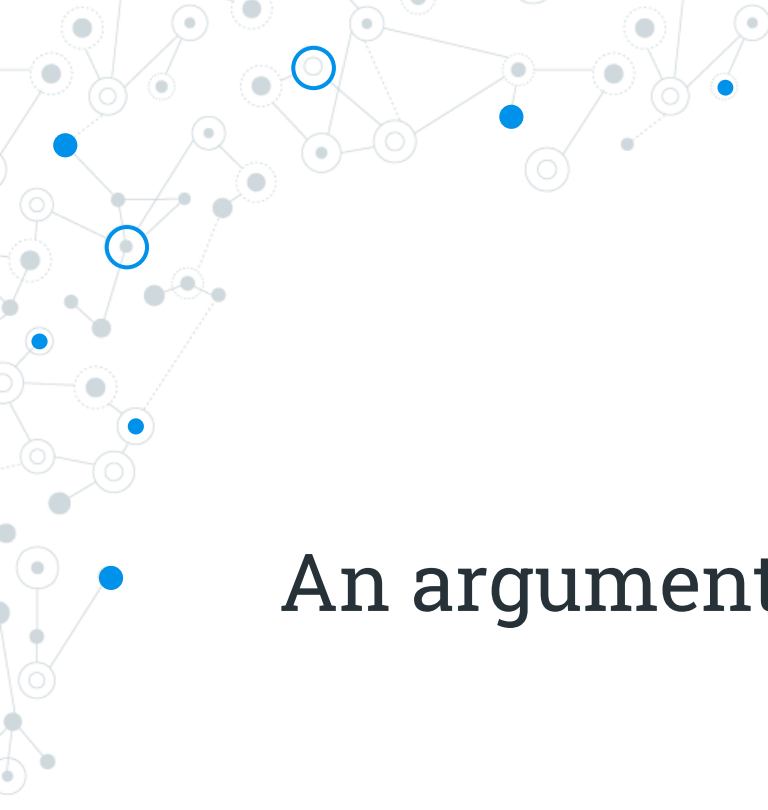
Lemma:

Any multiple of 2 is an even number.

A decorative network graph in the bottom-right corner, similar to the one in the top-left. It consists of a cluster of nodes connected by lines. Some nodes are solid blue, and some are hollow blue, set against a background of light gray lines and nodes.

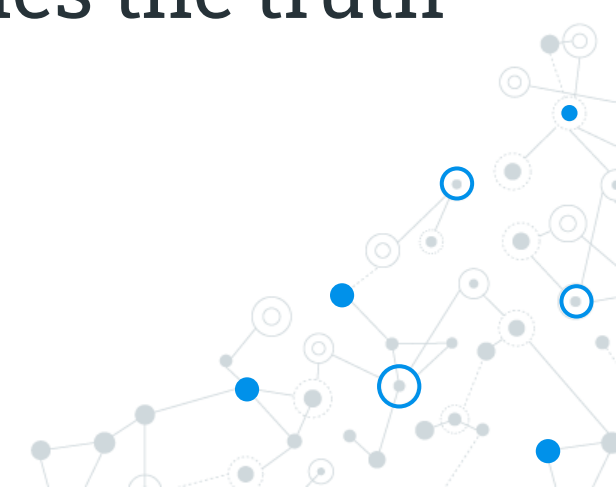
Corollary:

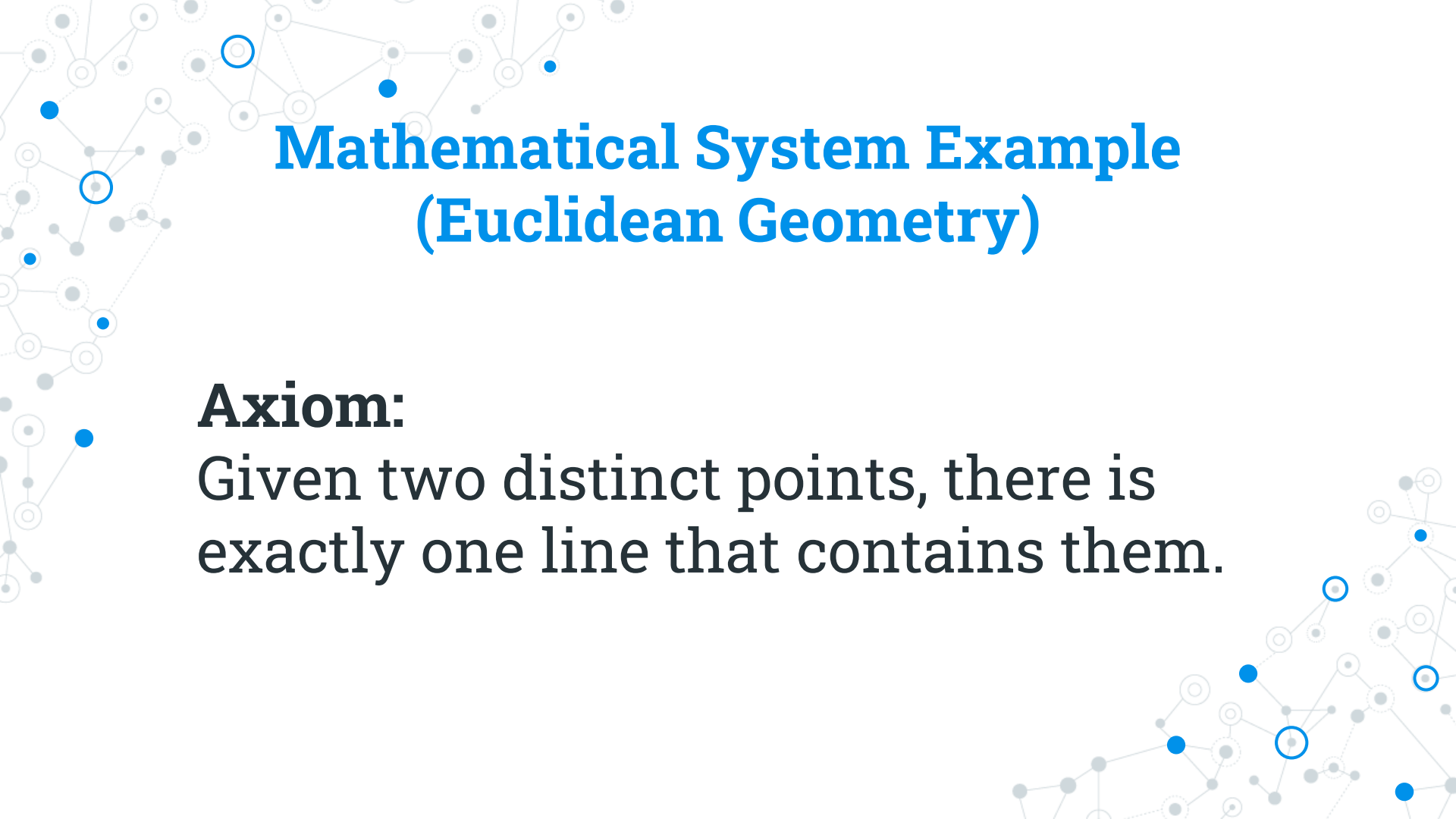
If you add two even numbers, the result is even.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles or dots, while others are grey. The edges are thin grey lines.

Proof

An argument that establishes the truth of a theorem.

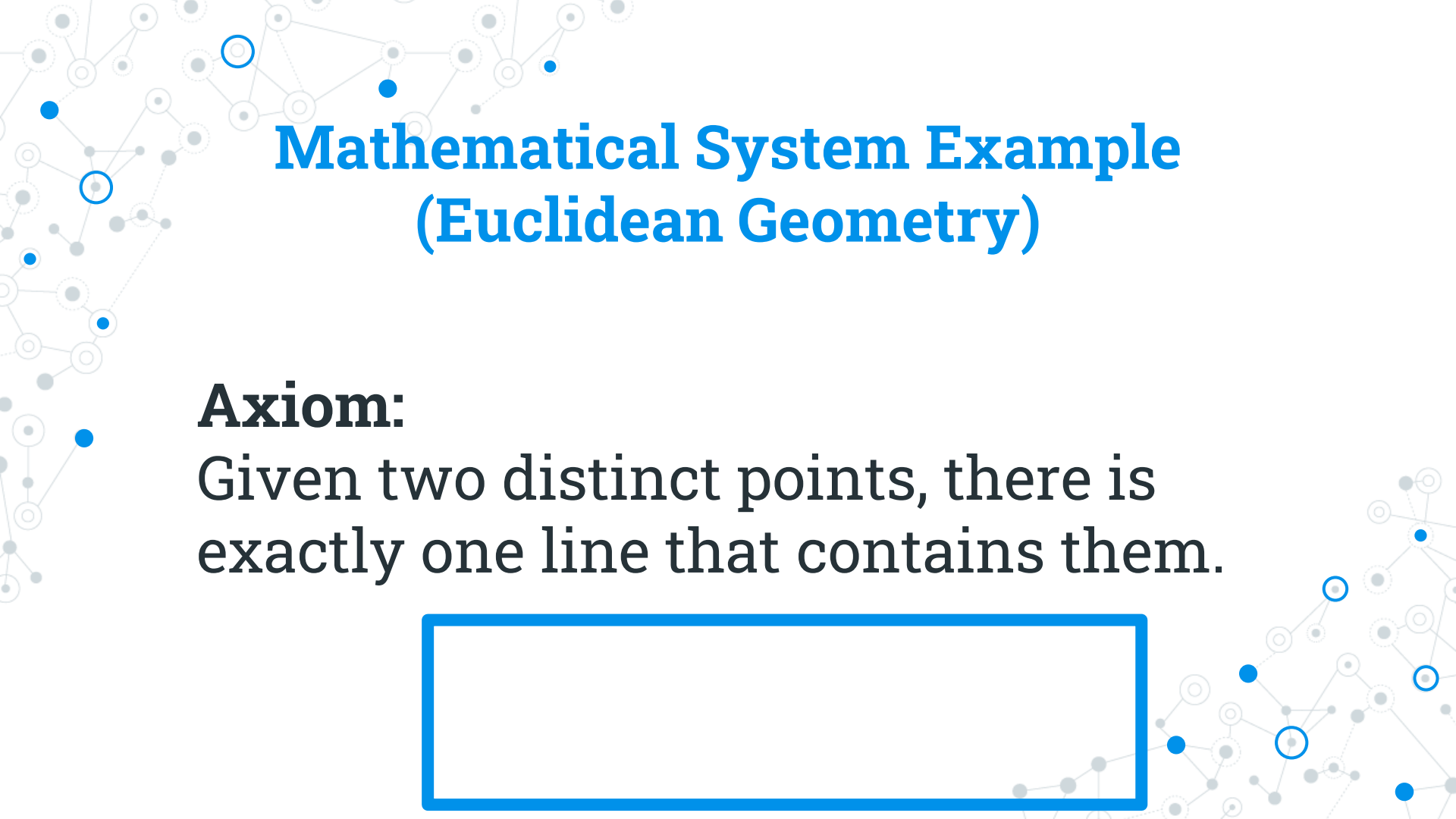
A decorative network diagram in the bottom-right corner, similar to the one in the top-left, showing a web of interconnected nodes and edges with some blue highlights.



Mathematical System Example (Euclidean Geometry)

Axiom:

Given two distinct points, there is exactly one line that contains them.

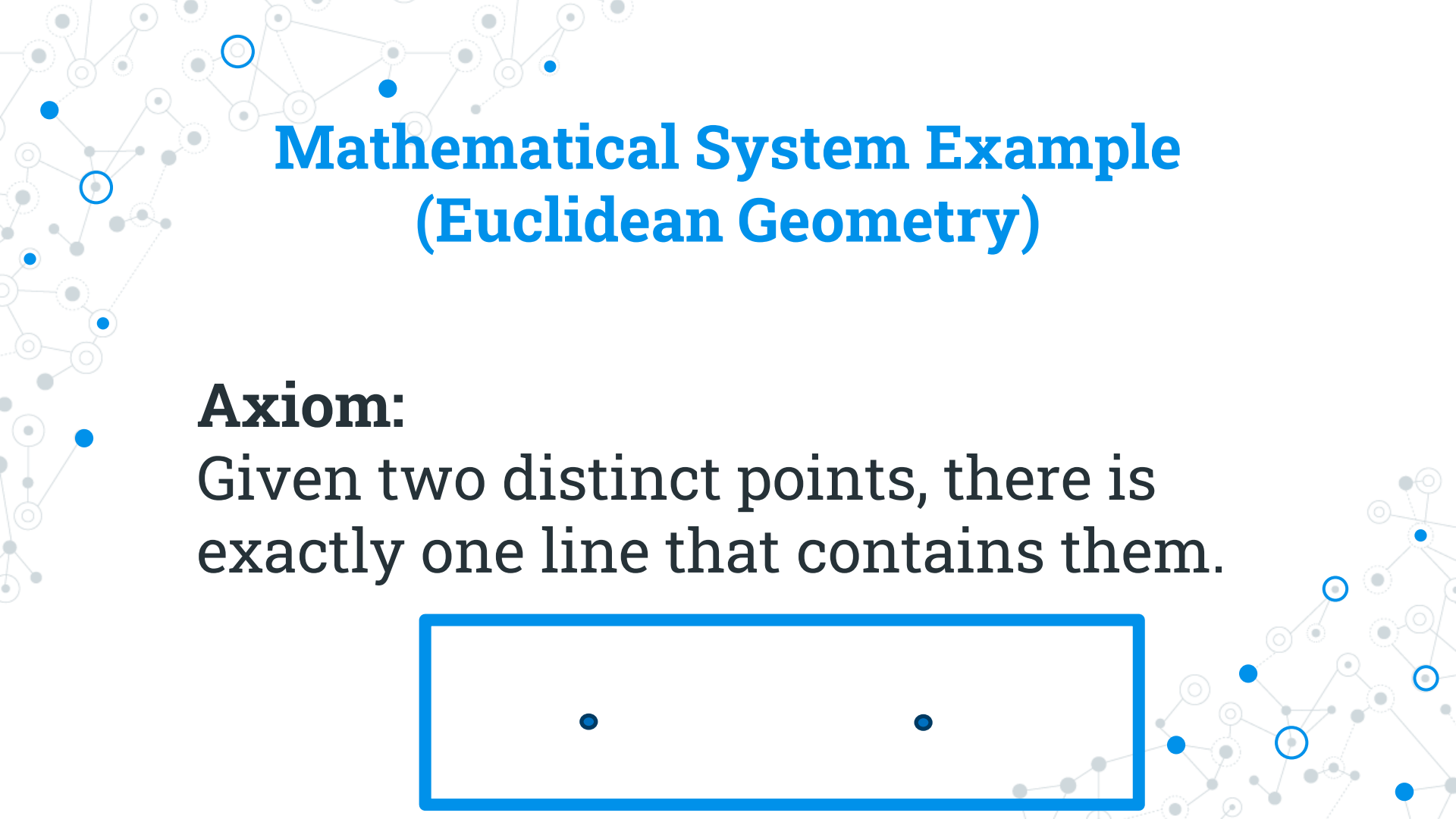
A decorative background featuring a network graph with nodes and edges. Some nodes are highlighted with blue circles or dots, while others are grey. The graph is composed of interconnected lines and dots, creating a complex, web-like structure.

Mathematical System Example (Euclidean Geometry)

Axiom:

Given two distinct points, there is exactly one line that contains them.



A decorative background consisting of a network of nodes and lines. The nodes are represented by small circles, some of which are blue and some are grey. The lines are thin and grey, connecting the nodes in a complex, web-like pattern. The overall aesthetic is clean and modern, with a focus on geometric and network structures.

Mathematical System Example (Euclidean Geometry)

Axiom:

Given two distinct points, there is exactly one line that contains them.

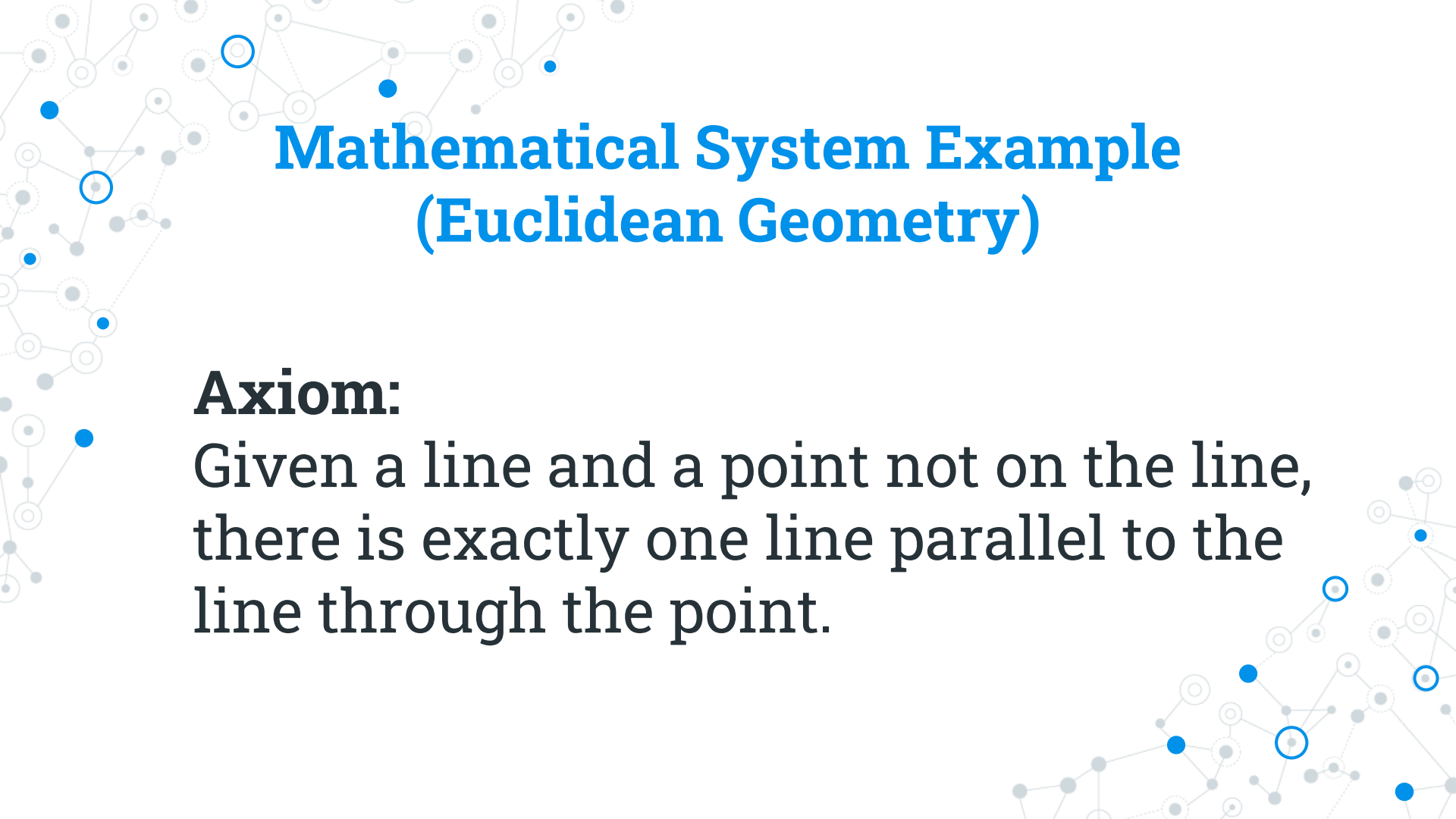


Mathematical System Example (Euclidean Geometry)

Axiom:

Given two distinct points, there is exactly one line that contains them.





Mathematical System Example (Euclidean Geometry)

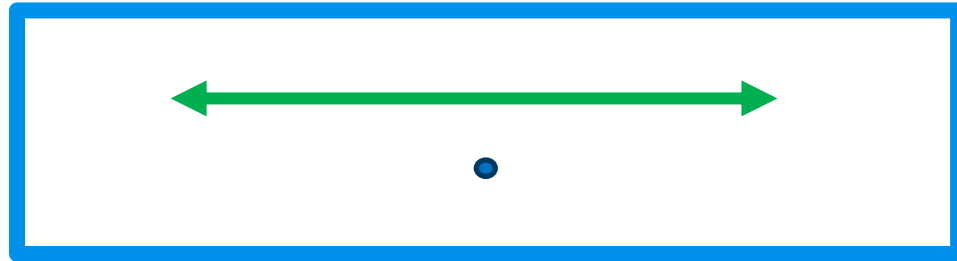
Axiom:

Given a line and a point not on the line, there is exactly one line parallel to the line through the point.

○ Mathematical System Example (Euclidean Geometry)

Axiom:

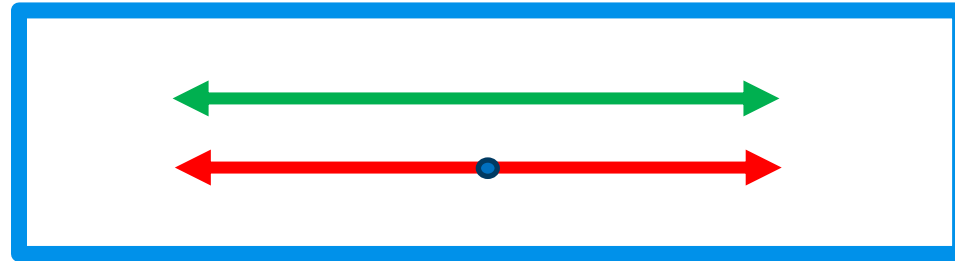
Given a line and a point not on the line, there is exactly one line parallel to the line through the point.

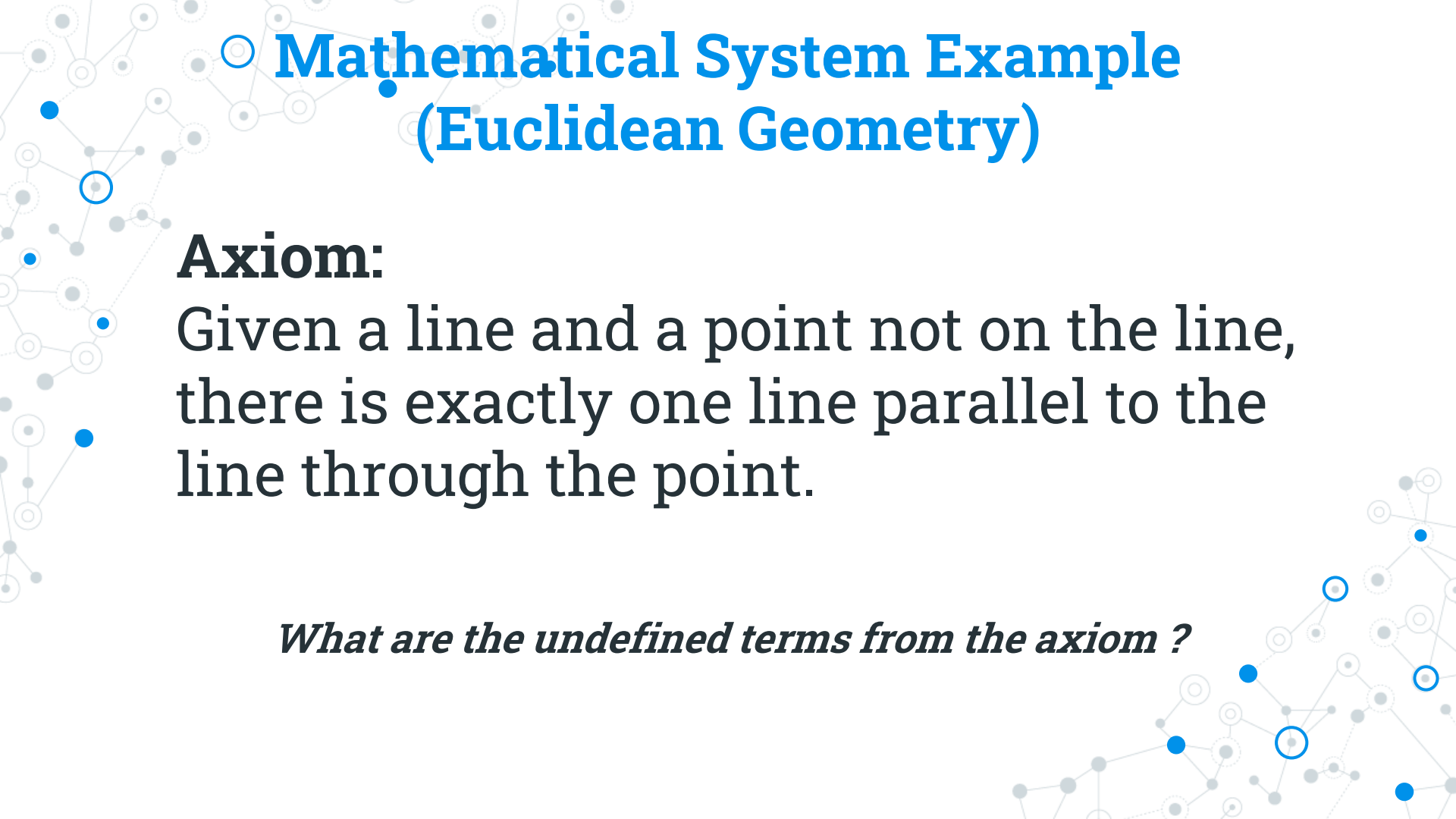


○ Mathematical System Example (Euclidean Geometry)

Axiom:

Given a line and a point not on the line, there is exactly one line parallel to the line through the point.



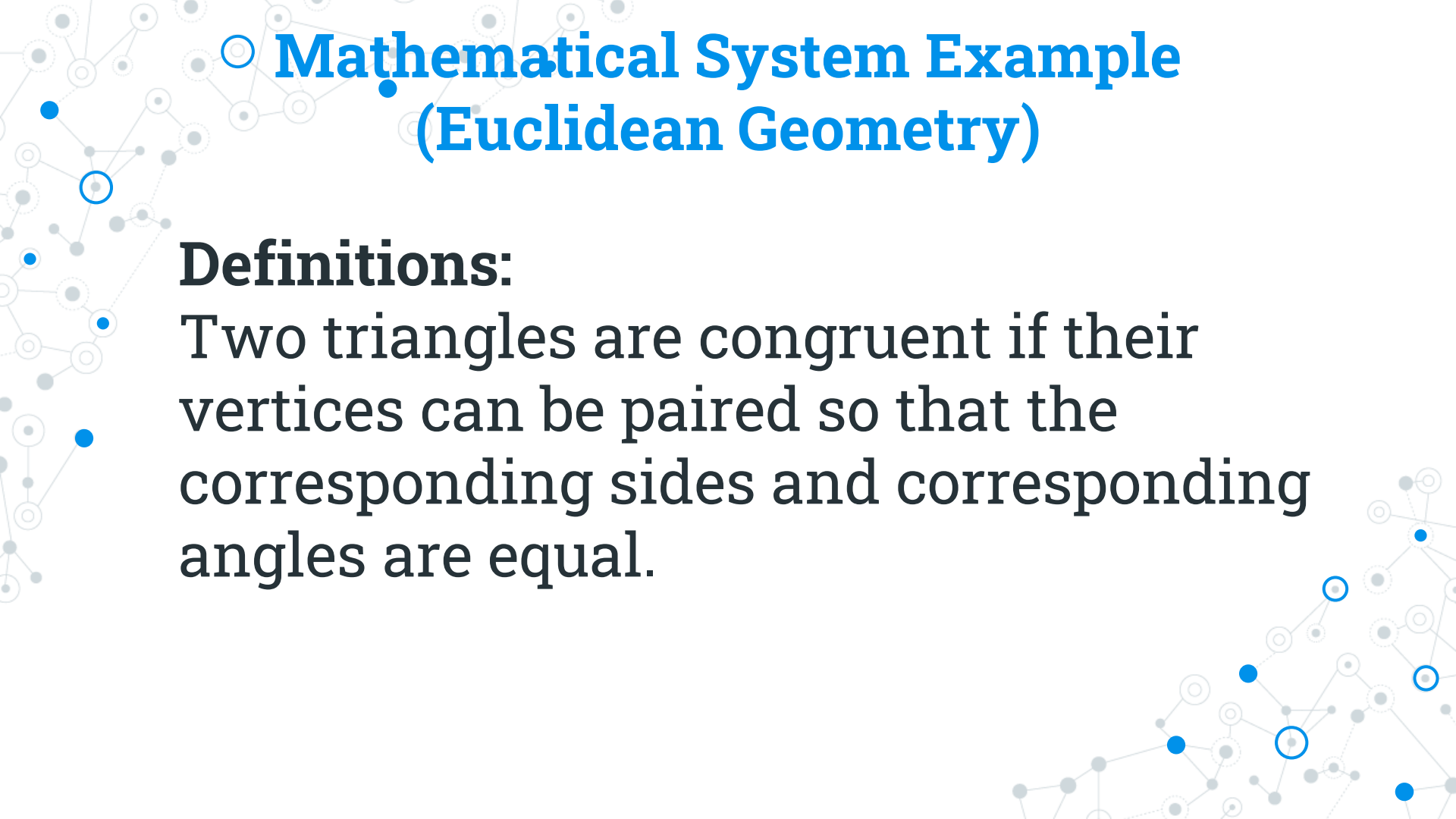


○ Mathematical System Example (Euclidean Geometry)

Axiom:

Given a line and a point not on the line, there is exactly one line parallel to the line through the point.

What are the undefined terms from the axiom ?



○ Mathematical System Example (Euclidean Geometry)

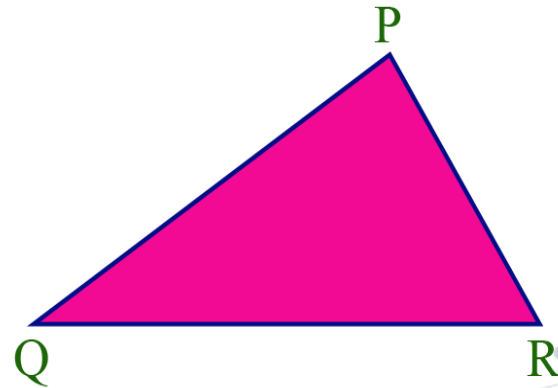
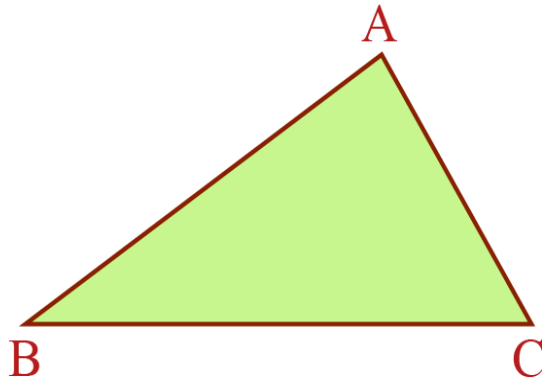
Definitions:

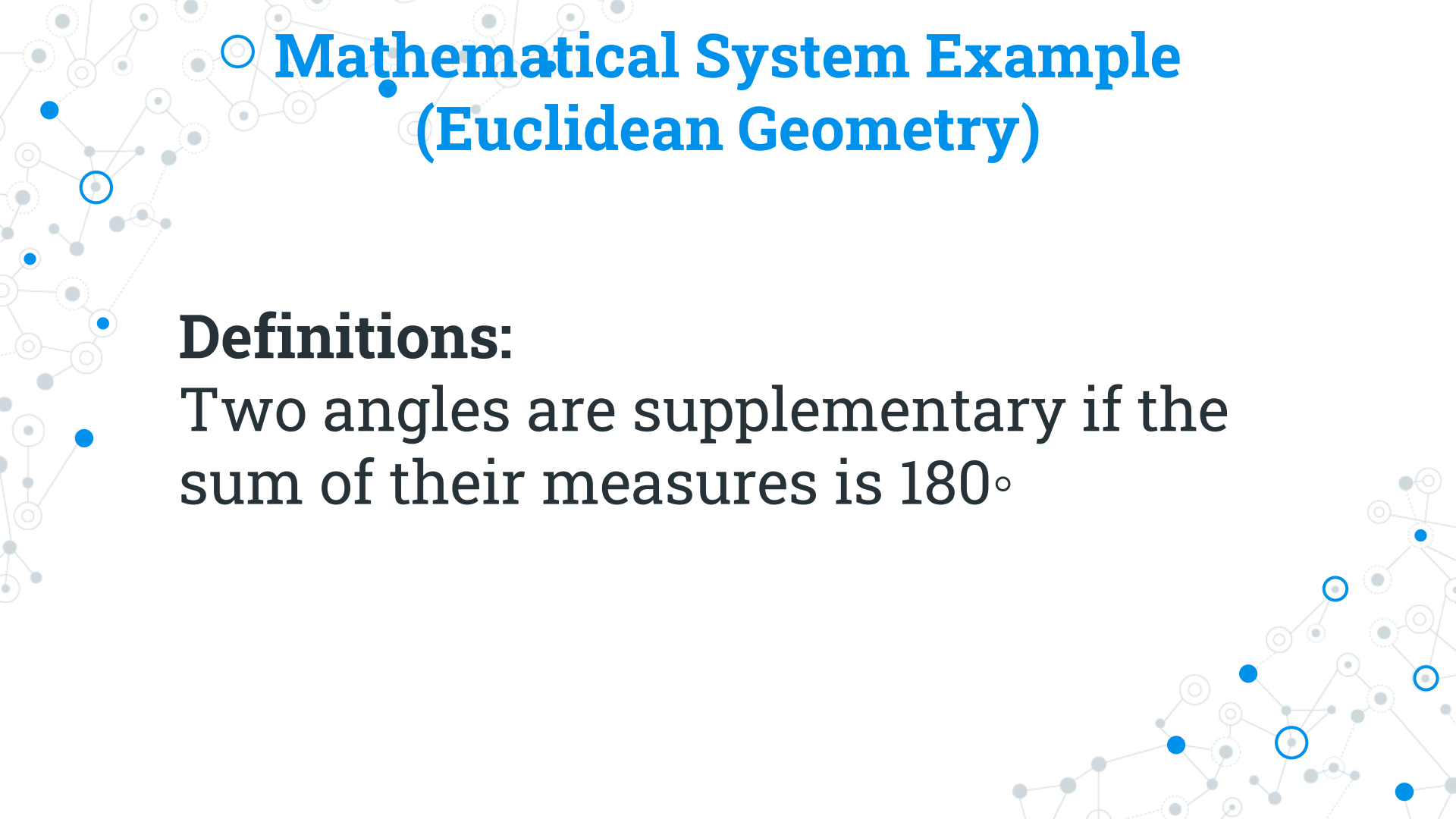
Two triangles are congruent if their vertices can be paired so that the corresponding sides and corresponding angles are equal.

Mathematical System Example (Euclidean Geometry)

Definitions:

Two triangles are congruent if their vertices can be paired so that the corresponding sides and corresponding angles are equal.





○ Mathematical System Example (Euclidean Geometry)

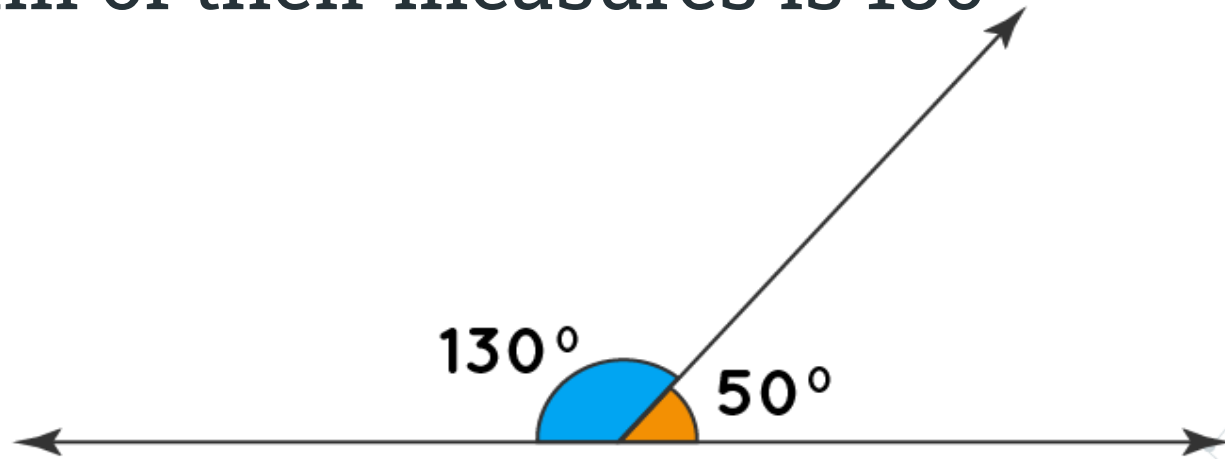
Definitions:

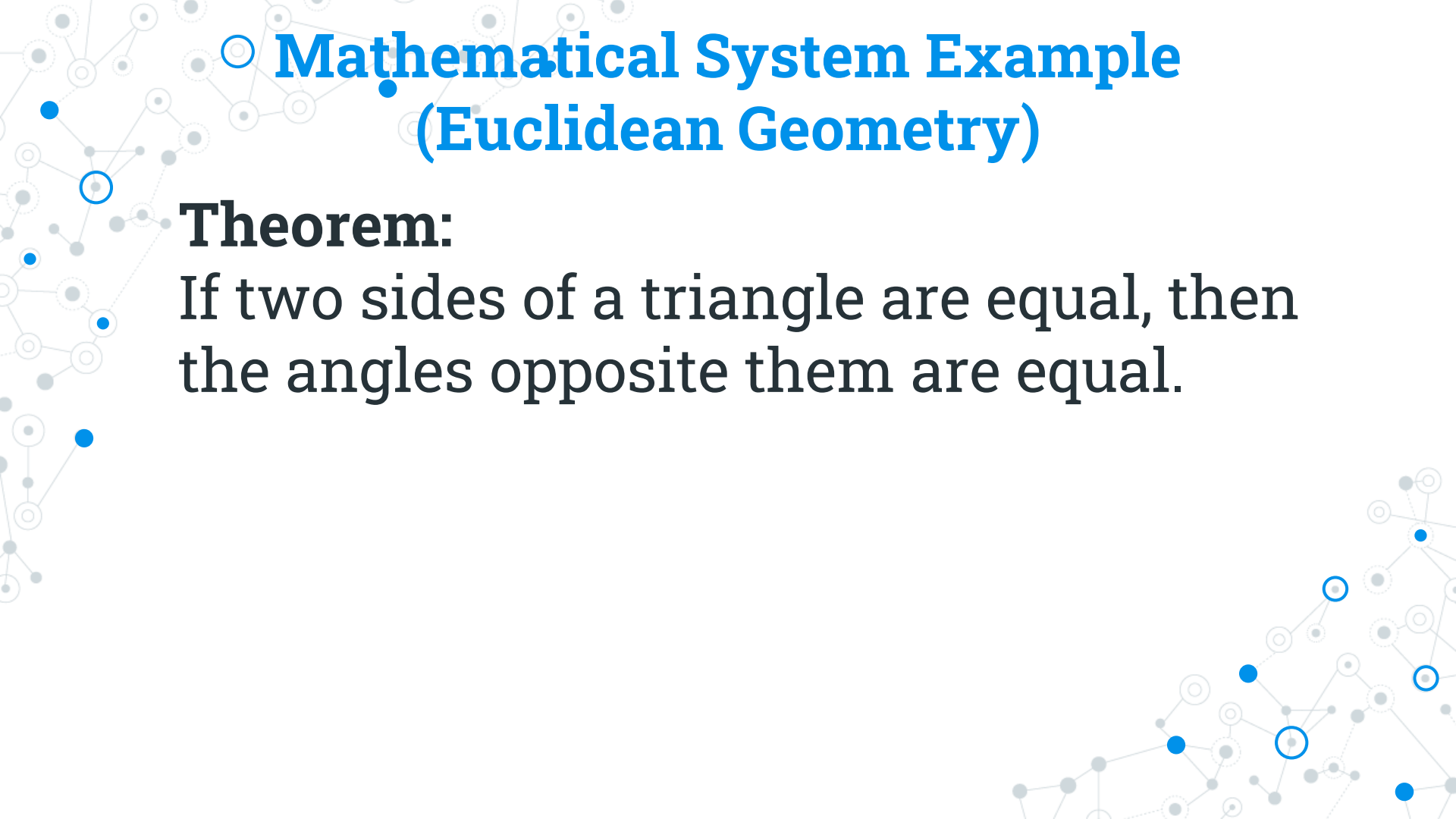
Two angles are supplementary if the sum of their measures is 180° .

Mathematical System Example (Euclidean Geometry)

Definitions:

Two angles are supplementary if the sum of their measures is 180° .





○ Mathematical System Example (Euclidean Geometry)

Theorem:

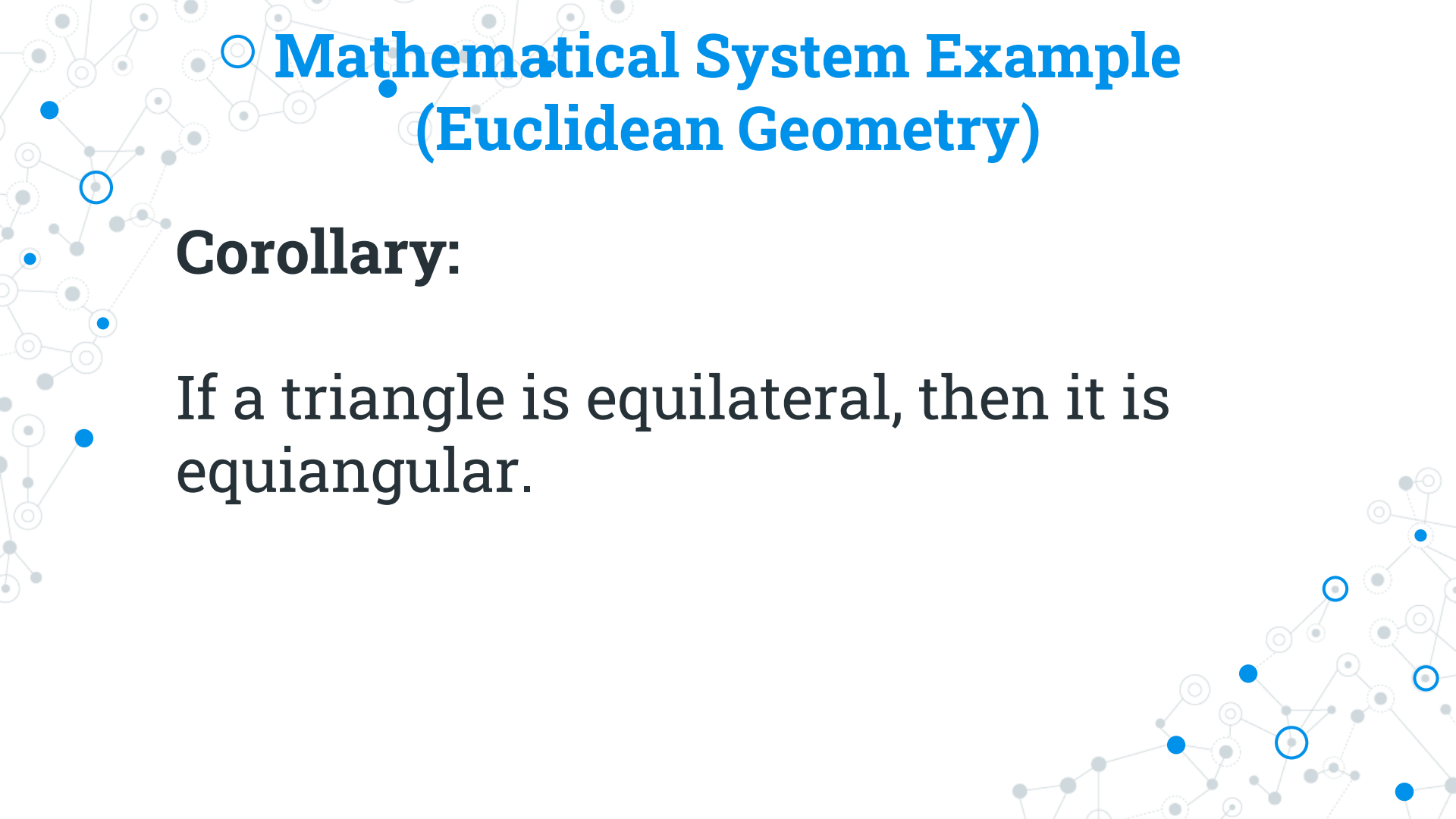
If two sides of a triangle are equal, then the angles opposite them are equal.

Mathematical System Example (Euclidean Geometry)

Theorem:

If two sides of a triangle are equal, then the angles opposite them are equal.





○ Mathematical System Example (Euclidean Geometry)

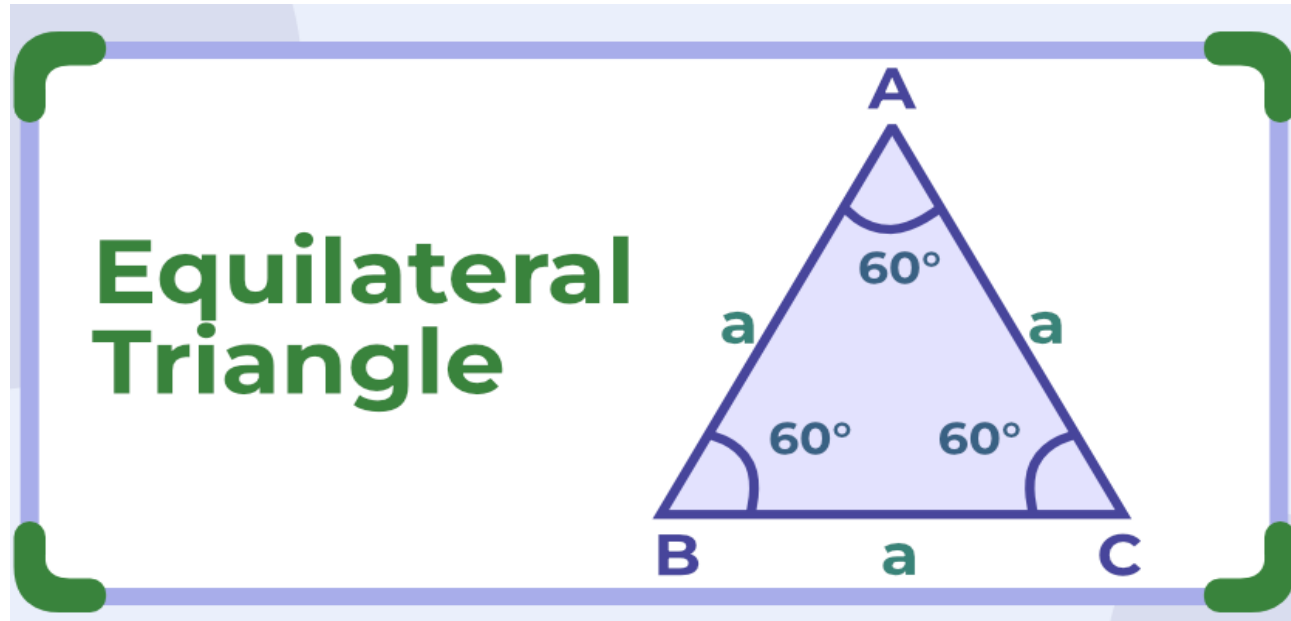
Corollary:

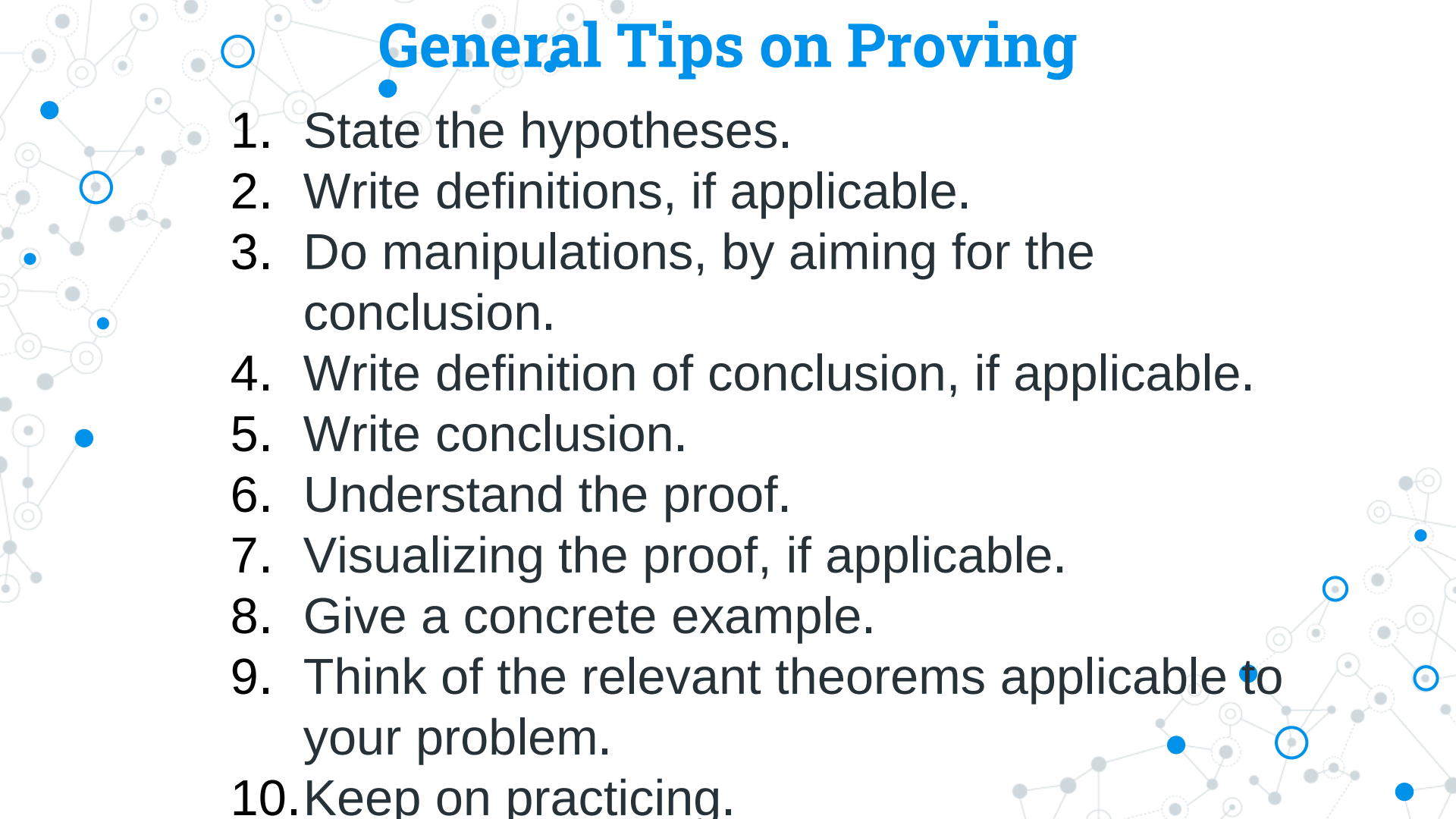
If a triangle is equilateral, then it is equiangular.

Mathematical System Example (Euclidean Geometry)

Corollary:

If a triangle is equilateral, then it is equiangular.





General Tips on Proving

1. State the hypotheses.
2. Write definitions, if applicable.
3. Do manipulations, by aiming for the conclusion.
4. Write definition of conclusion, if applicable.
5. Write conclusion.
6. Understand the proof.
7. Visualizing the proof, if applicable.
8. Give a concrete example.
9. Think of the relevant theorems applicable to your problem.
10. Keep on practicing.

Reduction to Definitions

.

If you are not sure how to start a proof,
convert all terms in the hypothesis to
their definitions.

Reduction to Definition Example

Hypothesis: A set S is finite.

Reduction to Definition Example

Hypothesis: A set S is finite.

A set S is *finite* if there exists an integer n such that S has exactly n elements.

Reduction to Definition Example

Hypothesis: A set S is finite.

A set S is *finite* if there exists an integer n such that S has exactly n elements.

$$|S| = n, n \in \mathbb{Z}^+$$

Reduction to Definition Example

(Writing to a mathematical expression)

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

Reduction to Definition Example

What is this statement trying to tell you ?

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

Reduction to Definition Example

What is this statement trying to tell you ?

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

Intuitively, this statement says that if you have an infinite supply of something (U), and you take a finite amount away (S), then you still have an infinite amount left.

Reduction to Definitions

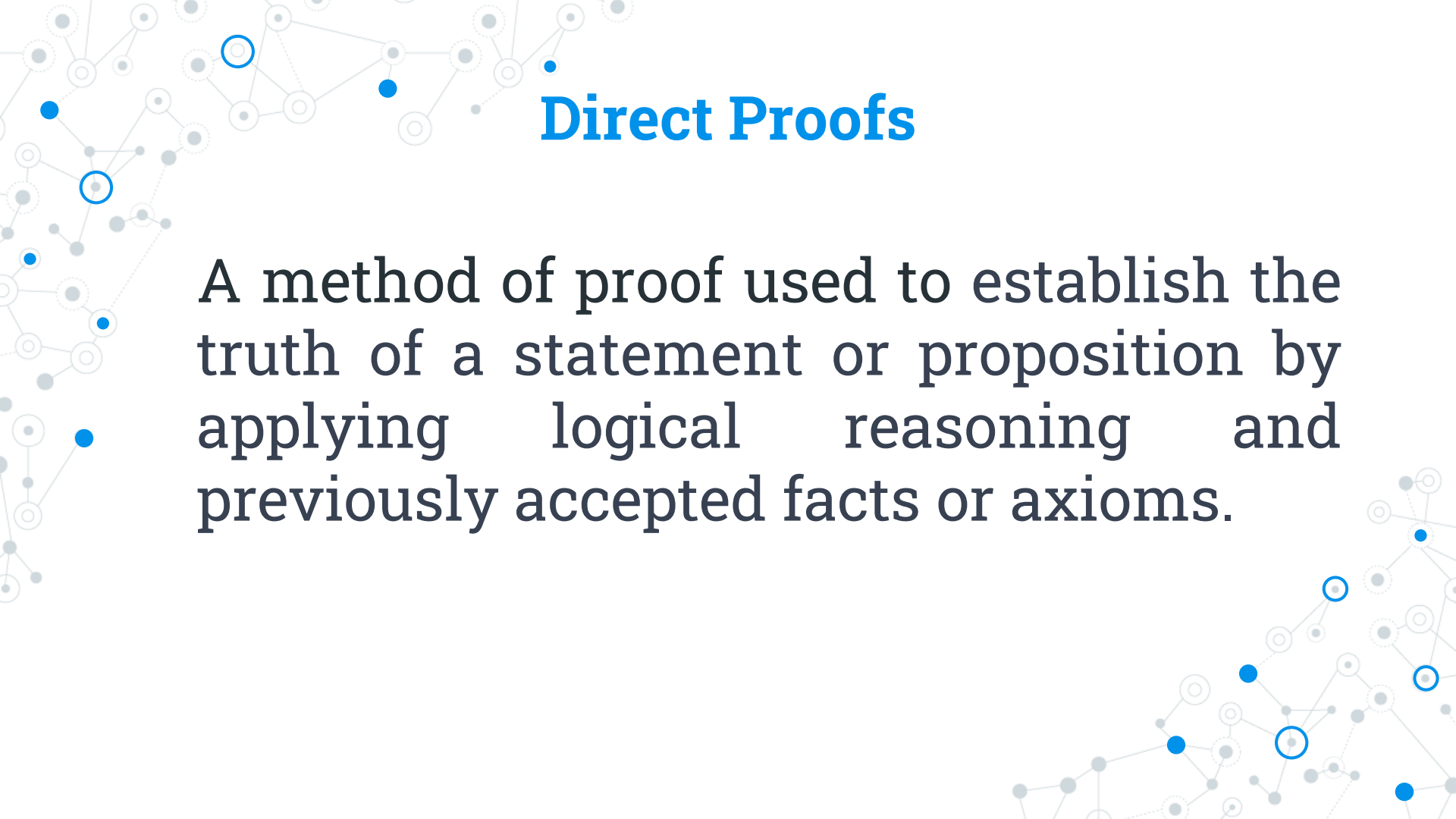
Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

Original Statement	New Statement
S is finite	There is a integer n such that $\ S\ = n$
U is infinite	For no integer p is $\ U\ = p$
T is the complement of S	$S \cup T = U$ and $S \cap T = \emptyset$



• Different Proof Methods

1. Direct Proof
 2. Proof by Contraposition
 3. Proof by Contradiction
 4. Counter Example
 5. Vacuous Proof
 6. Trivial Proof
 7. Proofs of Equivalence
 8. Existence Proof
 9. Proof by Induction
- 

A decorative background graphic consisting of a network of nodes and lines. The nodes are represented by small circles, some of which are solid blue, some are hollow blue, and others are solid grey. They are interconnected by thin grey lines, forming a complex web-like structure that is more dense on the left and right sides of the slide.

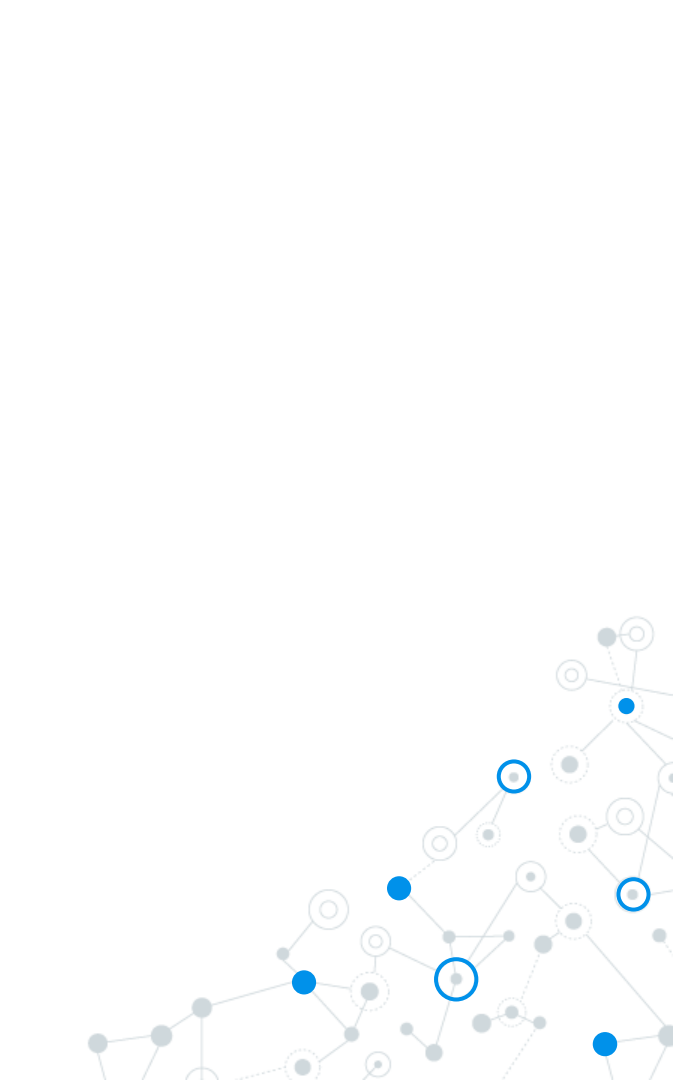
Direct Proofs

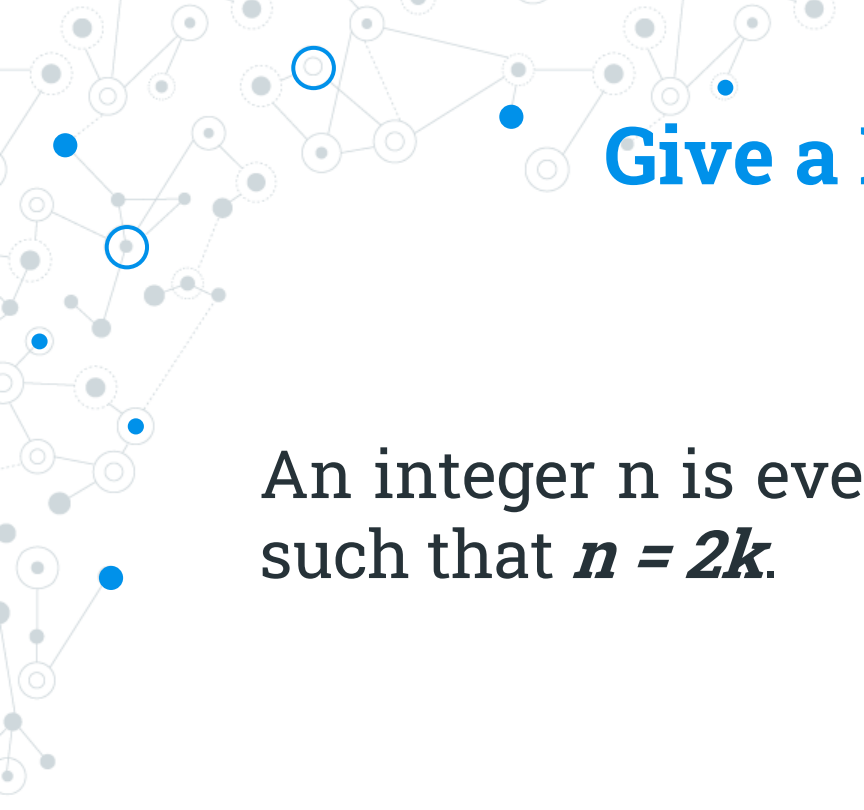
A method of proof used to establish the truth of a statement or proposition by applying logical reasoning and previously accepted facts or axioms.

A decorative network graph in the top-left corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots.

Direct Proofs

Assume p
Conclude q

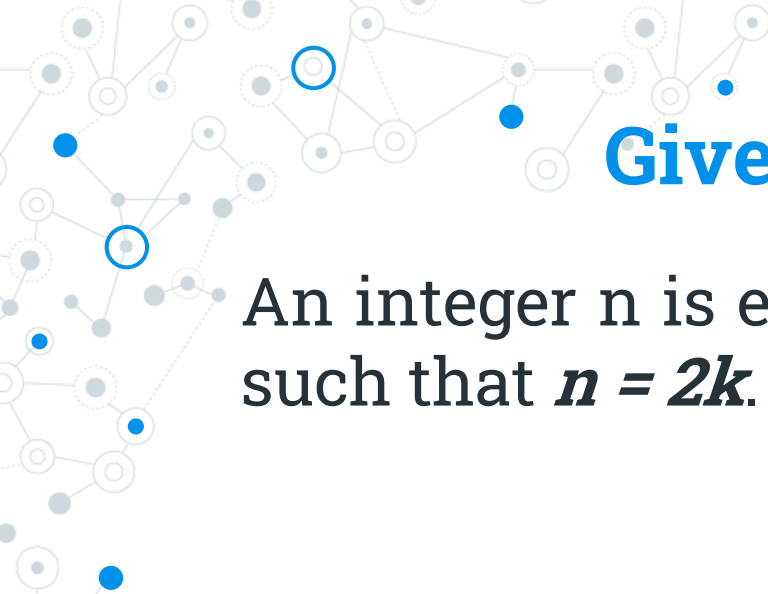
A decorative network graph in the bottom-right corner, featuring a complex web of interconnected nodes and edges. Some nodes are highlighted with blue circles, and others with blue dots.



Give a Direct Proof

An integer n is even if there exists an integer k such that $n = 2k$.



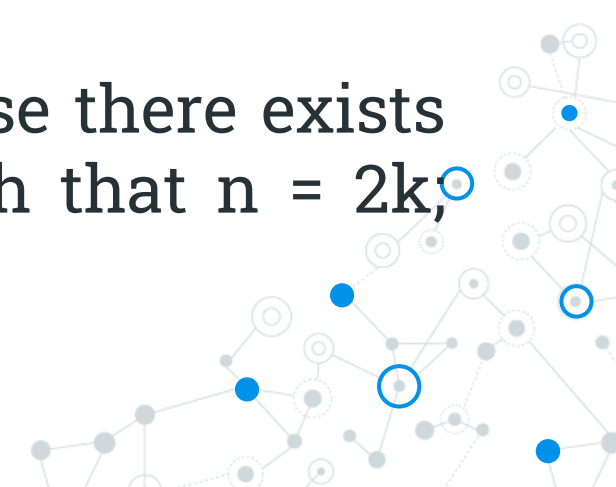


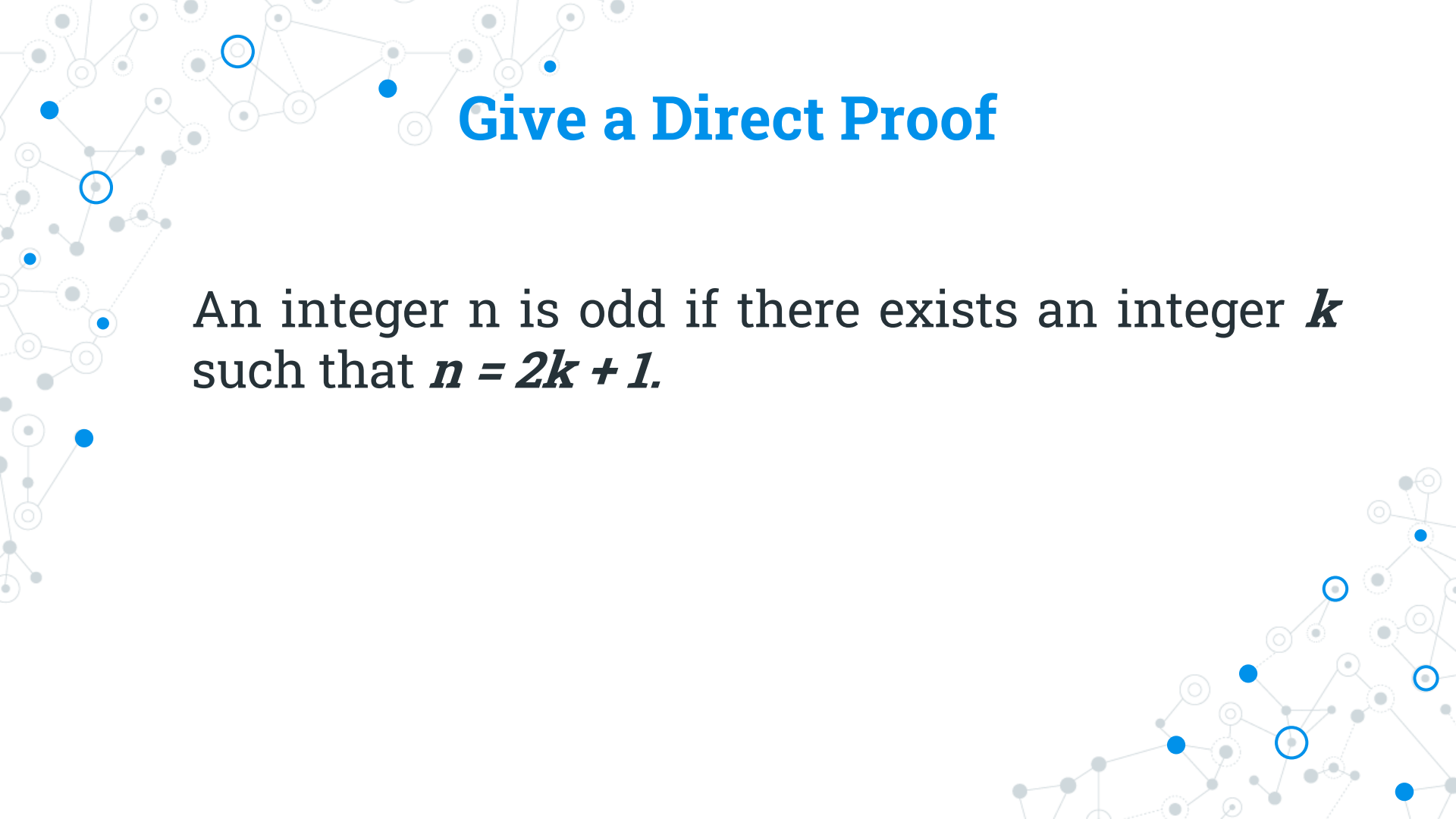
Give a Direct Proof

An integer n is even if there exists an integer k such that $n = 2k$.

Answer:

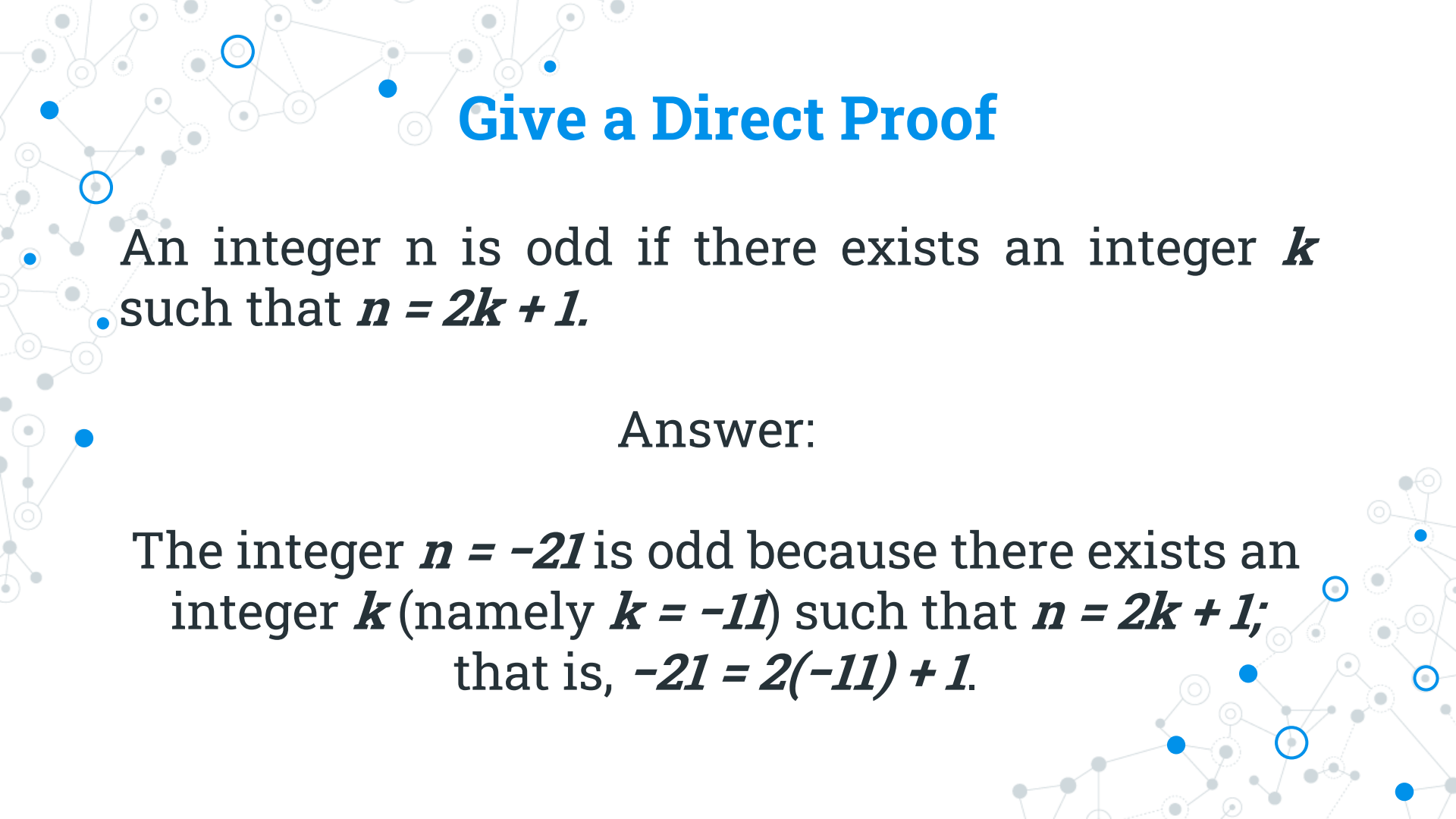
The integer $n = 12$ is even because there exists an integer k (namely $k = 6$) such that $n = 2k$, that is, $12 = 2 \cdot 6$.





Give a Direct Proof

An integer n is odd if there exists an integer k such that $n = 2k + 1$.

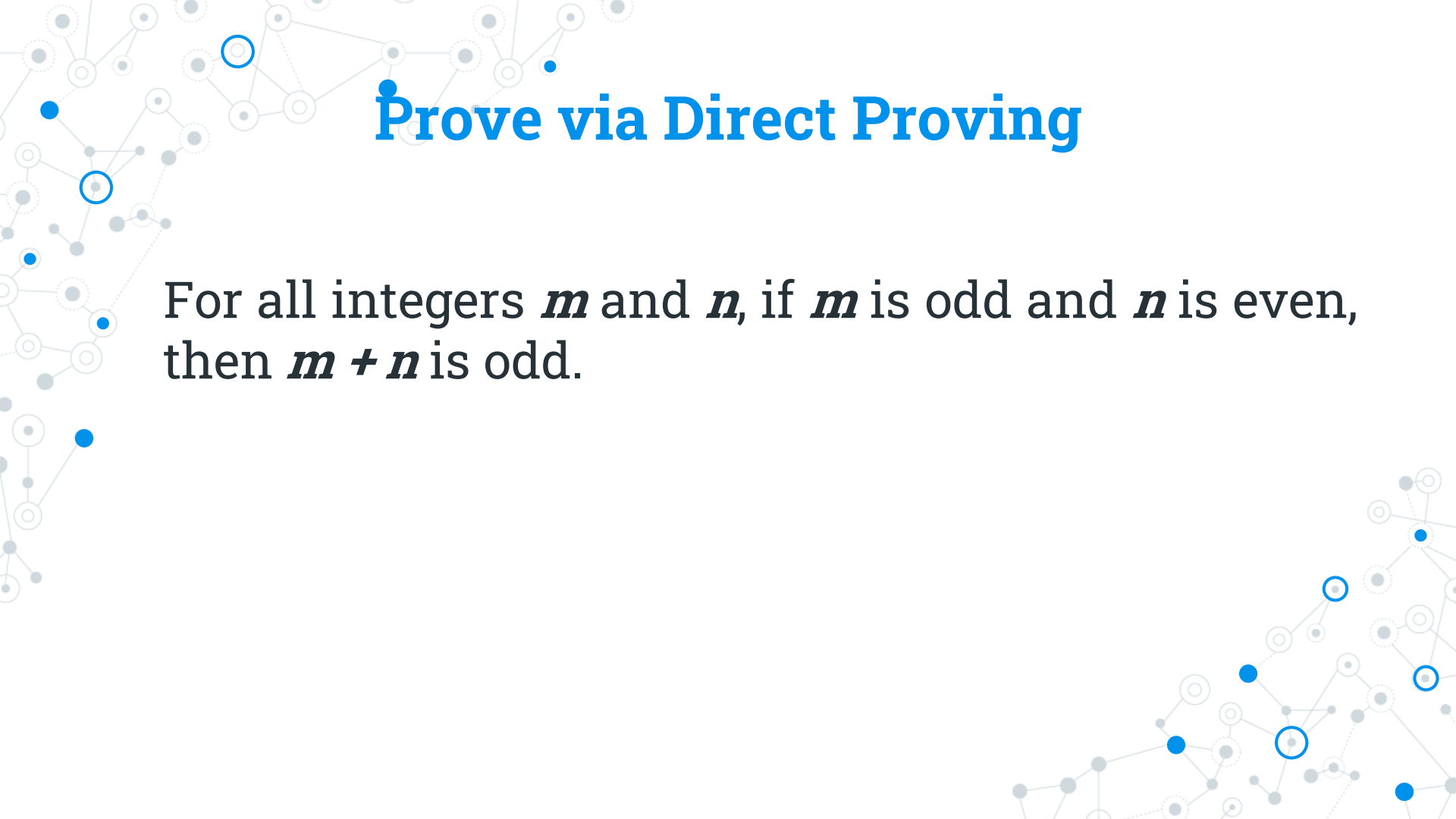


Give a Direct Proof

An integer n is odd if there exists an integer k such that $n = 2k + 1$.

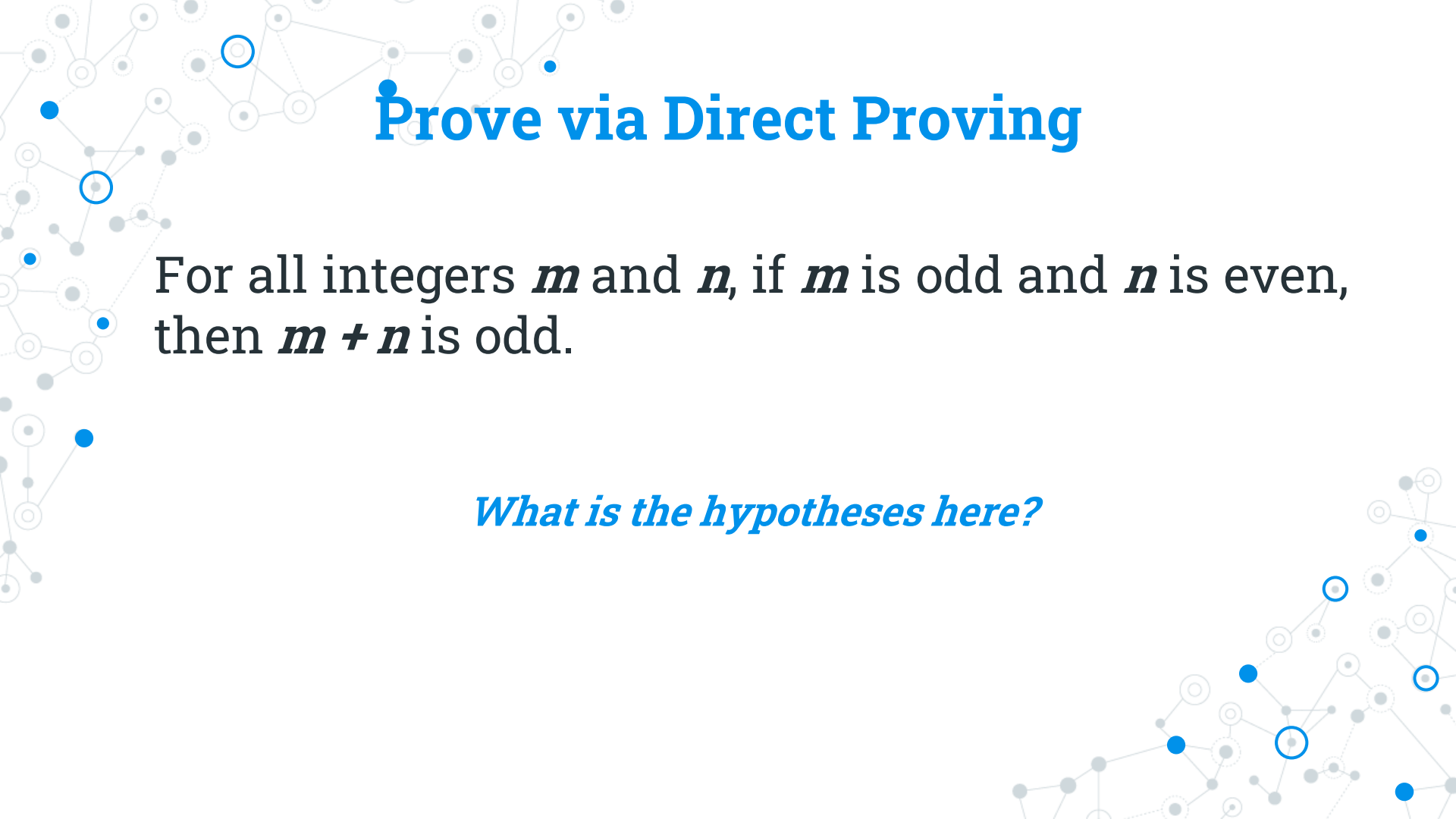
Answer:

The integer $n = -21$ is odd because there exists an integer k (namely $k = -11$) such that $n = 2k + 1$; that is, $-21 = 2(-11) + 1$.



Prove via Direct Proving

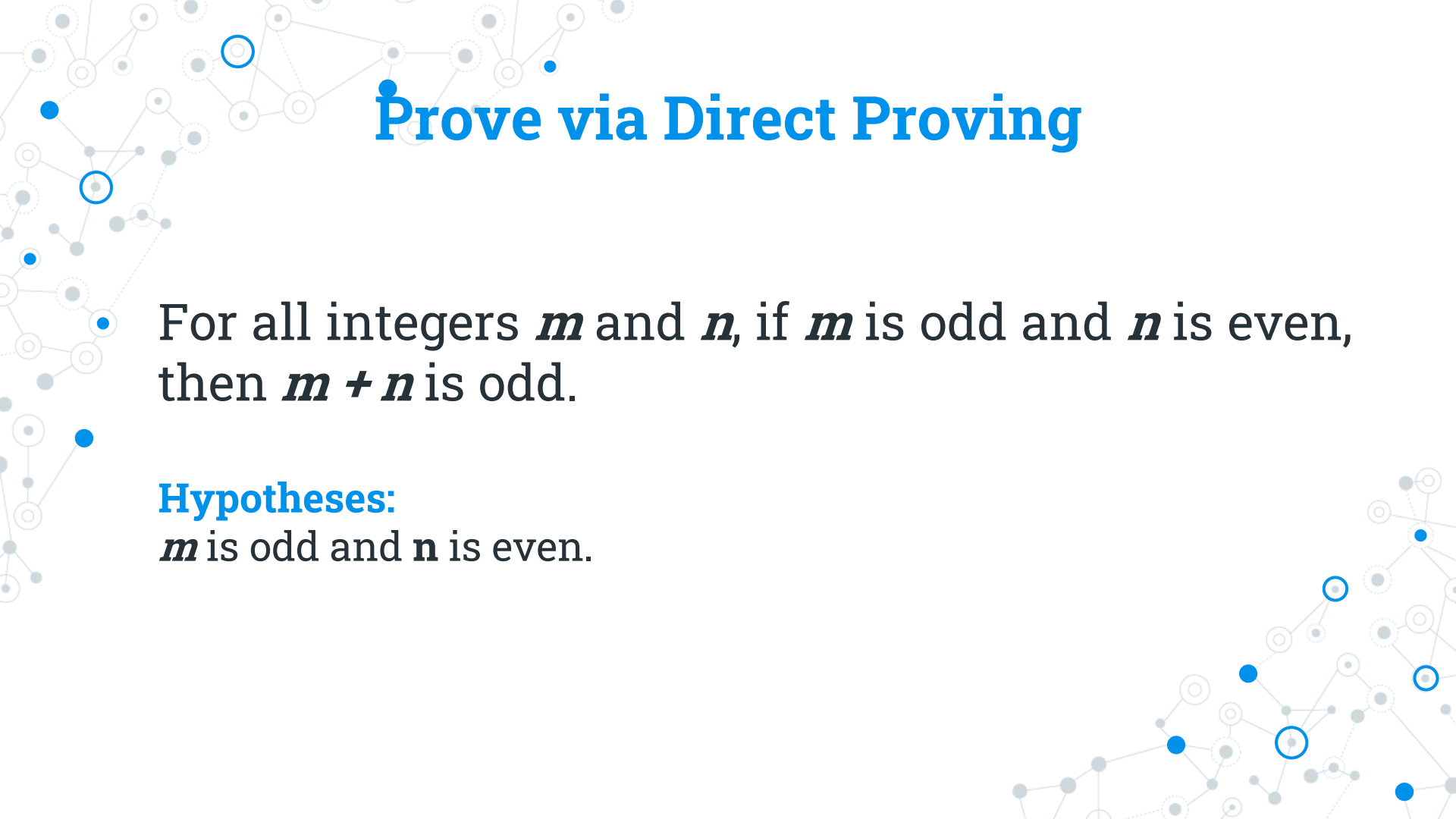
For all integers m and n , if m is odd and n is even, then $m + n$ is odd.



Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

What is the hypotheses here?

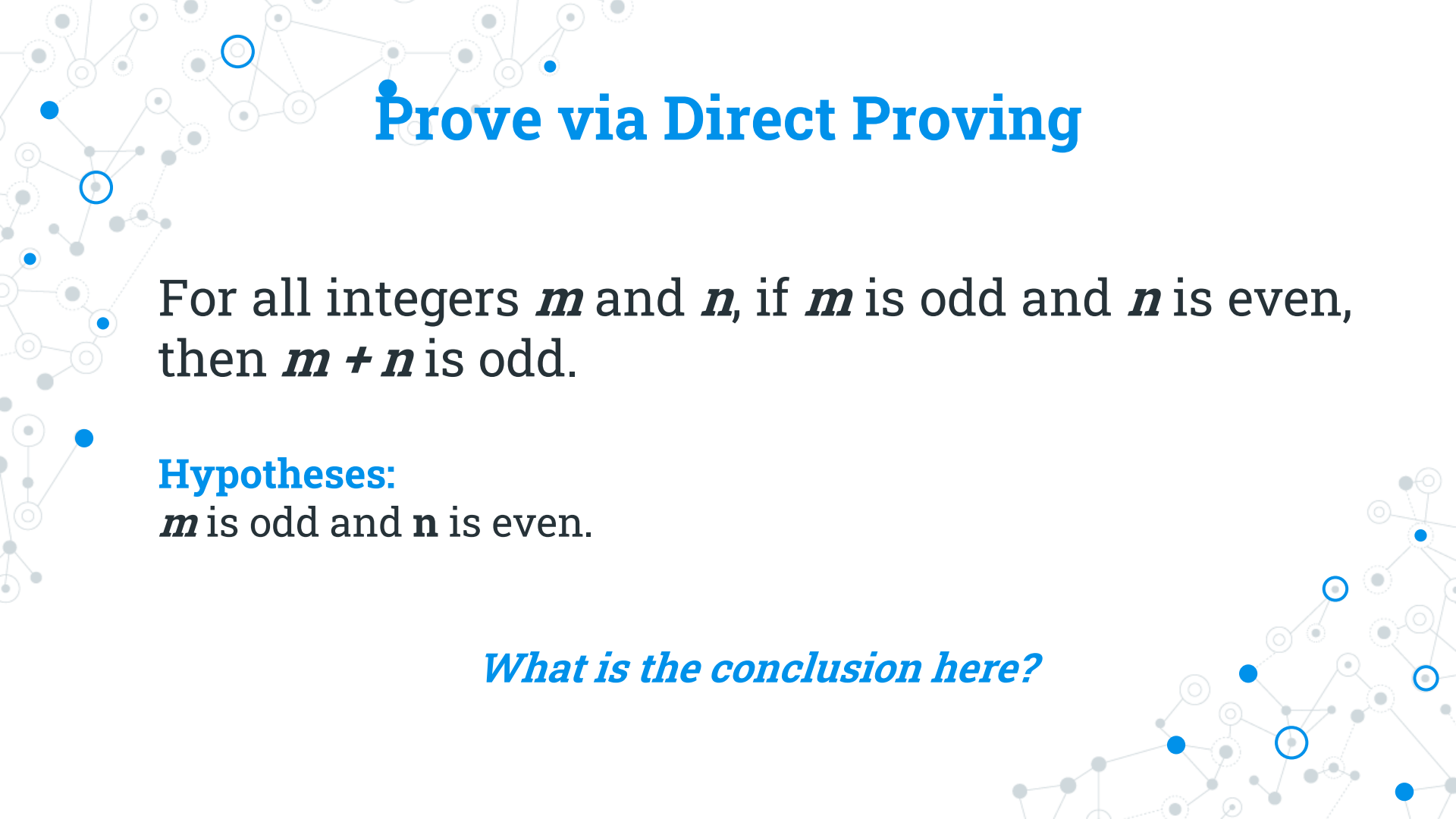


Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

m is odd and n is even.



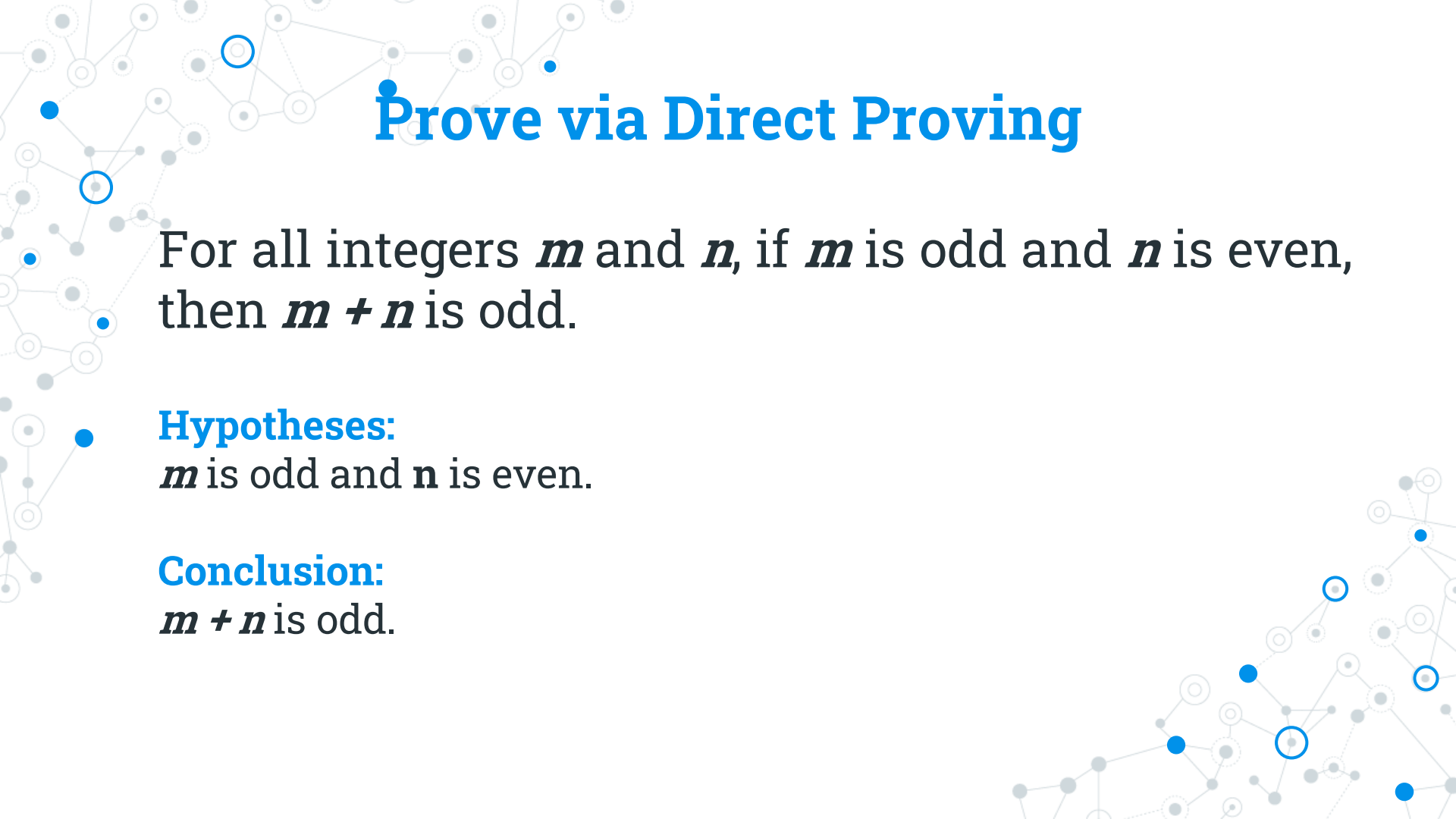
Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

m is odd and n is even.

What is the conclusion here?



Prove via Direct Proving

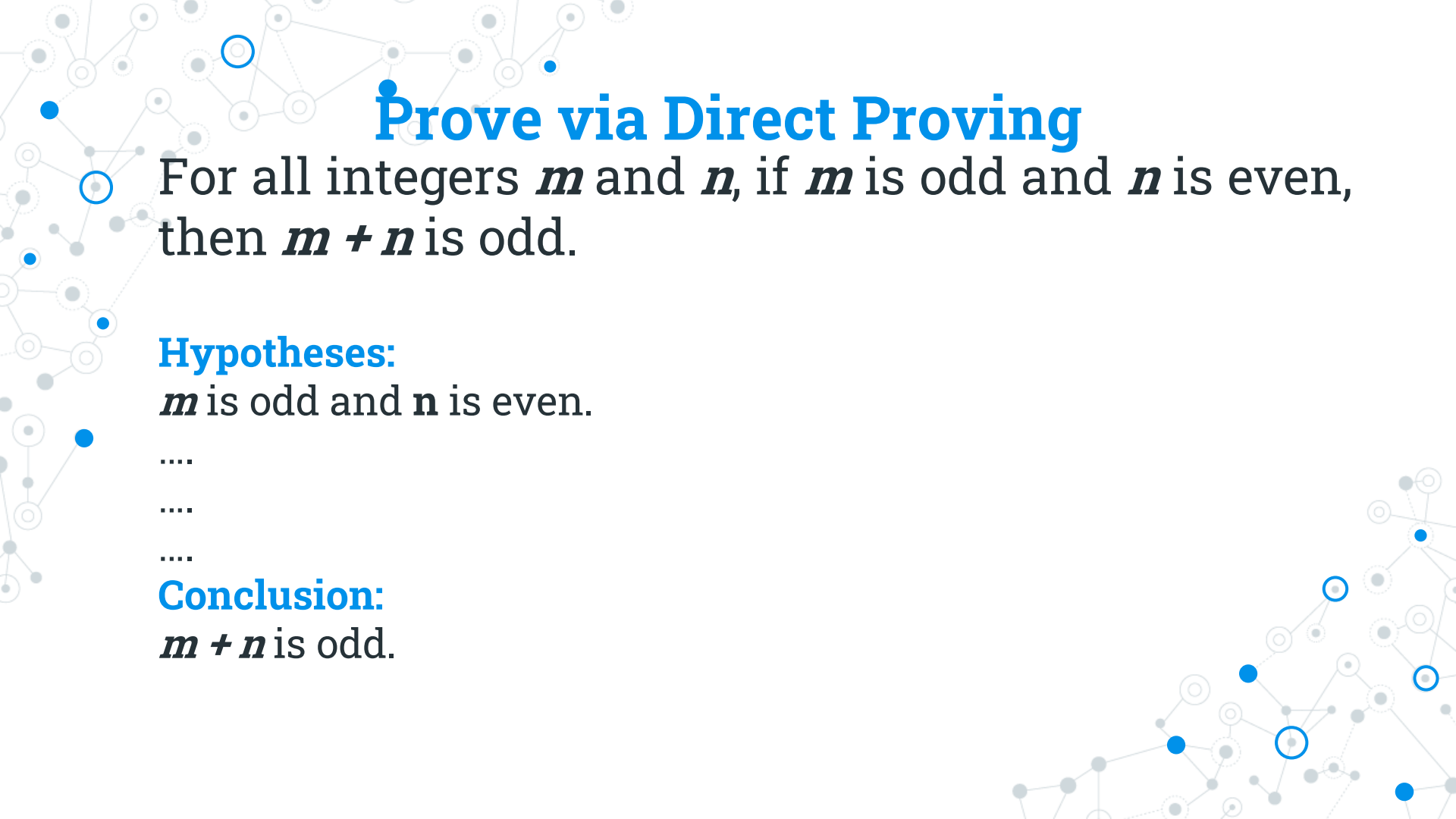
For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

m is odd and n is even.

Conclusion:

$m + n$ is odd.



Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

m is odd and n is even.

....

....

....

Conclusion:

$m + n$ is odd.

Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

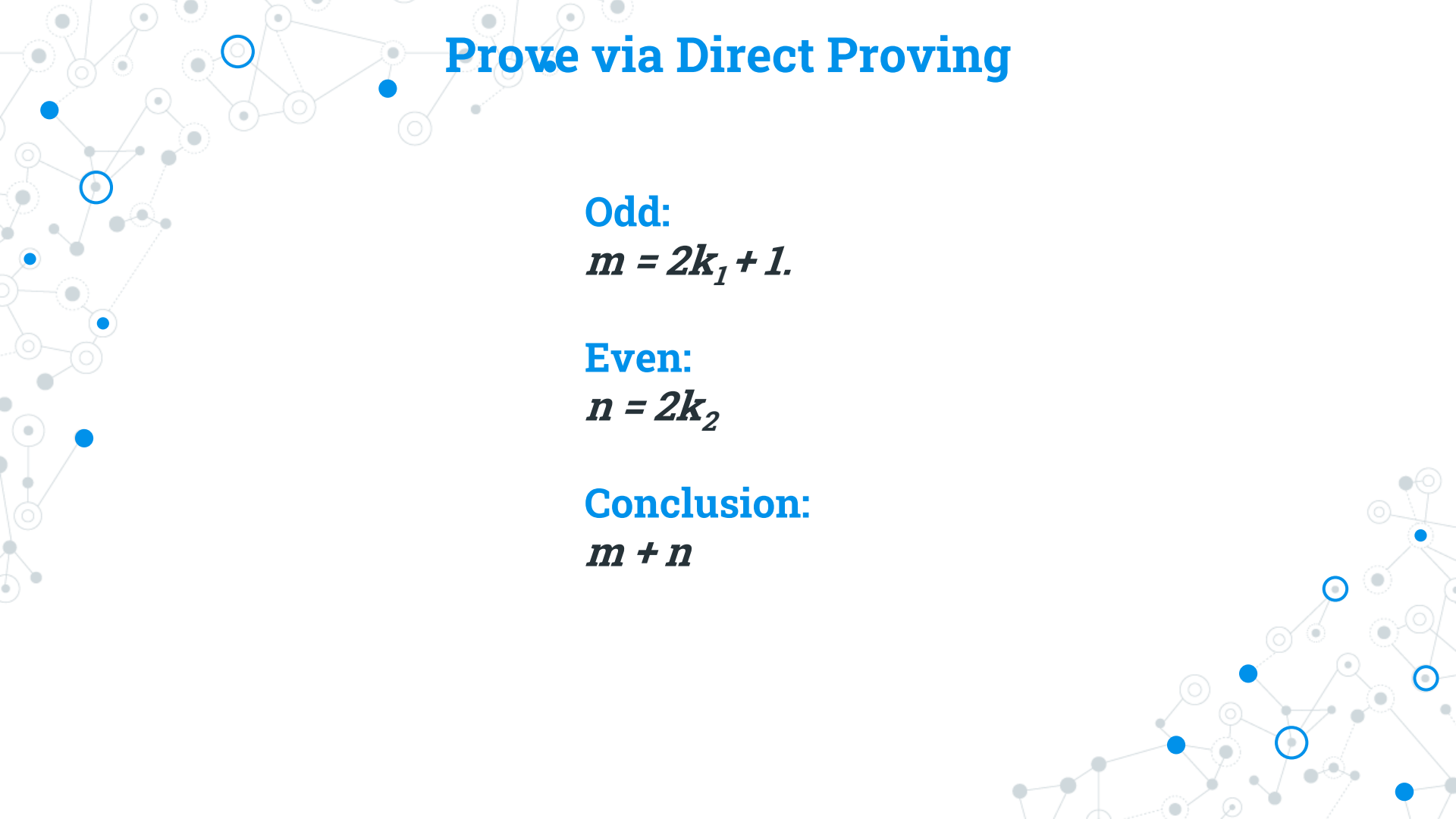
- m is odd and n is even.
- An integer m is odd if there exists an integer k_1 such that $m = 2k_1 + 1$

An integer n is even if there exists an integer k_2 such that $n = 2k_2$

...

Conclusion:

$m + n$ is odd.



Prove via Direct Proving

Odd:

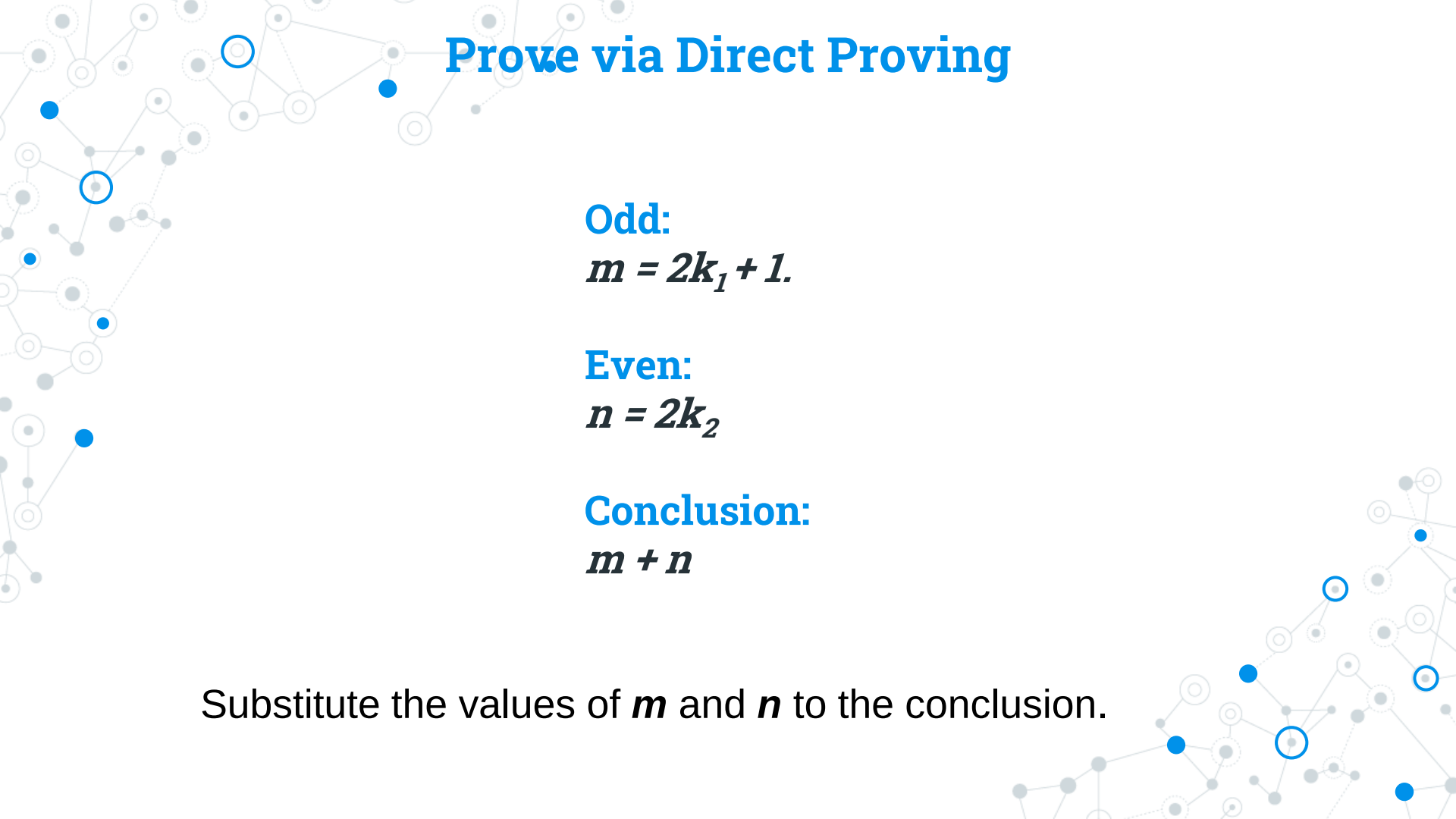
$$m = 2k_1 + 1.$$

Even:

$$n = 2k_2$$

Conclusion:

$$m + n$$



Prove via Direct Proving

Odd:

$$m = 2k_1 + 1.$$

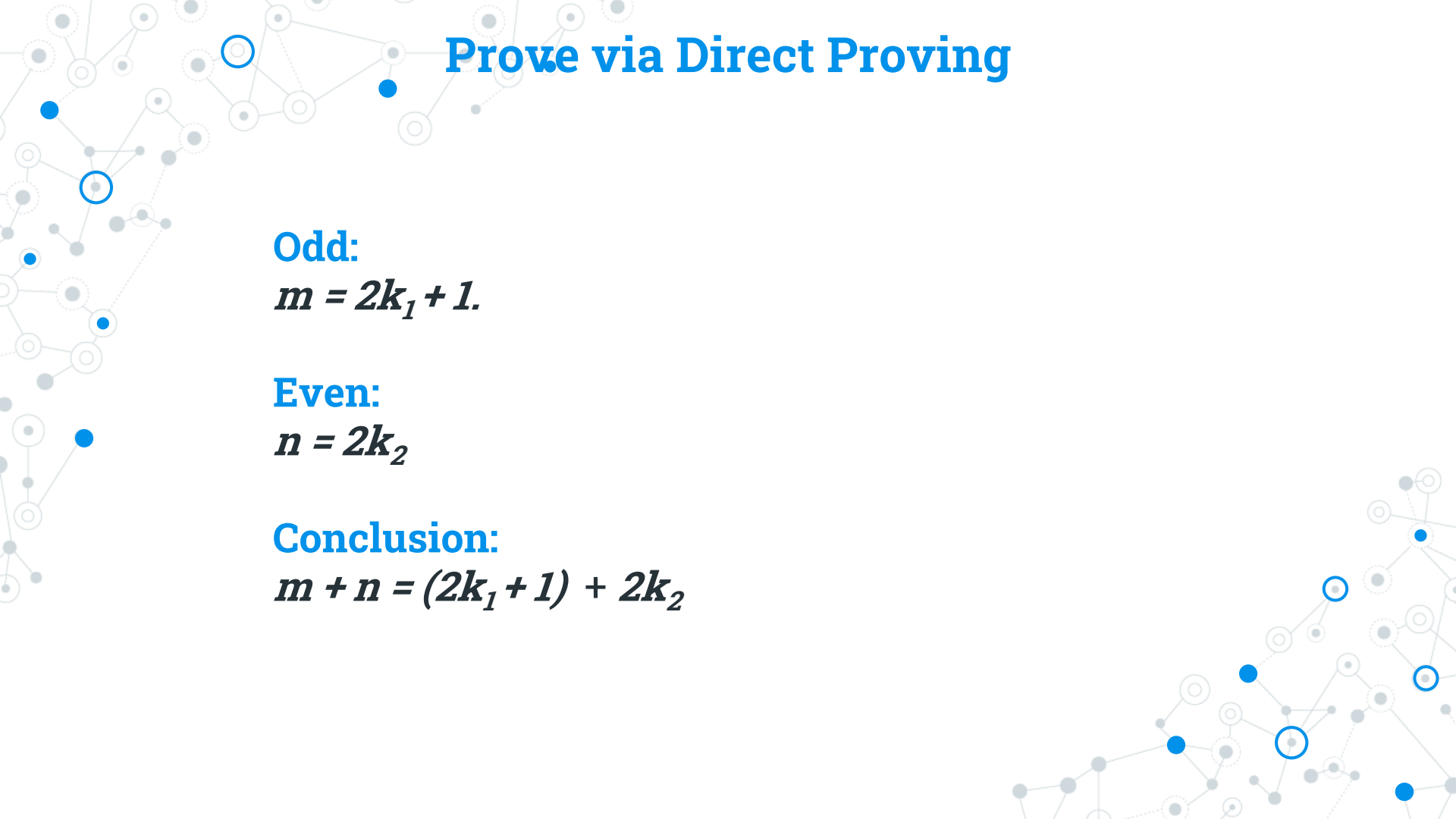
Even:

$$n = 2k_2$$

Conclusion:

$$m + n$$

Substitute the values of ***m*** and ***n*** to the conclusion.



Prove via Direct Proving

Odd:

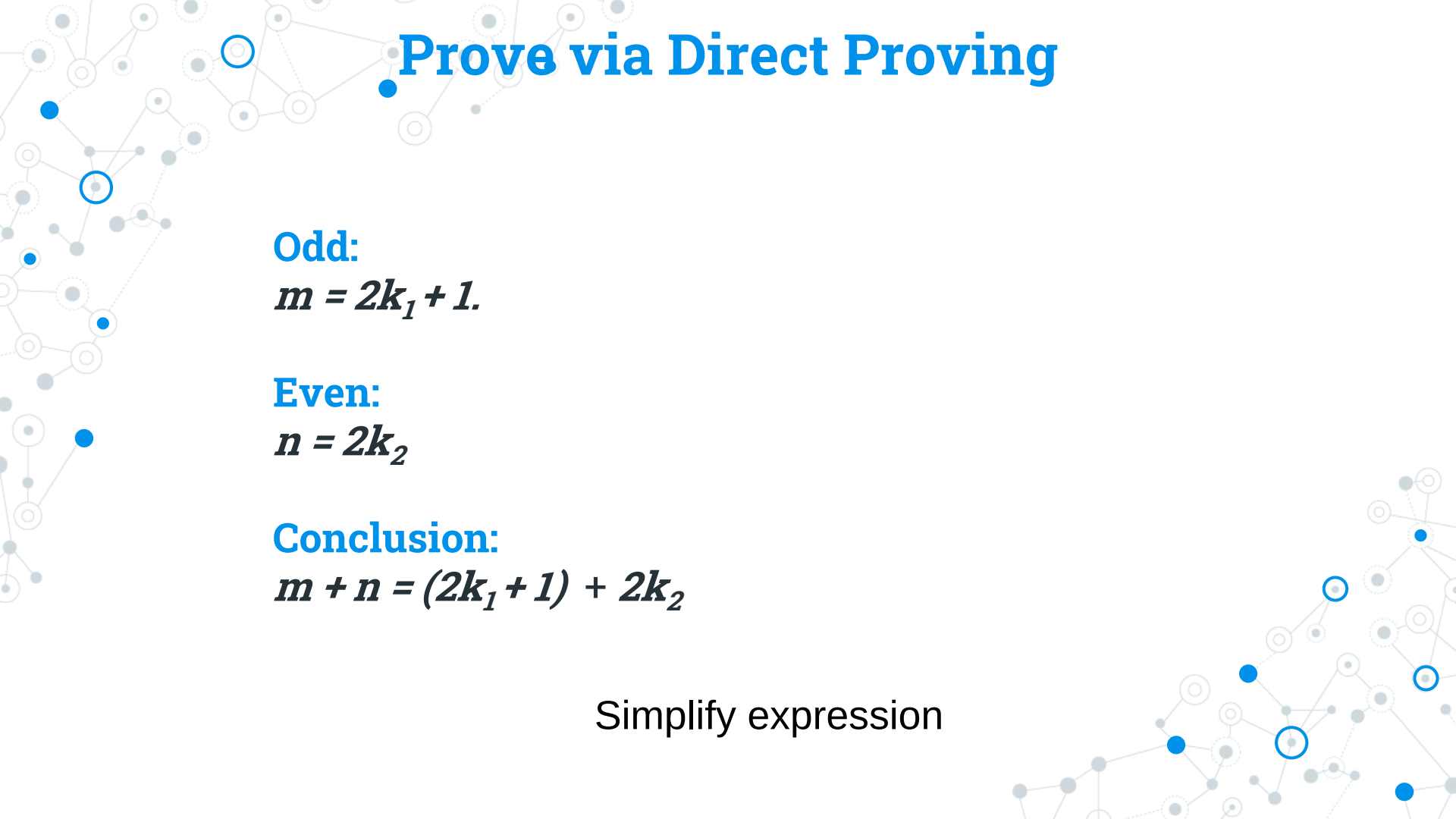
$$m = 2k_1 + 1.$$

Even:

$$n = 2k_2$$

Conclusion:

$$m + n = (2k_1 + 1) + 2k_2$$



Prove via Direct Proving

Odd:

$$m = 2k_1 + 1.$$

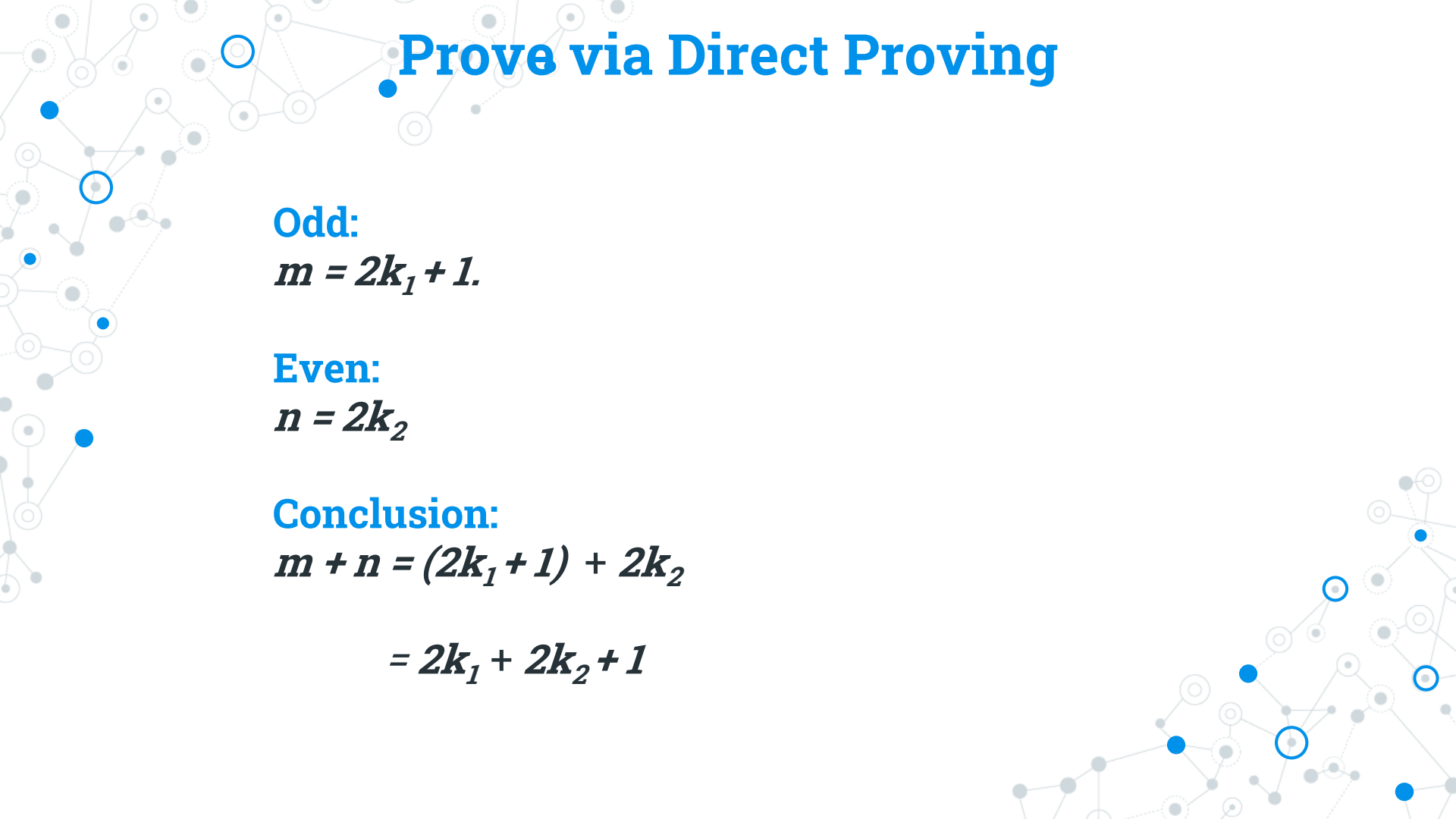
Even:

$$n = 2k_2$$

Conclusion:

$$m + n = (2k_1 + 1) + 2k_2$$

Simplify expression



Prove via Direct Proving

Odd:

$$m = 2k_1 + 1.$$

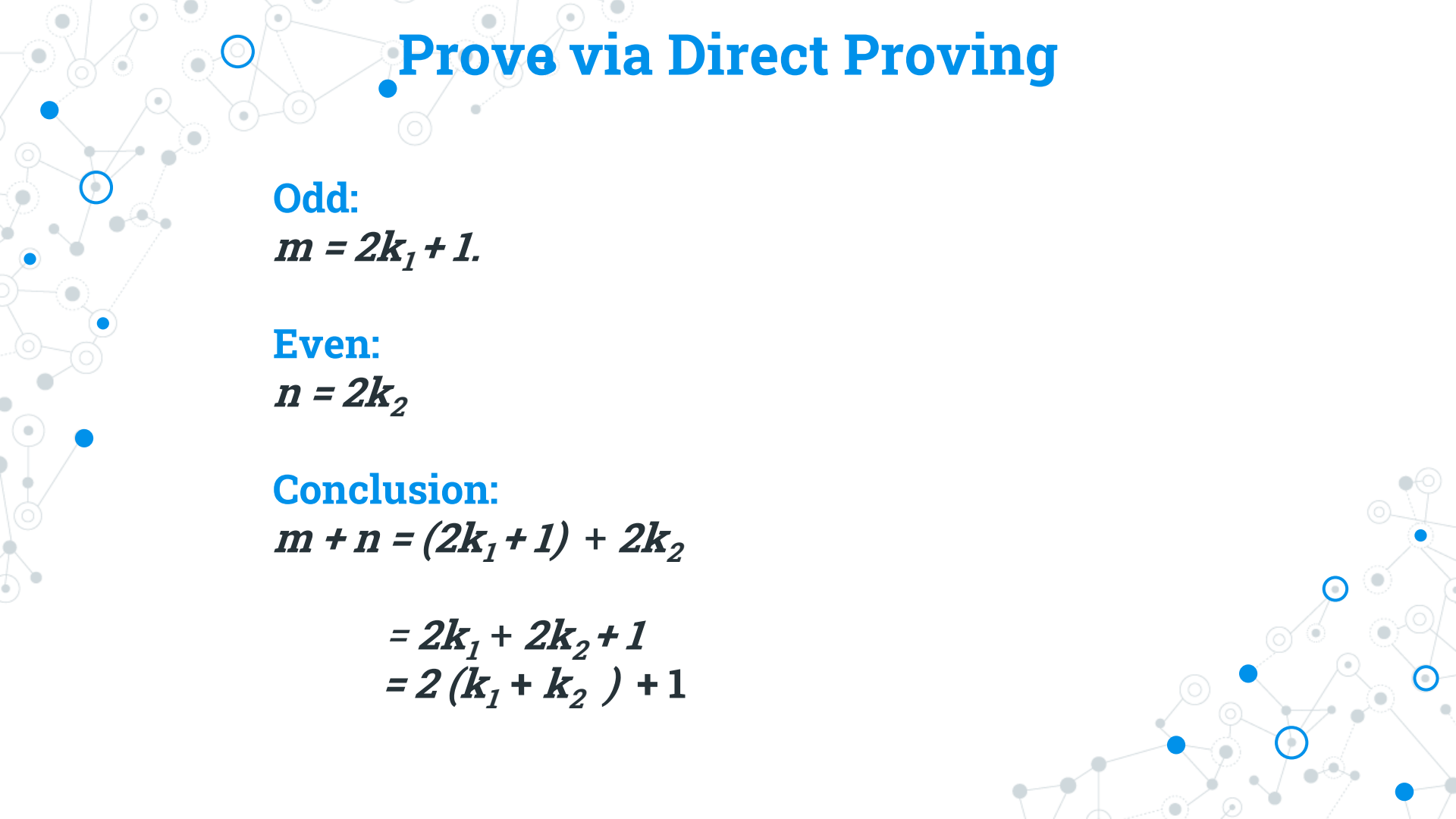
Even:

$$n = 2k_2$$

Conclusion:

$$m + n = (2k_1 + 1) + 2k_2$$

$$= 2k_1 + 2k_2 + 1$$



Prove via Direct Proving

Odd:

$$m = 2k_1 + 1.$$

Even:

$$n = 2k_2$$

Conclusion:

$$m + n = (2k_1 + 1) + 2k_2$$

$$= 2k_1 + 2k_2 + 1$$

$$= 2(k_1 + k_2) + 1$$

Prove via Direct Proving

Odd:

$$m = 2k_1 + 1.$$

Even:

$$n = 2k_2$$

Conclusion:

$$m + n = (2k_1 + 1) + 2k_2$$

$$\begin{aligned} &= 2k_1 + 2k_2 + 1 \\ &= 2(k_1 + k_2) + 1 \end{aligned}$$



Odd:

$$m = 2k + 1$$

Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

Hypotheses:

- m is odd and n is even.
- An integer m is odd if there exists an integer k_1 such that $m = 2k_1 + 1$

An integer n is even if there exists an integer k_2 such that $n = 2k_2$

$$m + n = (2k_1 + 1) + (2k_2) = 2(k_1 + k_2) + 1$$

Conclusion:

$m + n$ is odd.

Prove via Direct Proving

For all integers m and n , if m is odd and n is even, then $m + n$ is odd.

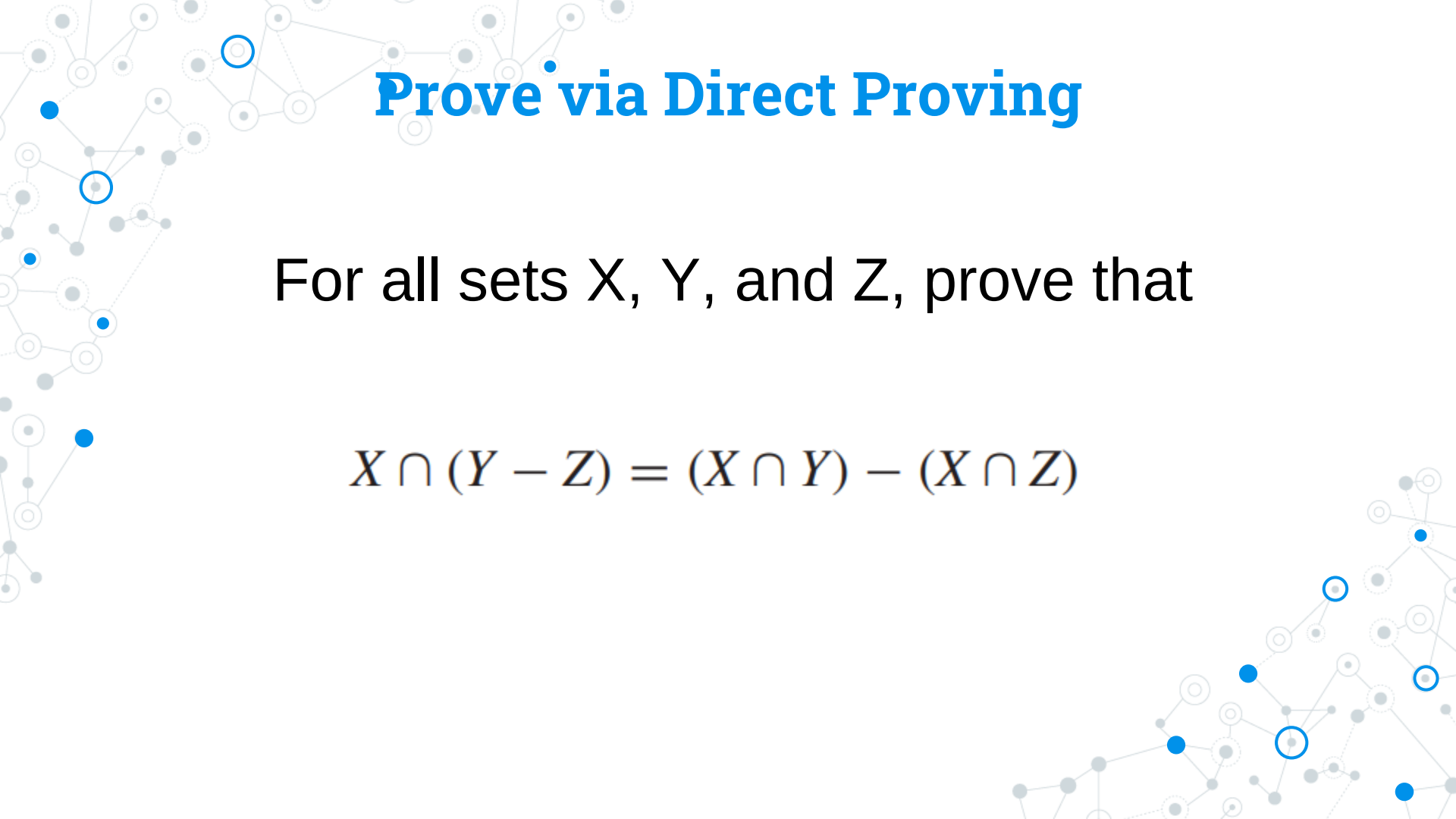
Hypotheses:

- m is odd and n is even.
- An integer m is odd if there exists an integer k_1 such that $m = 2k_1 + 1$

An integer n is even if there exists an integer k_2 such that $n = 2k_2$

$$m + n = (2k_1 + 1) + (2k_2) = 2(k_1 + k_2) + 1$$

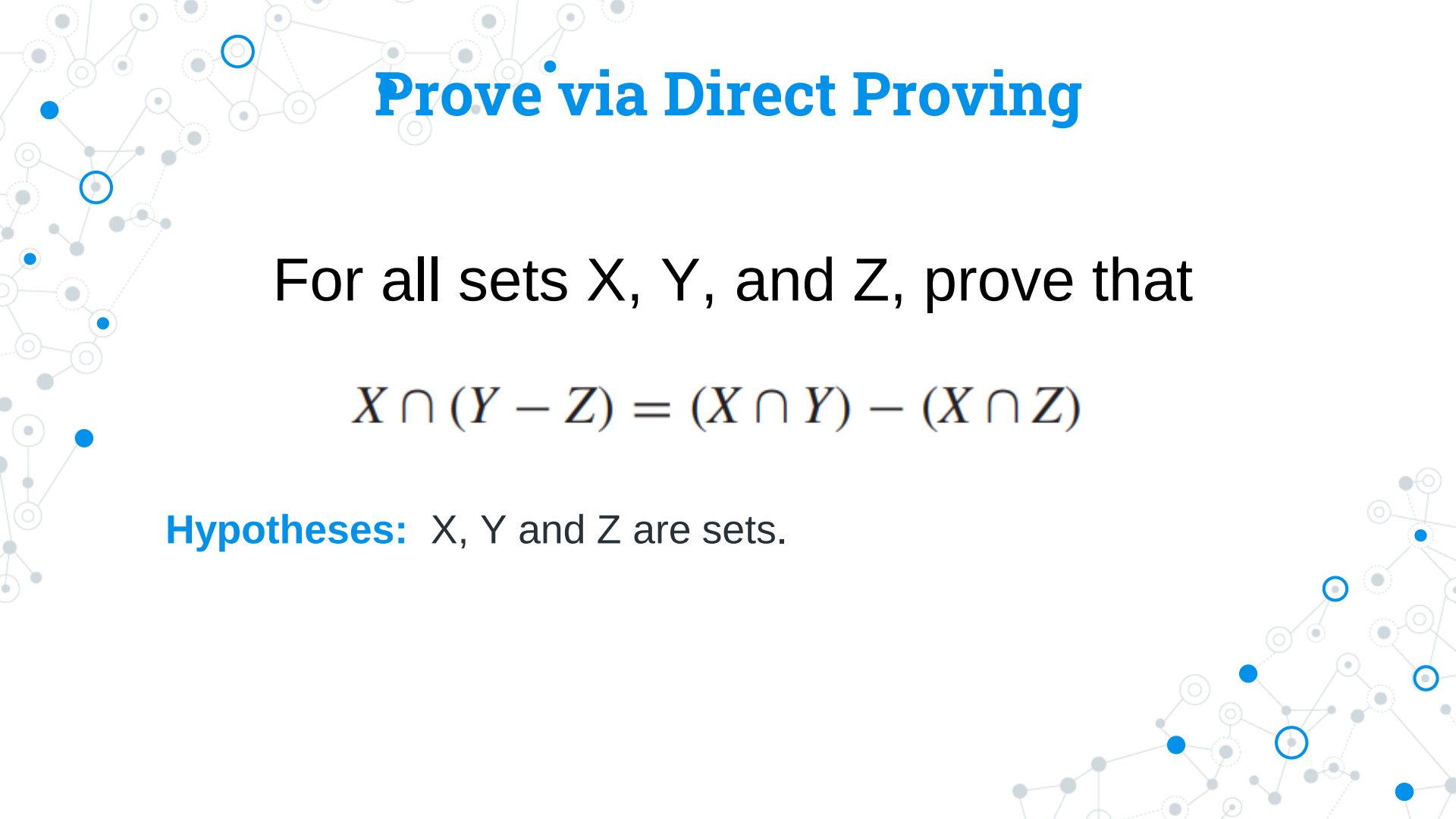
Thus, there exists an integer k (namely $k = k_1 + k_2$) such that $m + n = 2k + 1$. Therefore, $m + n$ is odd.



Prove via Direct Proving

For all sets X , Y , and Z , prove that

$$X \cap (Y - Z) = (X \cap Y) - (X \cap Z)$$

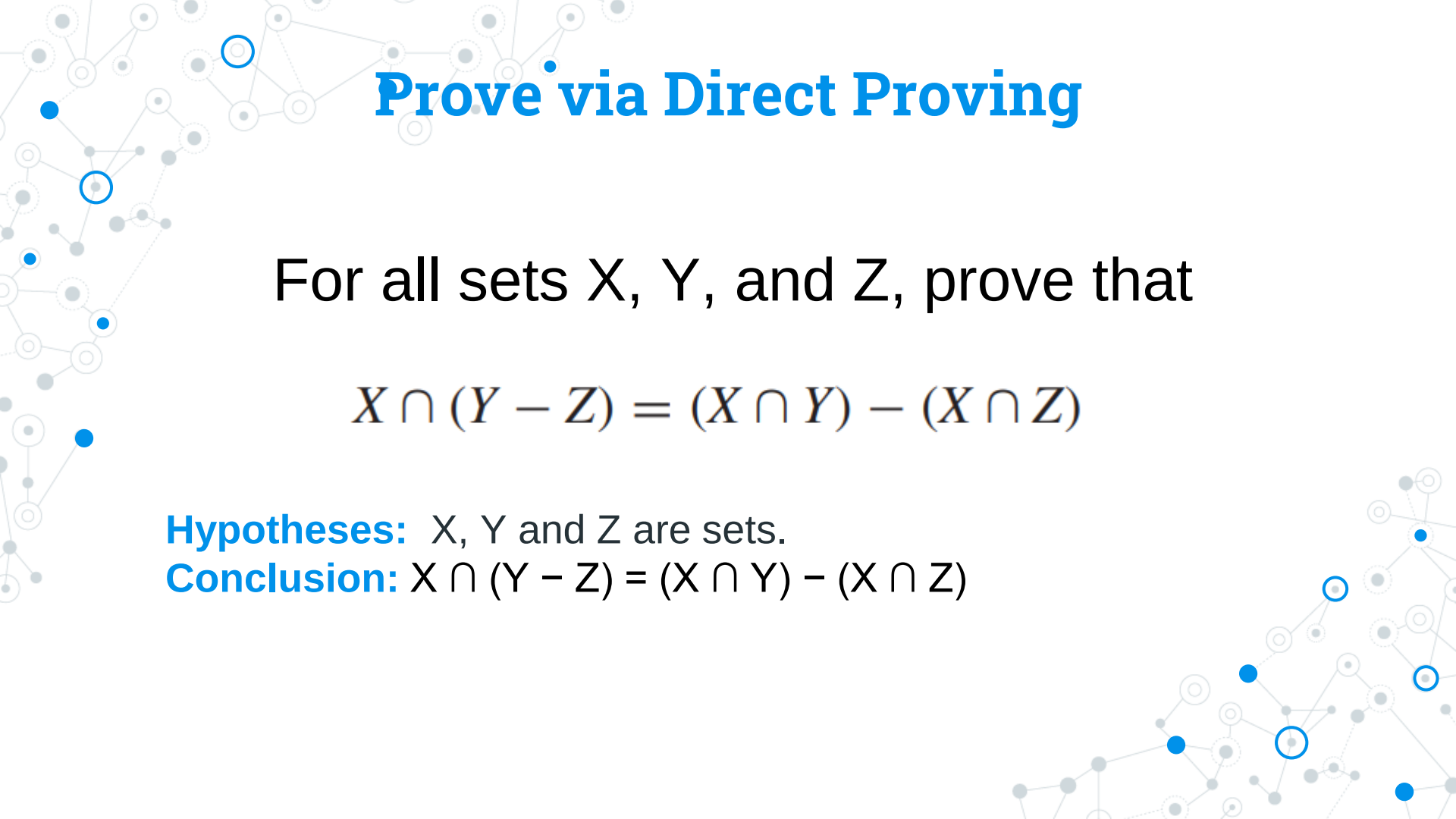


Prove via Direct Proving

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Hypotheses: X , Y and Z are sets.



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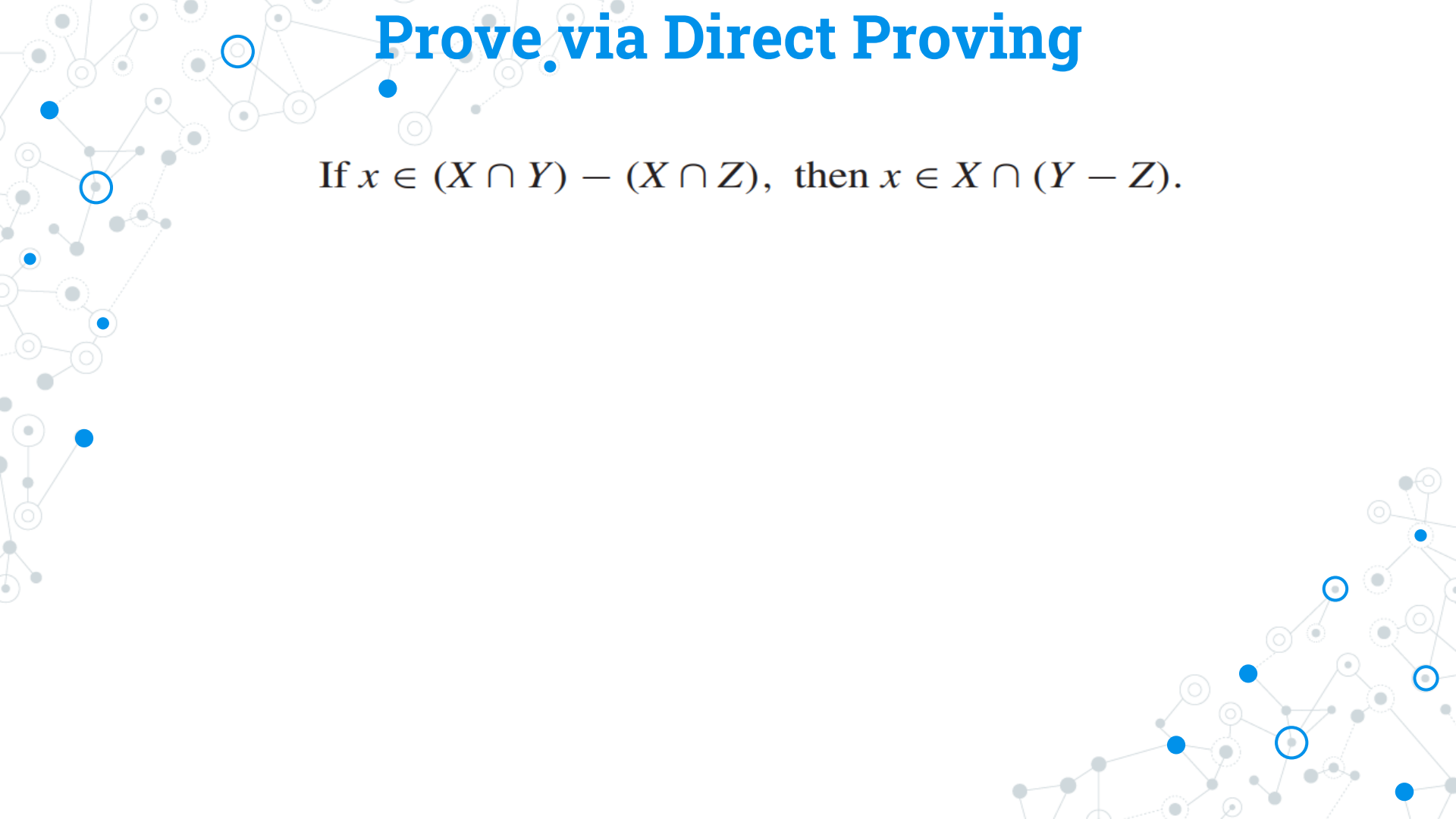
Hypotheses: X , Y and Z are sets.



If $x \in X \cap (Y - Z)$, then $x \in (X \cap Y) - (X \cap Z)$.

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Prove via Direct Proving

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Prove via Direct Proving

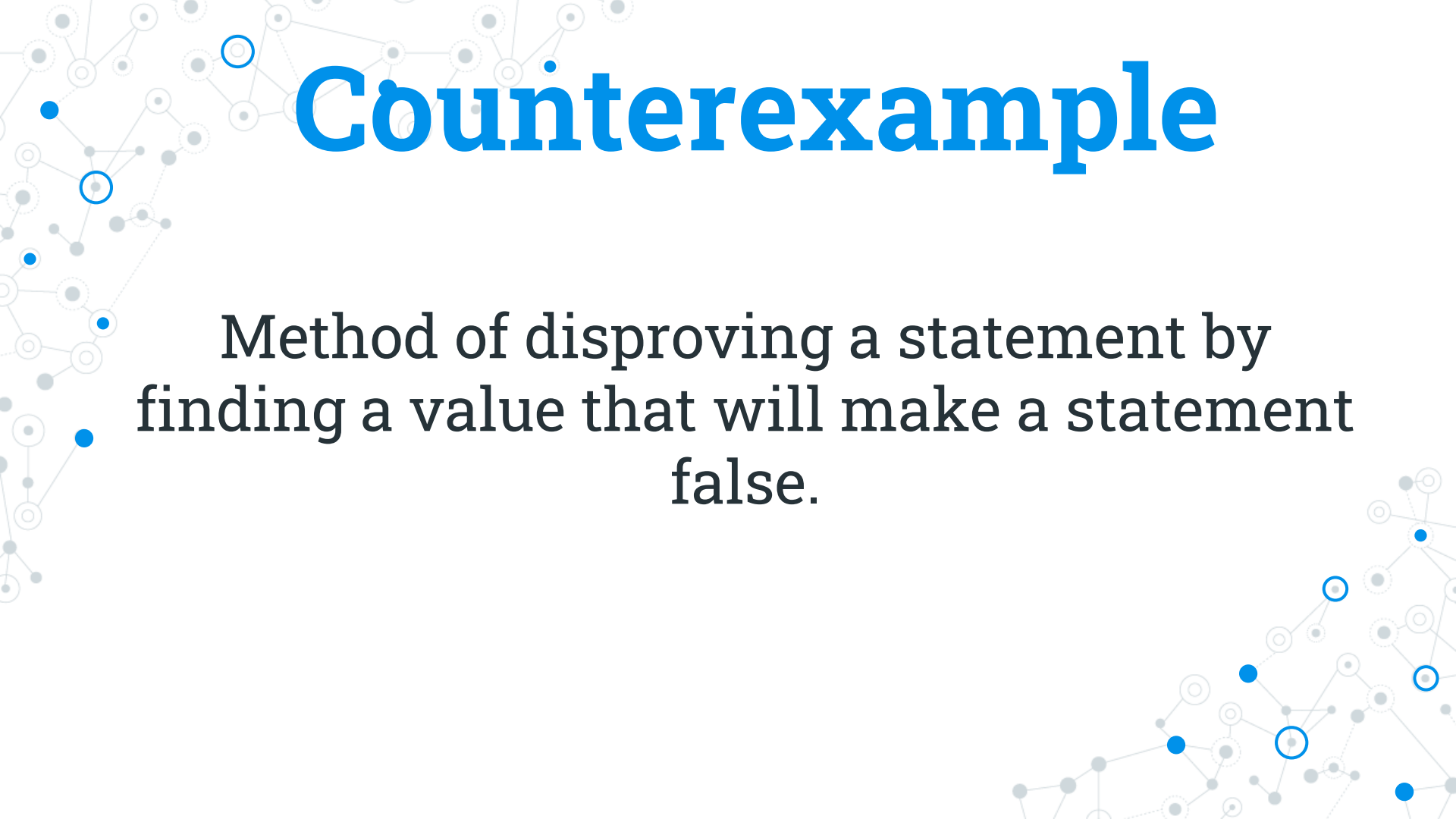
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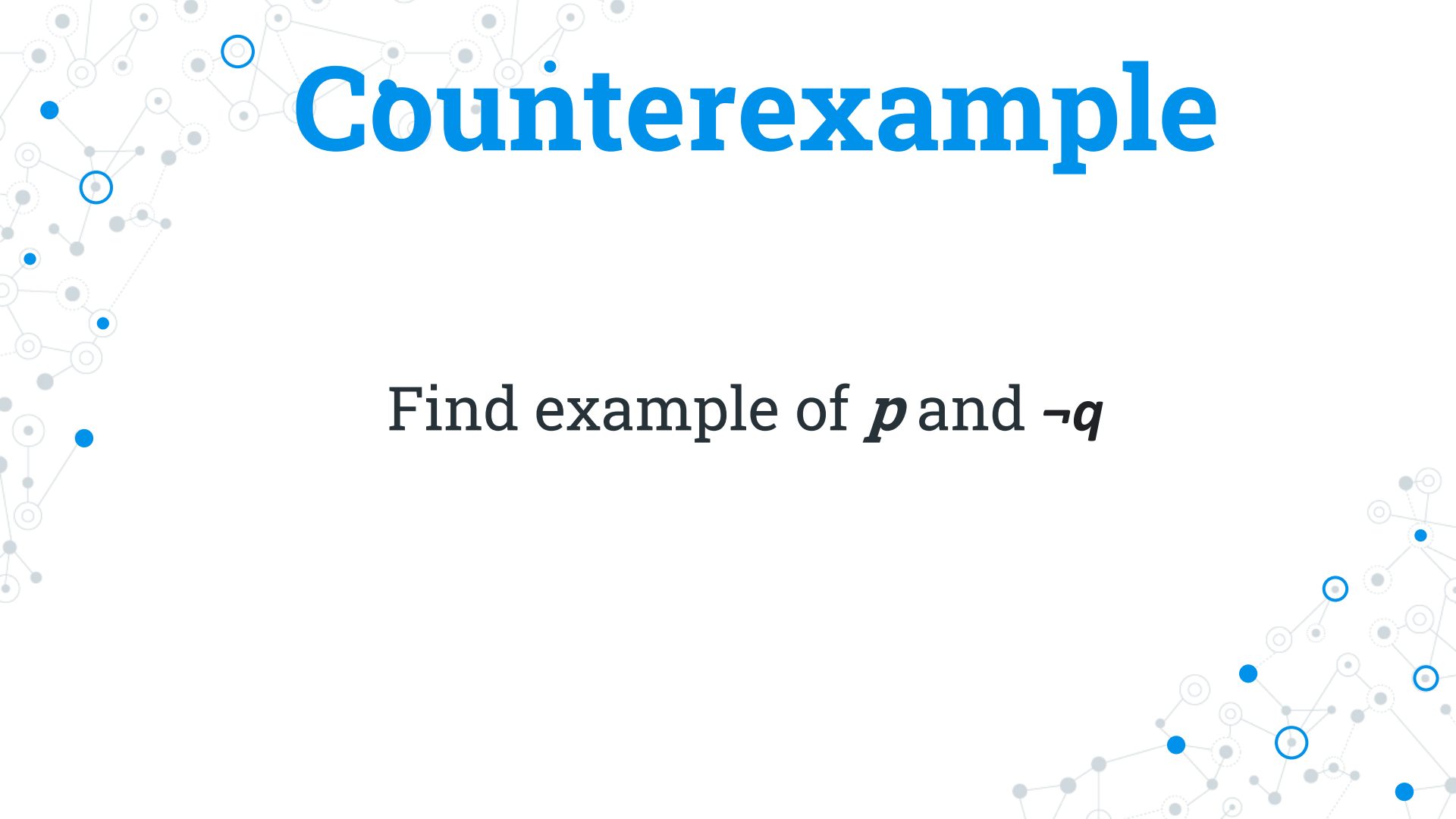
- ✓ If $x \in X \cap (Y - Z)$, then $x \in (X \cap Y) - (X \cap Z)$.
- ✓ If $x \in (X \cap Y) - (X \cap Z)$, then $x \in X \cap (Y - Z)$.

Conclusion: $X \cap (Y - Z) = (X \cap Y) - (X \cap Z)$

A decorative background featuring a network diagram with nodes and connecting lines. The nodes are represented by circles of varying shades of gray and blue, some with concentric circles. The lines are thin and gray, creating a web-like structure that spans the entire slide.

Counterexample

Method of disproving a statement by finding a value that will make a statement false.



Counterexample

Find example of p and $\neg q$

Counterexample

$$\forall n \in \mathbf{Z}^+ \\ (2^n + 1 \text{ is prime})$$

Counterexample

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$$n = 3$$

Counterexample

$$\forall n \in \mathbf{Z}^+$$

($2^n + 1$ is prime)

$$n = 3$$

$$2^3 + 1 = 9,$$

A decorative network graph pattern in the top-left corner, consisting of various sized grey and blue circles connected by thin grey lines. Some circles are solid blue, while others are hollow with blue outlines. The pattern is dense and extends towards the top-left corner.

Counterexample

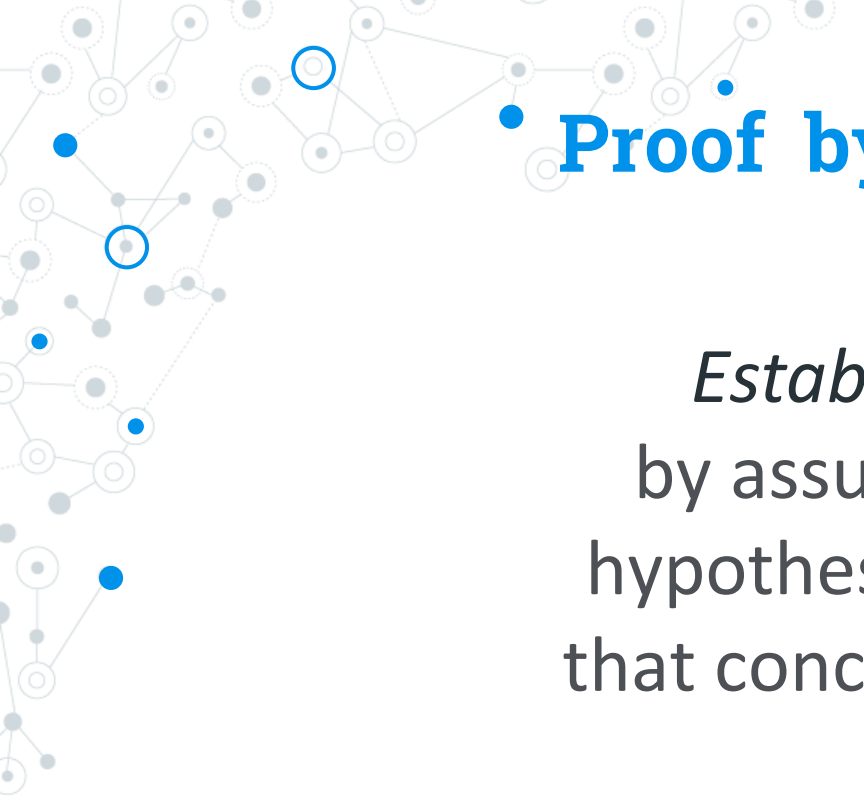
All primes are odd.

A decorative network graph pattern in the bottom-right corner, similar to the one in the top-left, featuring grey and blue circles connected by thin grey lines. It is positioned in the bottom-right corner of the slide.

Counterexample

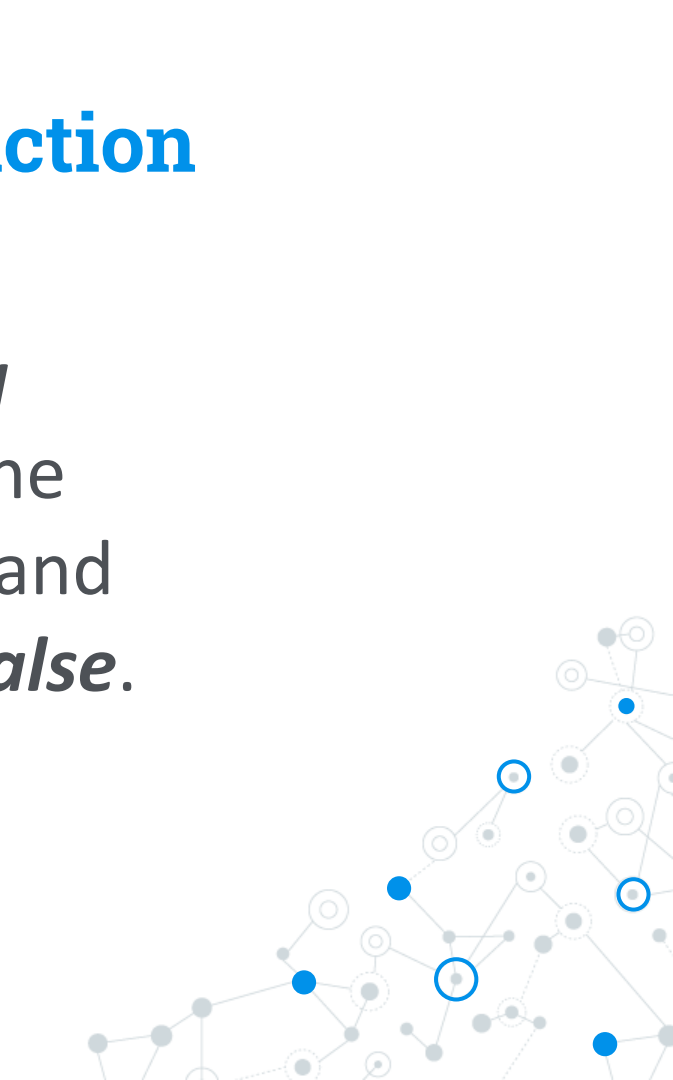
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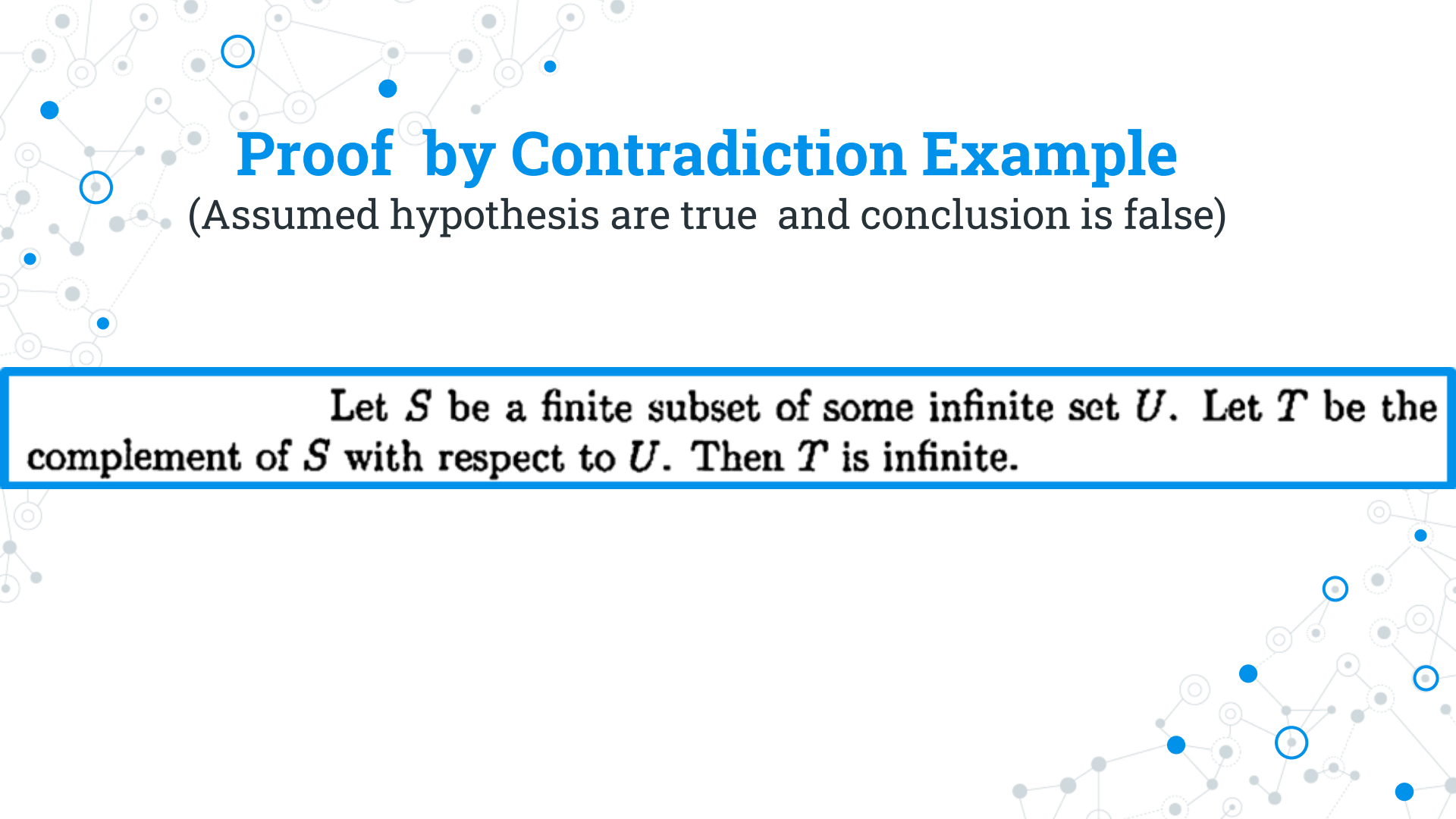
The integer 2 is a prime, but 2 is even.



• Proof by Contradiction

Establishes $p \rightarrow q$
by assuming that the
hypothesis p is **true** and
that conclusion q is **false**.

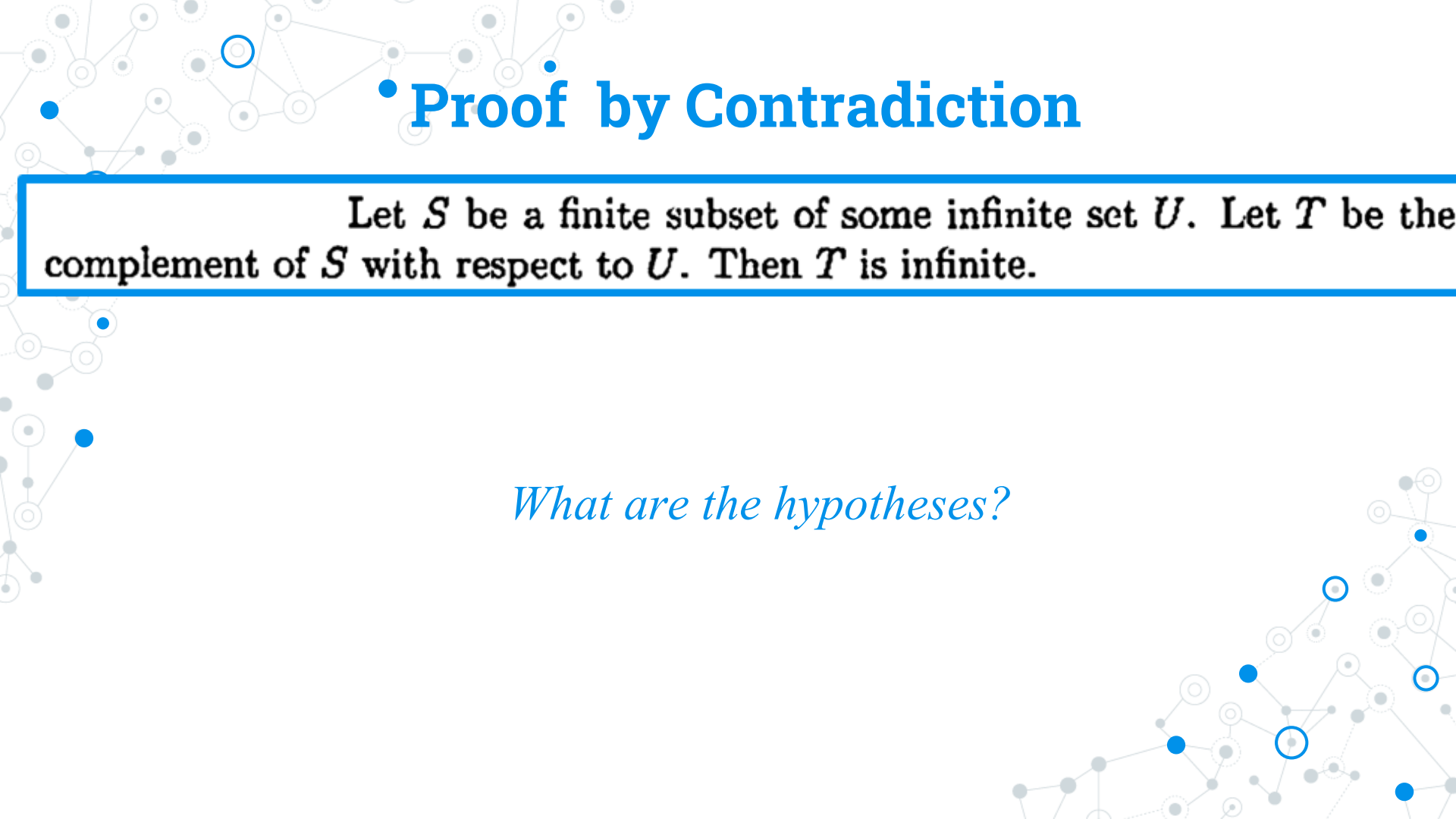




Proof by Contradiction Example

(Assumed hypothesis are true and conclusion is false)

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.



• Proof by Contradiction

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

What are the hypotheses?

• Proof by Contradiction

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

Hypotheses:

$|S| = s, s \in \mathbb{Z}^+$ (S is finite)

No integer u is $|U| = u$ (U is infinite)

$T \cup S = U$

$S \cap T = \emptyset$ (T is complement of S with respect to U)

Assume:

T is finite

$|T| = t, t \in \mathbb{Z}^+$

• Proof by Contradiction

Let S be a finite subset of some infinite set U . Let T be the complement of S with respect to U . Then T is infinite.

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• No integer u is $|U| = u$

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Assume:

T is finite

$$|T| = t, t \in \mathbb{Z}^+$$

Proving:

$$|S| = s \text{ and } |T| = t$$

Because $S \cup T = U$ and $S \cap T = \emptyset$, then $|U| = s + t$

Therefore, U is finite, because there is an integer $(s+t)$ that describes the number of elements in set U .

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Proving contradicted to initial hypothesis, by assuming conclusion false.

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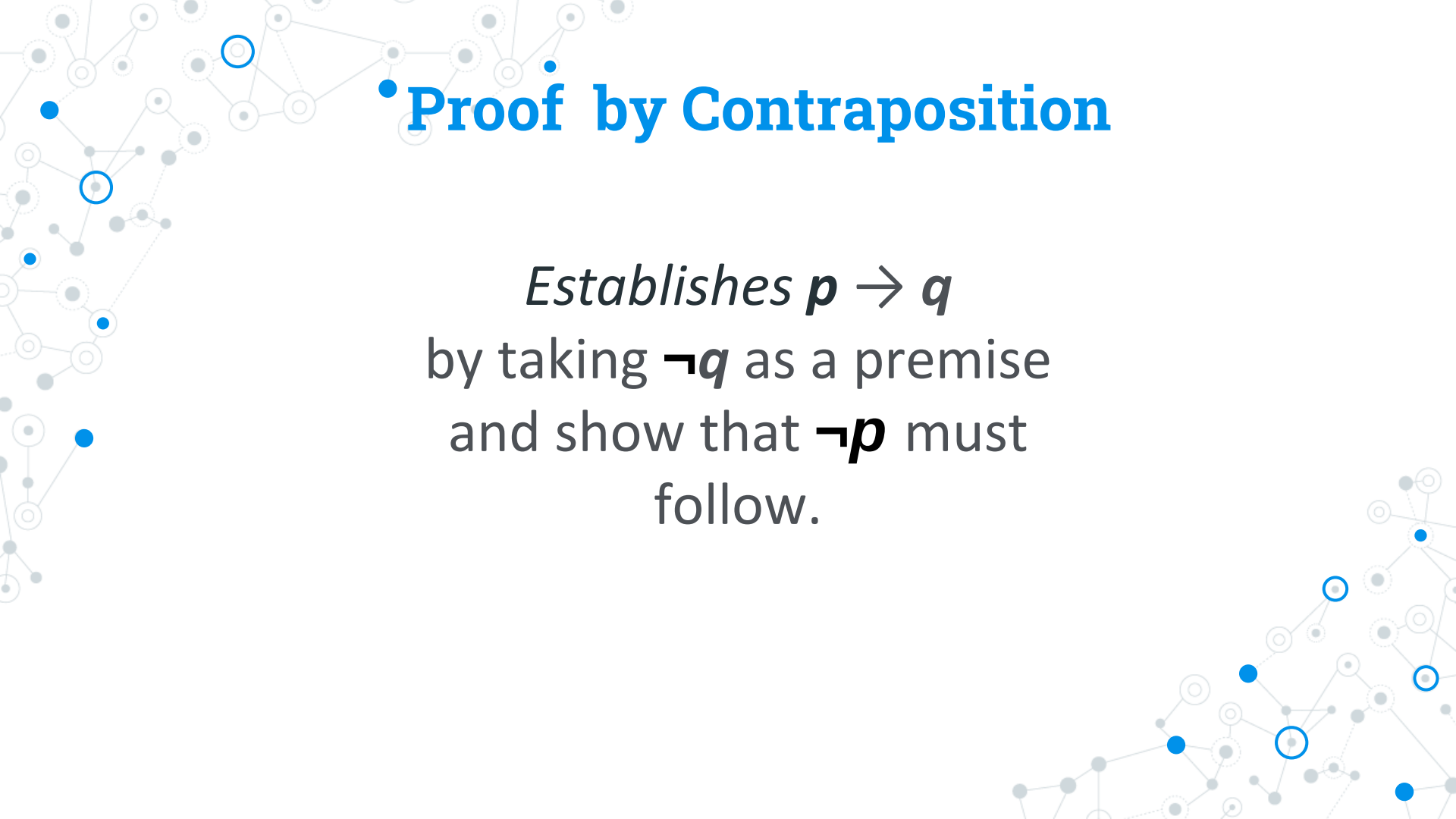
Proving:

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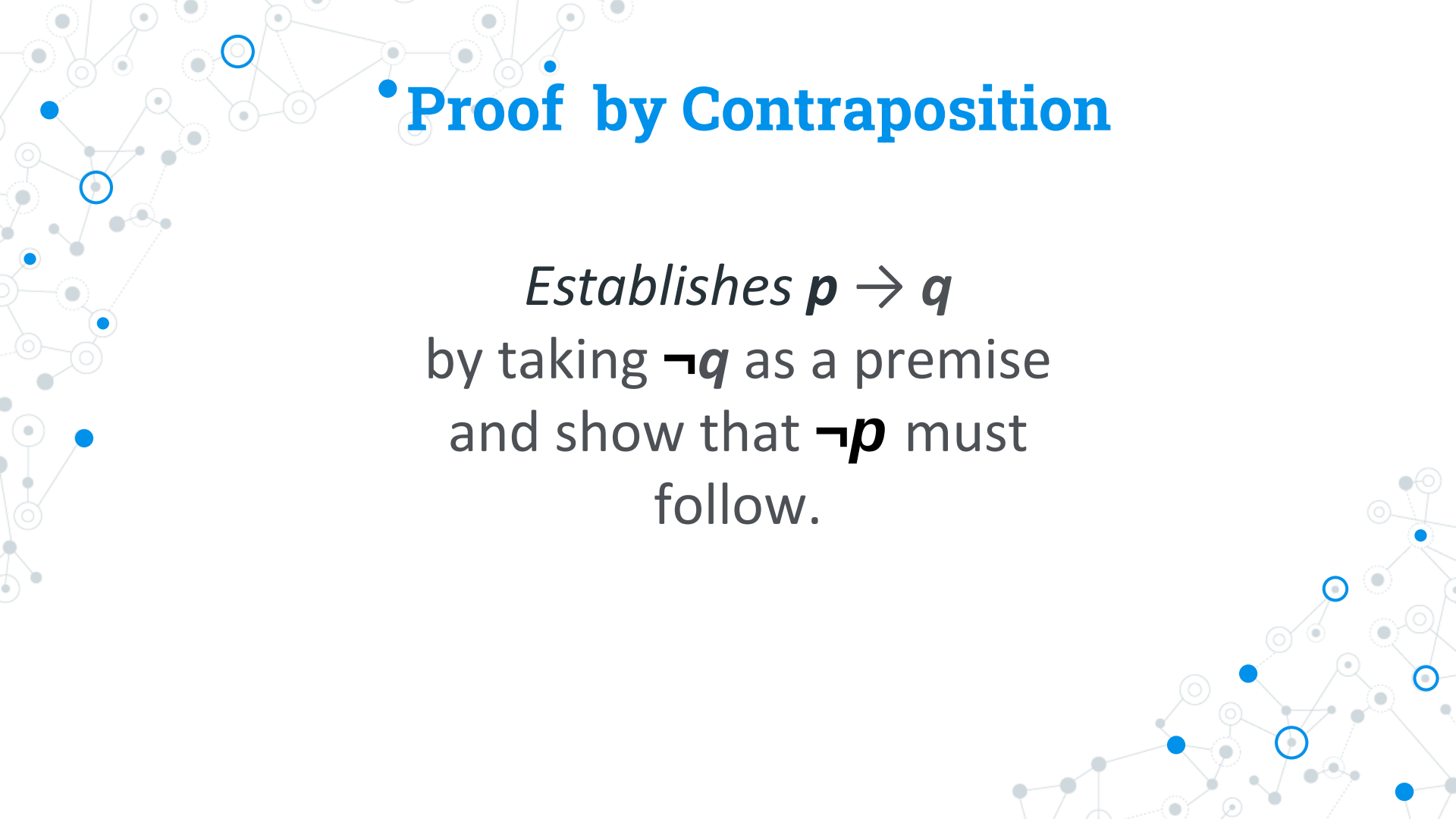
Proving contradicted to initial hypothesis, by assuming conclusion false. Therefore, conclusion is true.



• Proof by Contraposition

Establishes $p \rightarrow q$

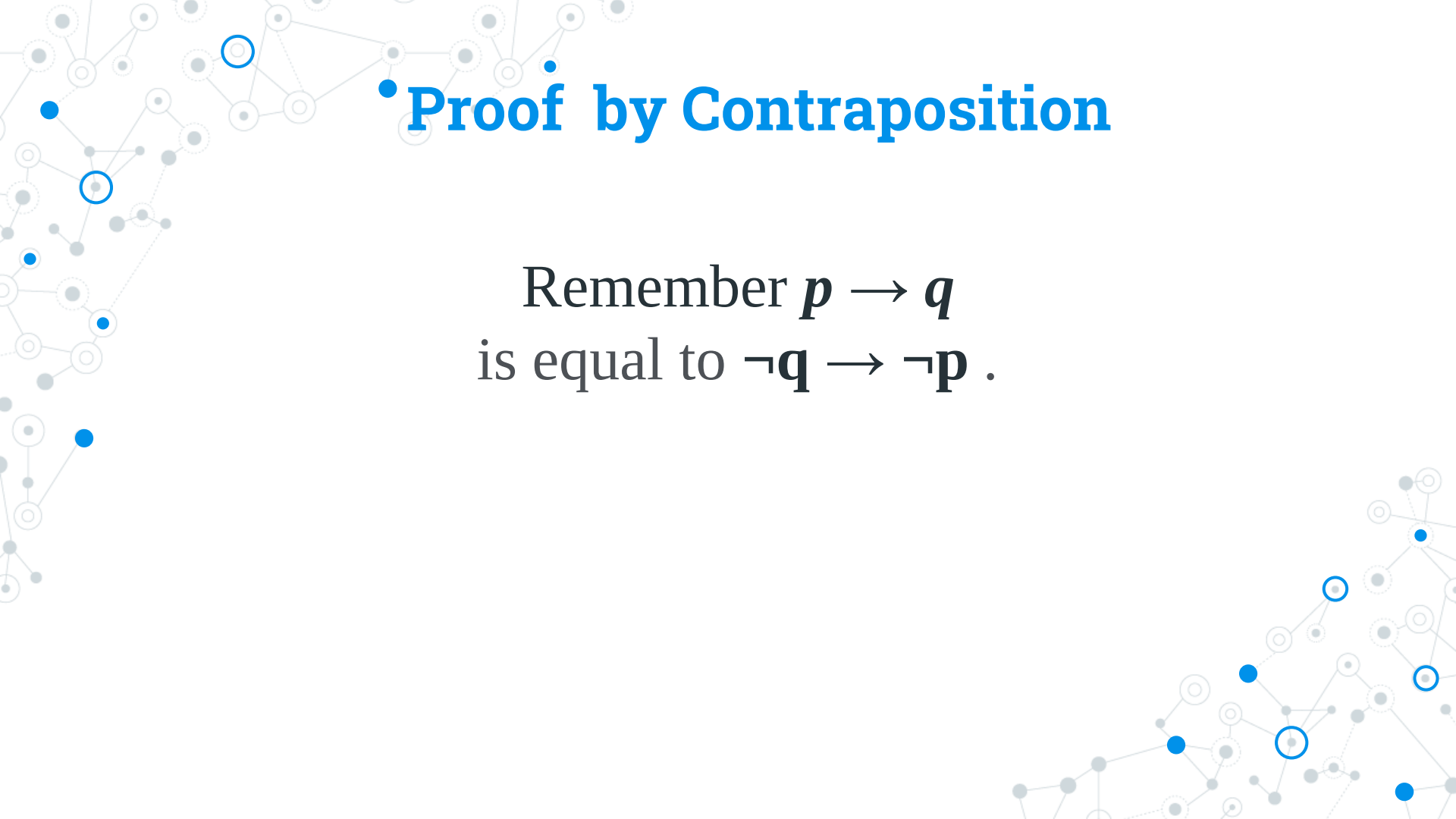
by taking $\neg q$ as a premise
and show that $\neg p$ must
follow.



• Proof by Contraposition

Establishes $p \rightarrow q$

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• Proof by Contraposition

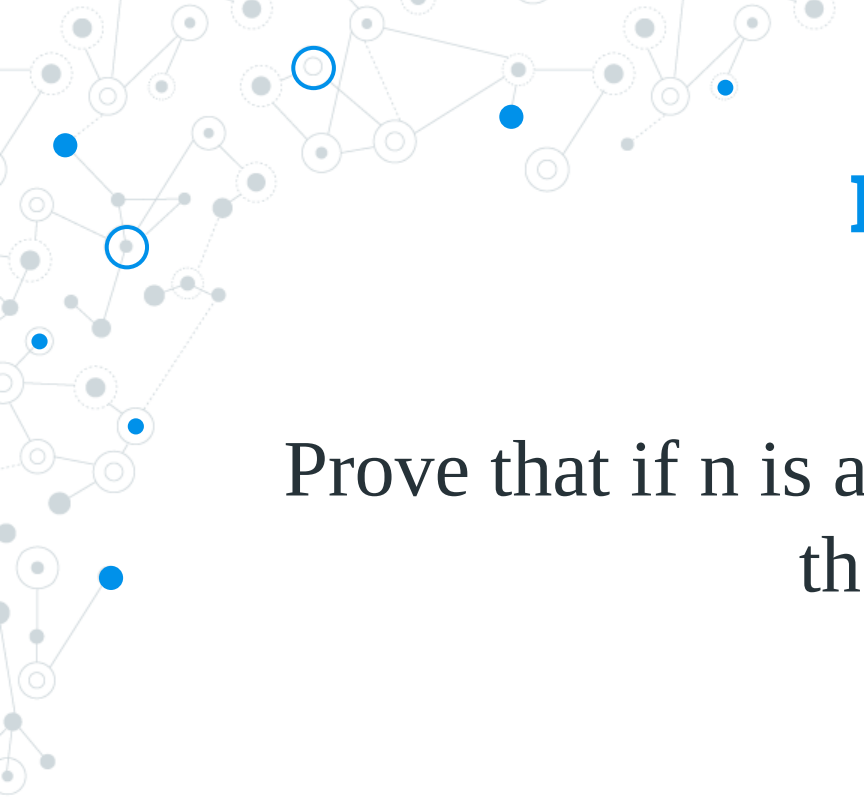
Remember $p \rightarrow q$
is equal to $\neg q \rightarrow \neg p$.



Proof by Contraposition

Assume $\neg q$
Conclude $\neg p$

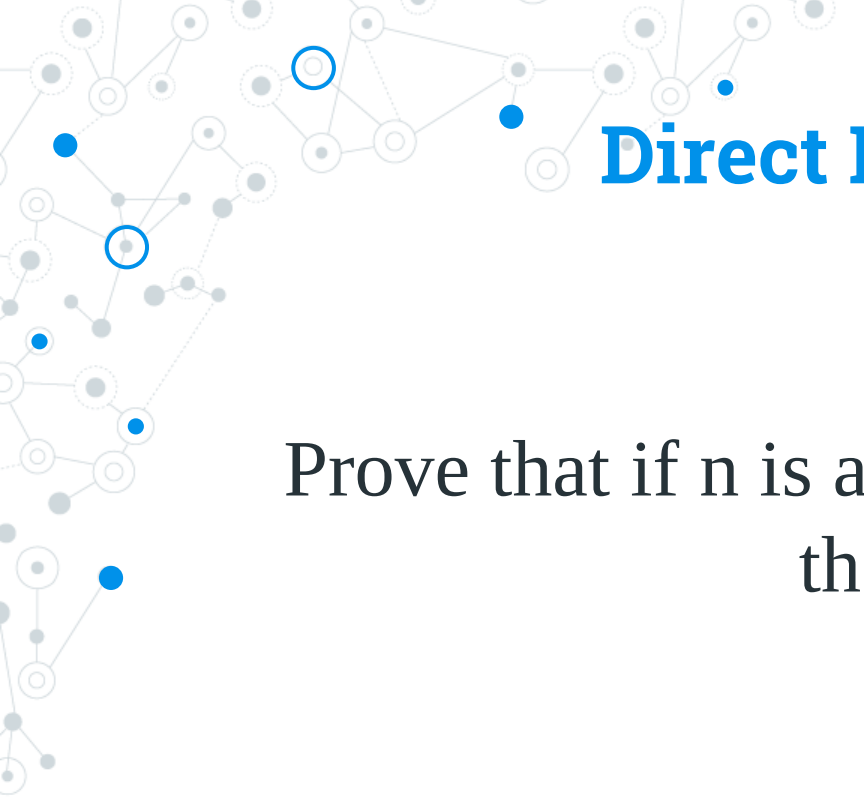




Prove

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.



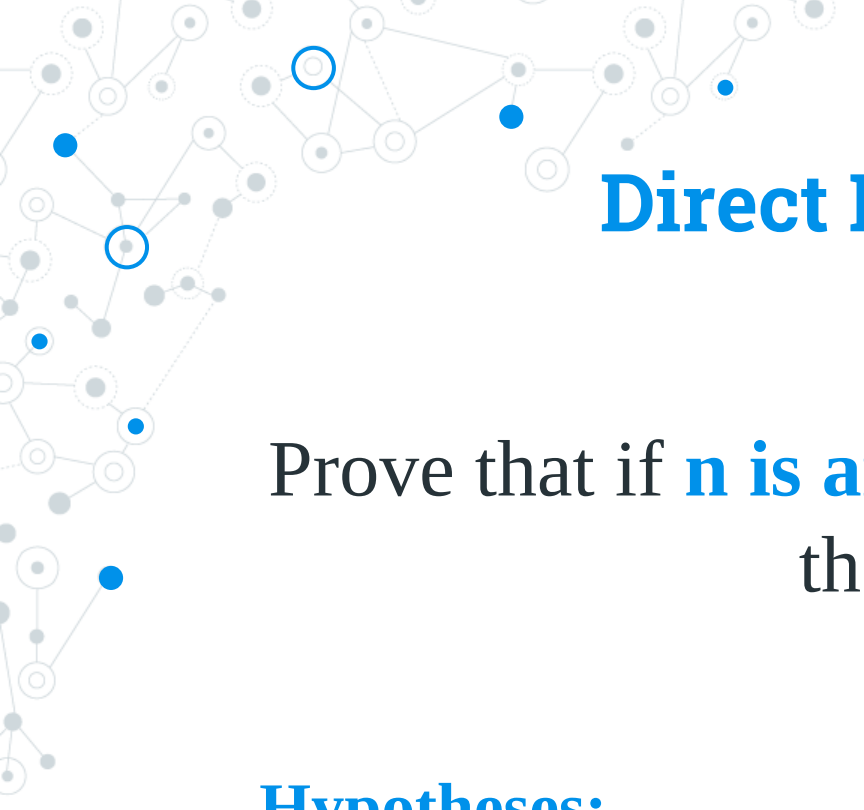


Direct Proof ($p \rightarrow q$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

What are the hypotheses ?



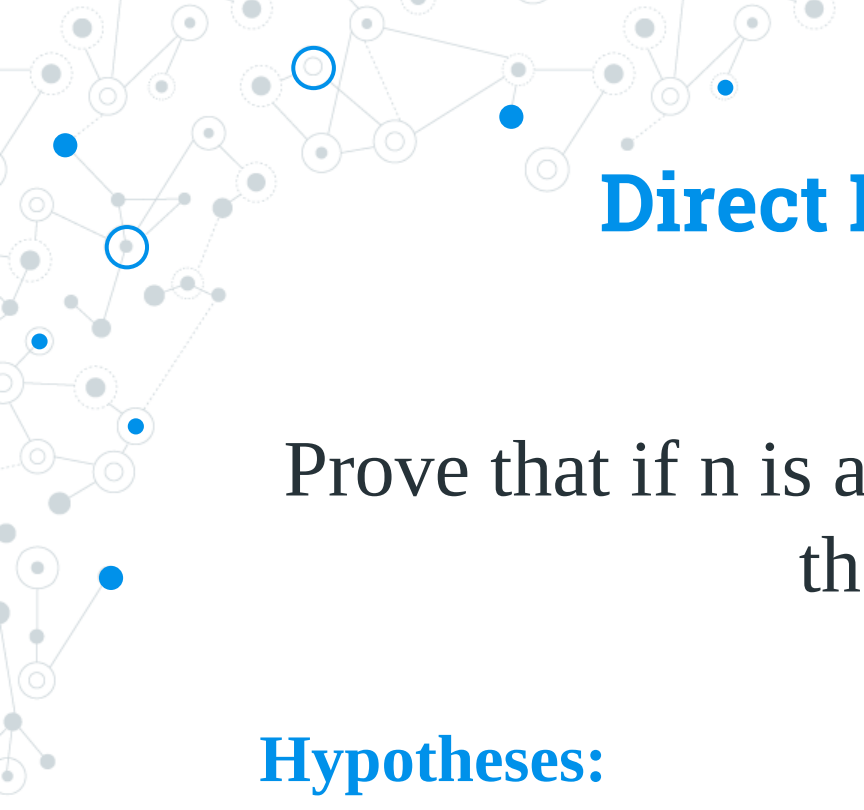


Direct Proof ($p \rightarrow q$)

Prove that if **n is an integer** and $3n + 2$ is odd, then n is odd.

Hypotheses:



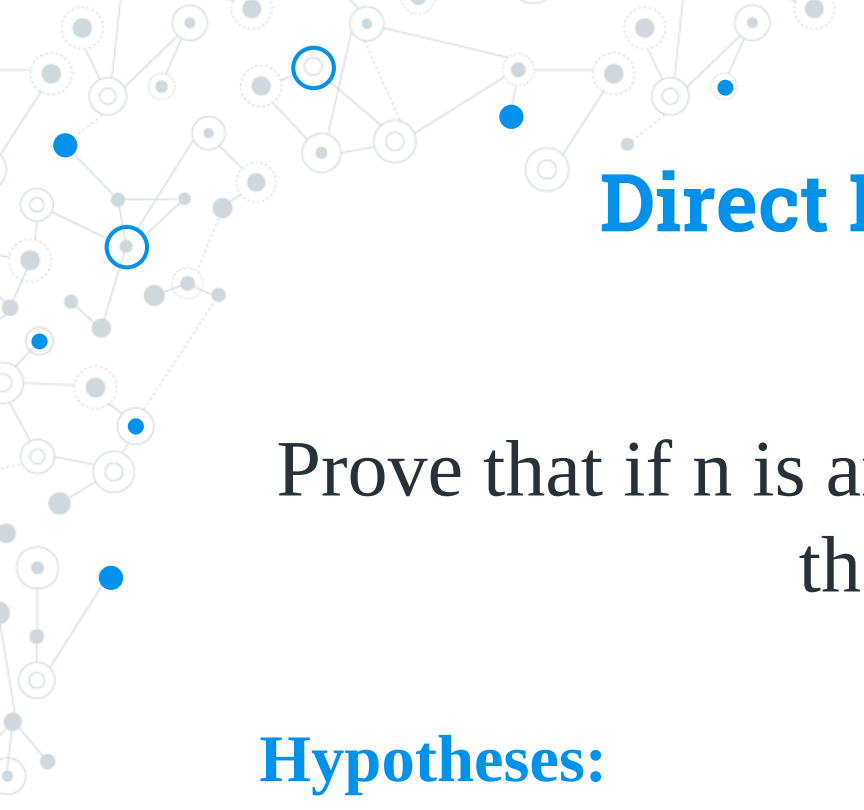


Direct Proof ($p \rightarrow q$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Hypotheses:

$$n \in \mathbb{Z}$$

Direct Proof ($p \rightarrow q$)

Prove that if n is an integer and **$3n + 2$ is odd**,
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- **Direct Proof ($p \rightarrow q$)**

Prove that if n is an integer and **$3n + 2$ is odd**,
then n is odd.

Hypotheses:

$$n \in \mathbb{Z}$$

$$3n + 2 = 2k + 1, \text{ where } k \in \mathbb{Z}$$

What is the conclusion ?

- **Direct Proof ($p \rightarrow q$)**

Prove that if n is an integer and **$3n + 2$ is odd**,
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Hypotheses:

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Conclusion:

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Manipulations:

$$3n + 2 = 2k + 1$$

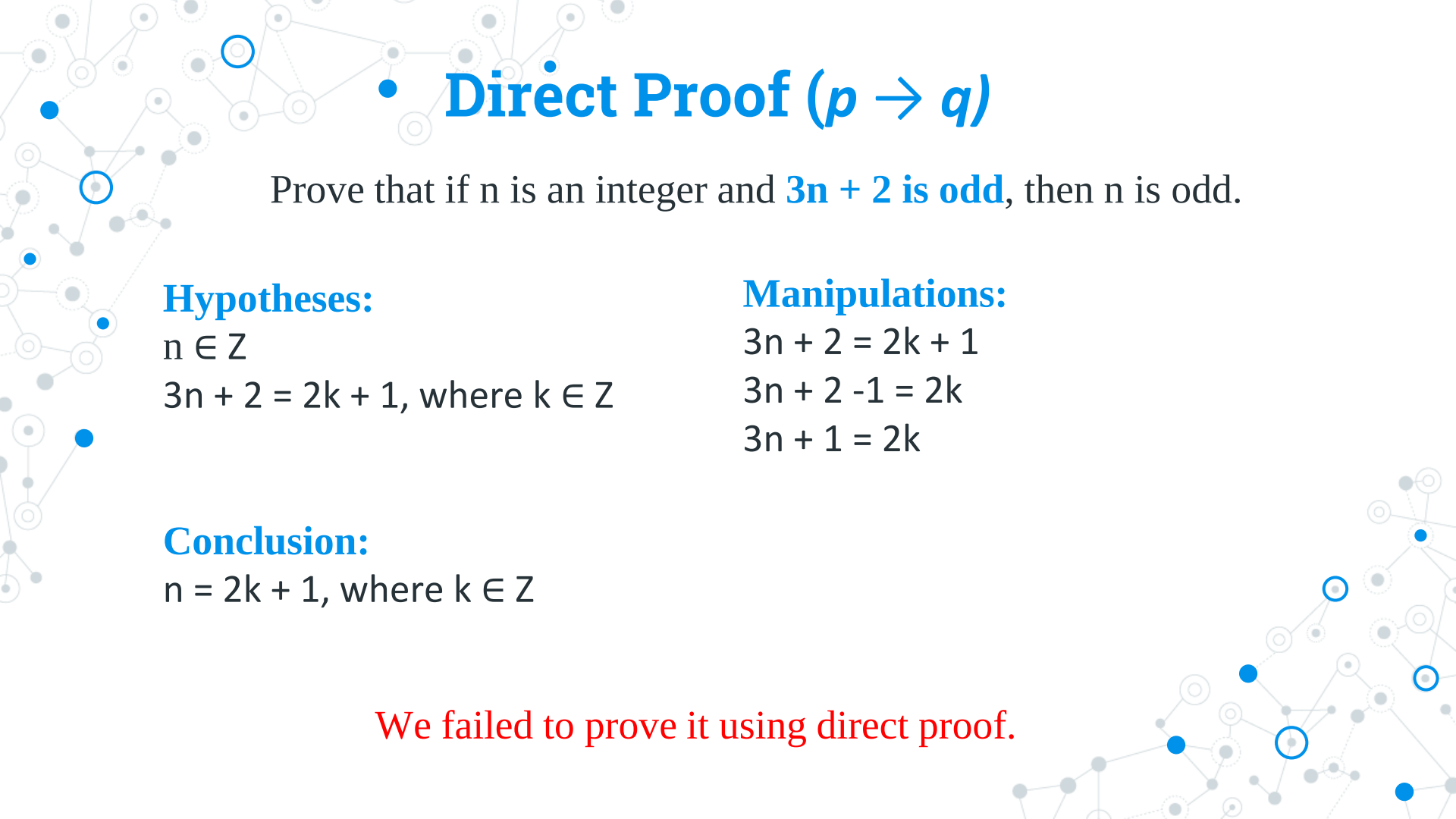
$$3n + 2 - 1 = 2k$$

$$3n + 1 = 2k$$

$$n = \frac{2k - 1}{3}$$

$$2k$$

$$2k + 1$$



• Direct Proof ($p \rightarrow q$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

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Manipulations:

$$3n + 2 = 2k + 1$$

$$3n + 2 - 1 = 2k$$

$$3n + 1 = 2k$$

We failed to prove it using direct proof.

Contraposition ($\neg q \rightarrow \neg p$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

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Conclusion:

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Assume conclusion is false.

Contraposition ($\neg q \rightarrow \neg p$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Hypotheses:

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Conclusion:

$$n = 2k + 1, \text{ where } k \in \mathbb{Z}$$

Assumed conclusion ($\neg q$):

n is even

$$n = 2k, \text{ where } k \in \mathbb{Z}$$

Solution:

Let $n = 2k$ in $3n + 2$

$$3(2k) + 2$$

$$6k + 2$$

$$2(3k + 1)$$

Contraposition ($\neg q \rightarrow \neg p$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Hypotheses:

$$n \in \mathbb{Z}$$

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Let $n = 2k$ in $3n + 2$

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$$6k + 2$$

$$2(3k + 1)$$

A multiple of 2, therefore $3n + 2$ is even

2k

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Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Hypotheses:

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$$3(2k) + 2$$

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$$2(3k + 1)$$

A multiple of 2, therefore $3n + 2$ is even

Contraposition ($\neg q \rightarrow \neg p$)

Prove that if n is an integer and $3n + 2$ is odd, then n is odd.

Hypotheses:

$$n \in \mathbb{Z}$$

$$3n + 2 = 2k + 1, \text{ where } k \in \mathbb{Z}$$

Conclusion:

$$n = 2k + 1, \text{ where } k \in \mathbb{Z}$$

Assumed conclusion ($\neg q$):

n is even

$$n = 2k, \text{ where } k \in \mathbb{Z}$$

Solution: ($\neg p$)

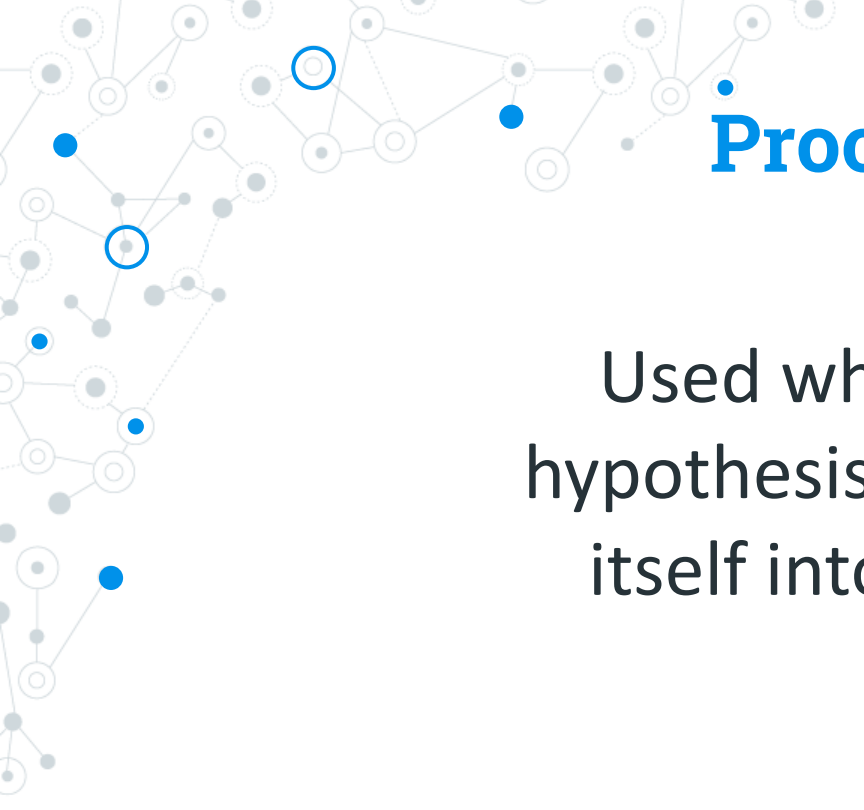
Let $n = 2k$ in $3n + 2$

$$3(2k) + 2$$

$$6k + 2$$

$$2(3k + 1)$$

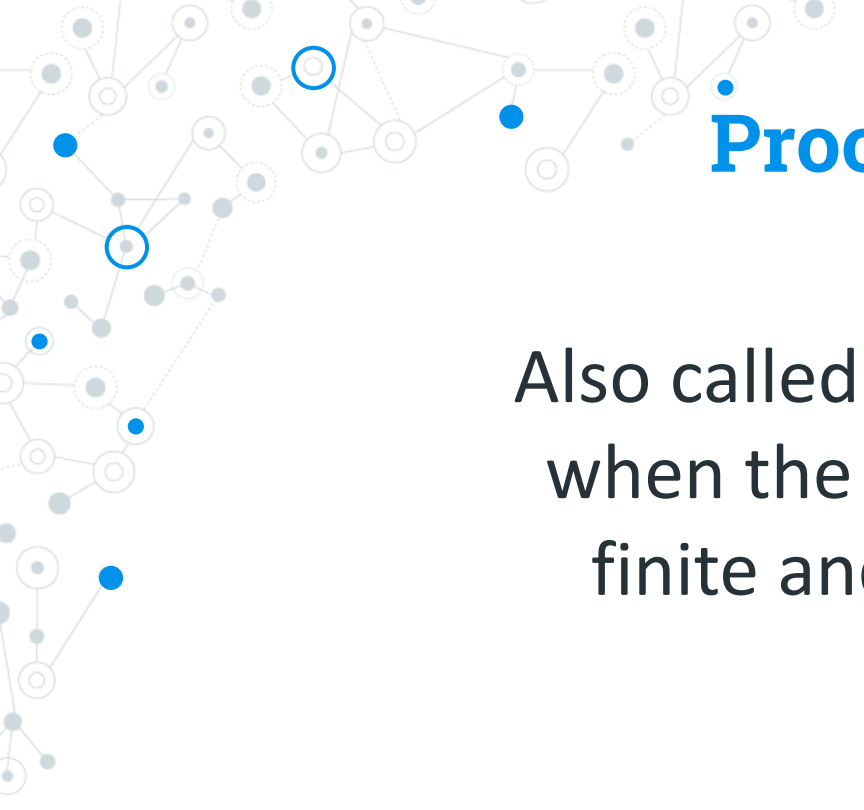
We have proven that $p \rightarrow q$, using $\neg p \rightarrow \neg q$

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a branching, tree-like structure.

Proof by Cases

Used when the original hypothesis naturally divides itself into various cases.

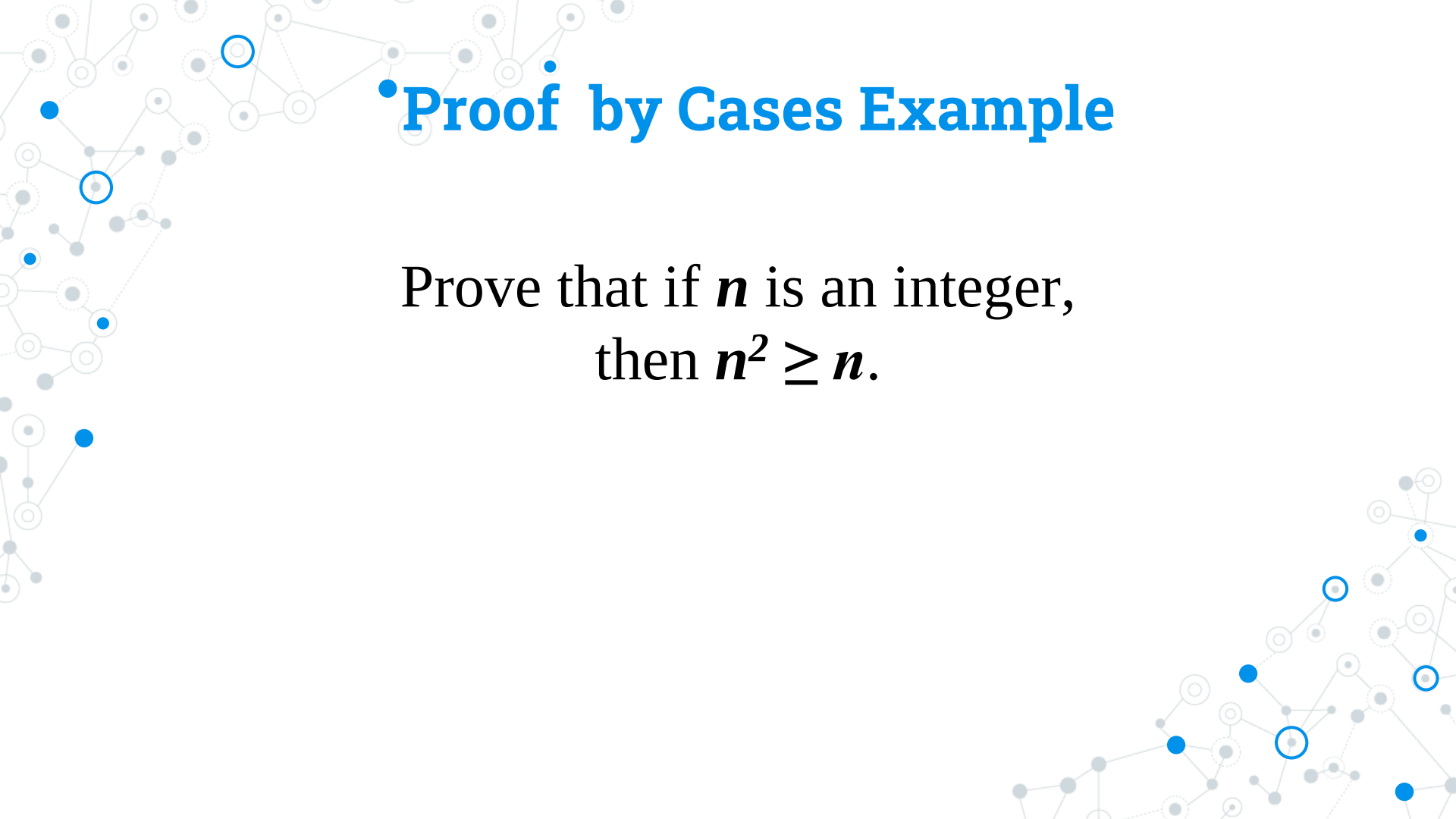
A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It shows a cluster of nodes, with some highlighted as solid blue circles and others as white circles with blue outlines, all interconnected by light gray lines.

A decorative network diagram in the top-left corner, featuring a complex web of interconnected nodes. Some nodes are solid blue circles, while others are white circles with blue outlines. The nodes are connected by thin, light gray lines, creating a mesh-like structure that extends towards the center of the slide.

Proof by Cases

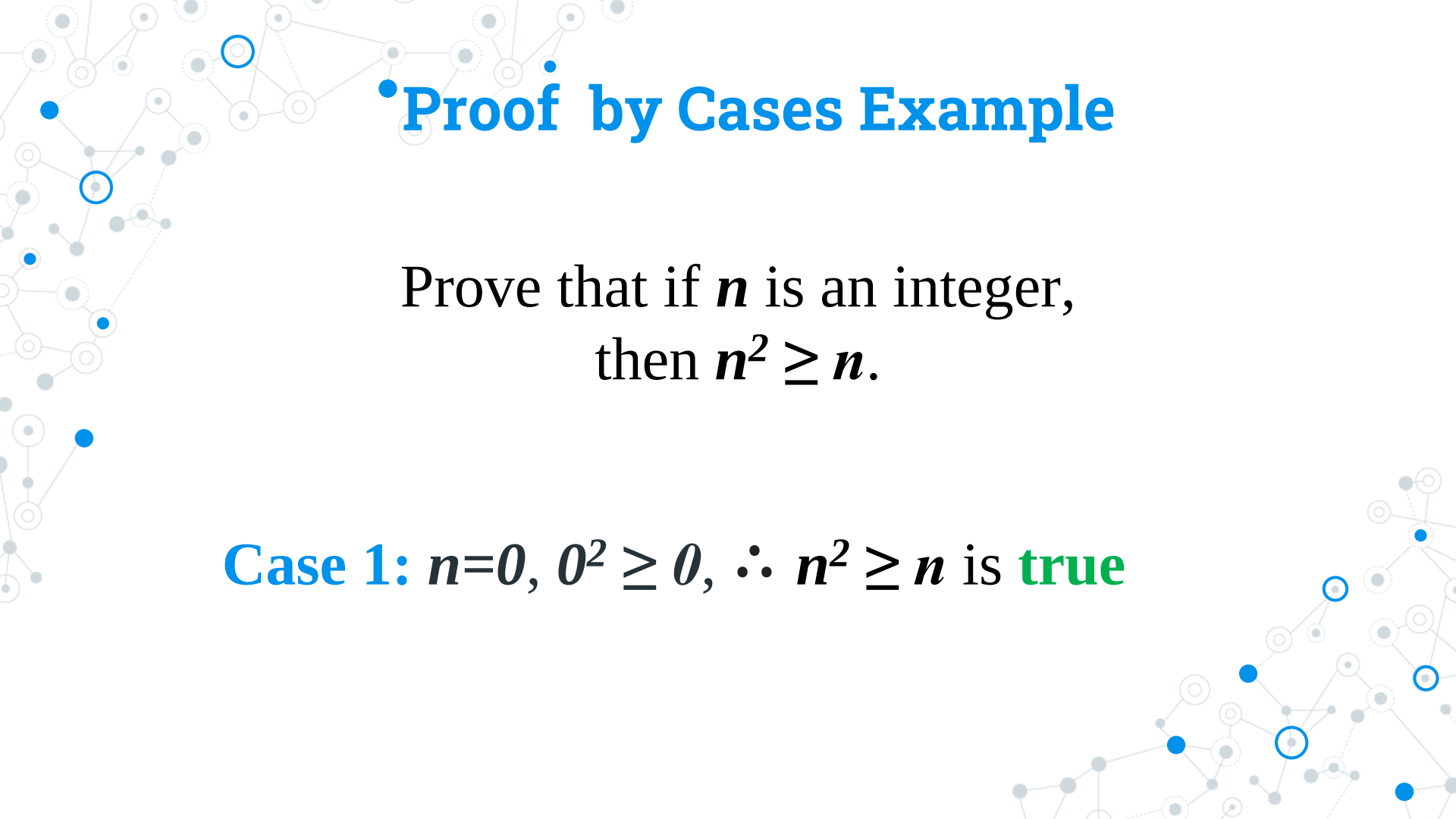
Also called exhaustive proof,
when the cases to prove is
finite and not too large.

A decorative network diagram in the bottom-right corner, similar to the one in the top-left. It consists of a cluster of interconnected nodes, with some nodes highlighted as solid blue circles and others as white circles with blue outlines, all connected by light gray lines.



• Proof by Cases Example

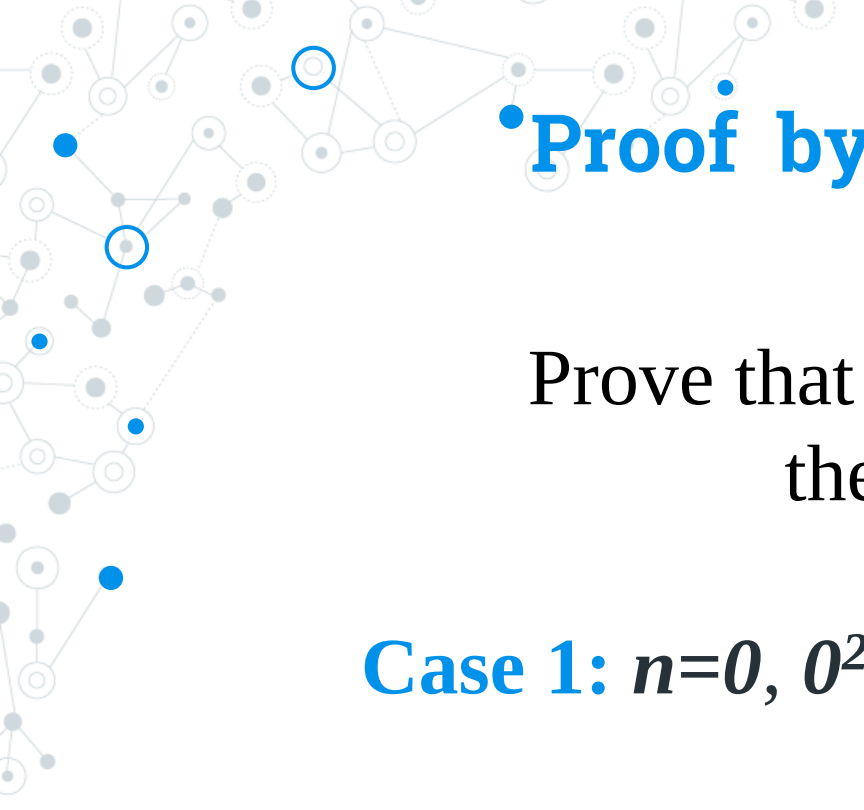
Prove that if n is an integer,
then $n^2 \geq n$.



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Case 1: $n=0$, $0^2 \geq 0$, $\therefore n^2 \geq n$ is **true**

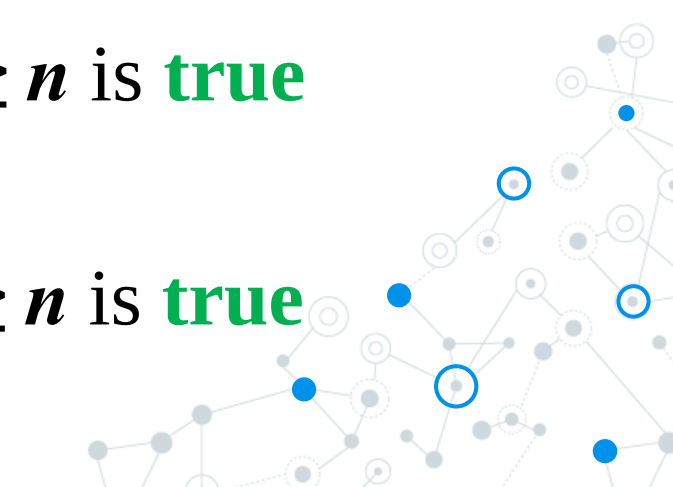


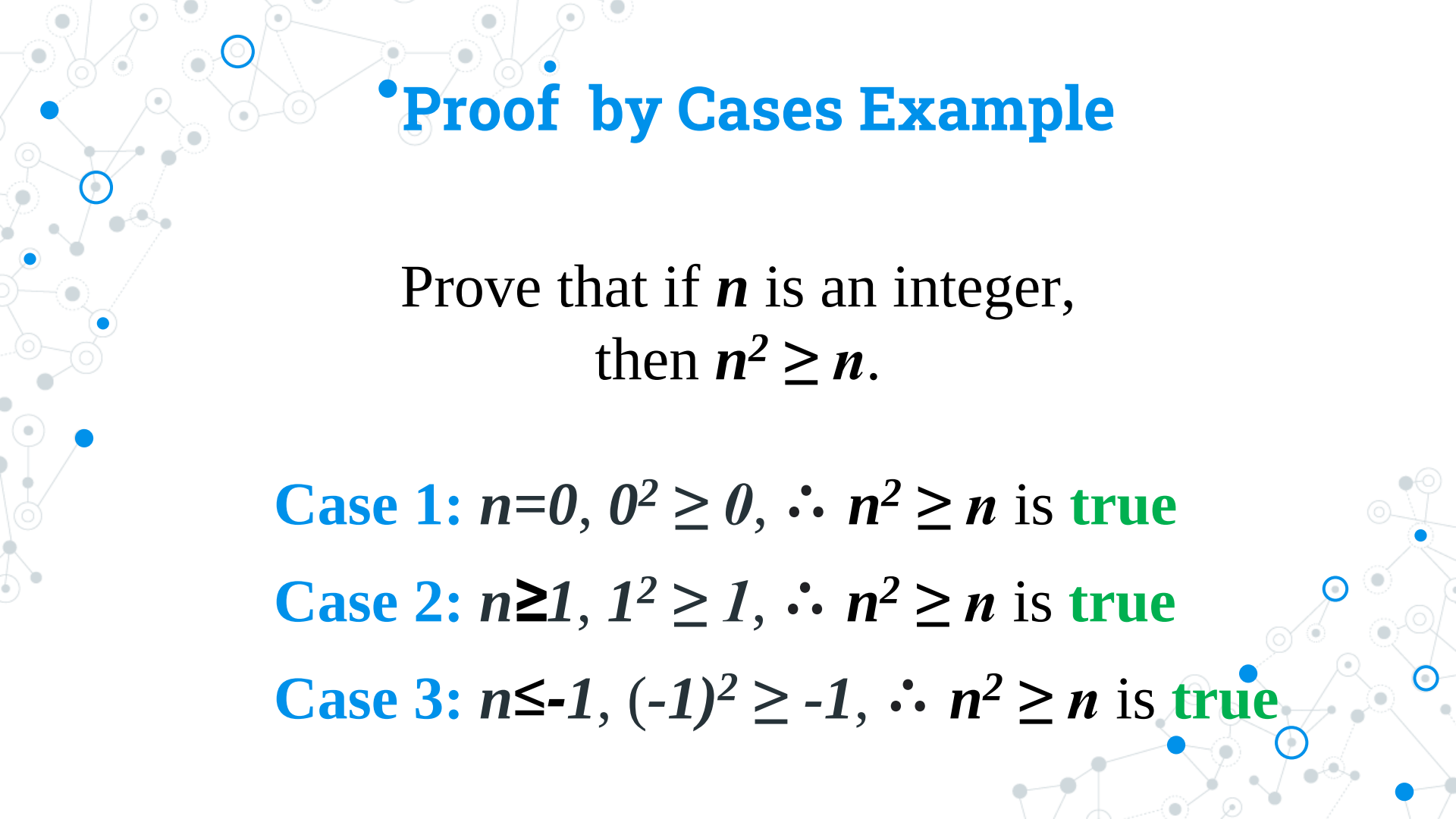
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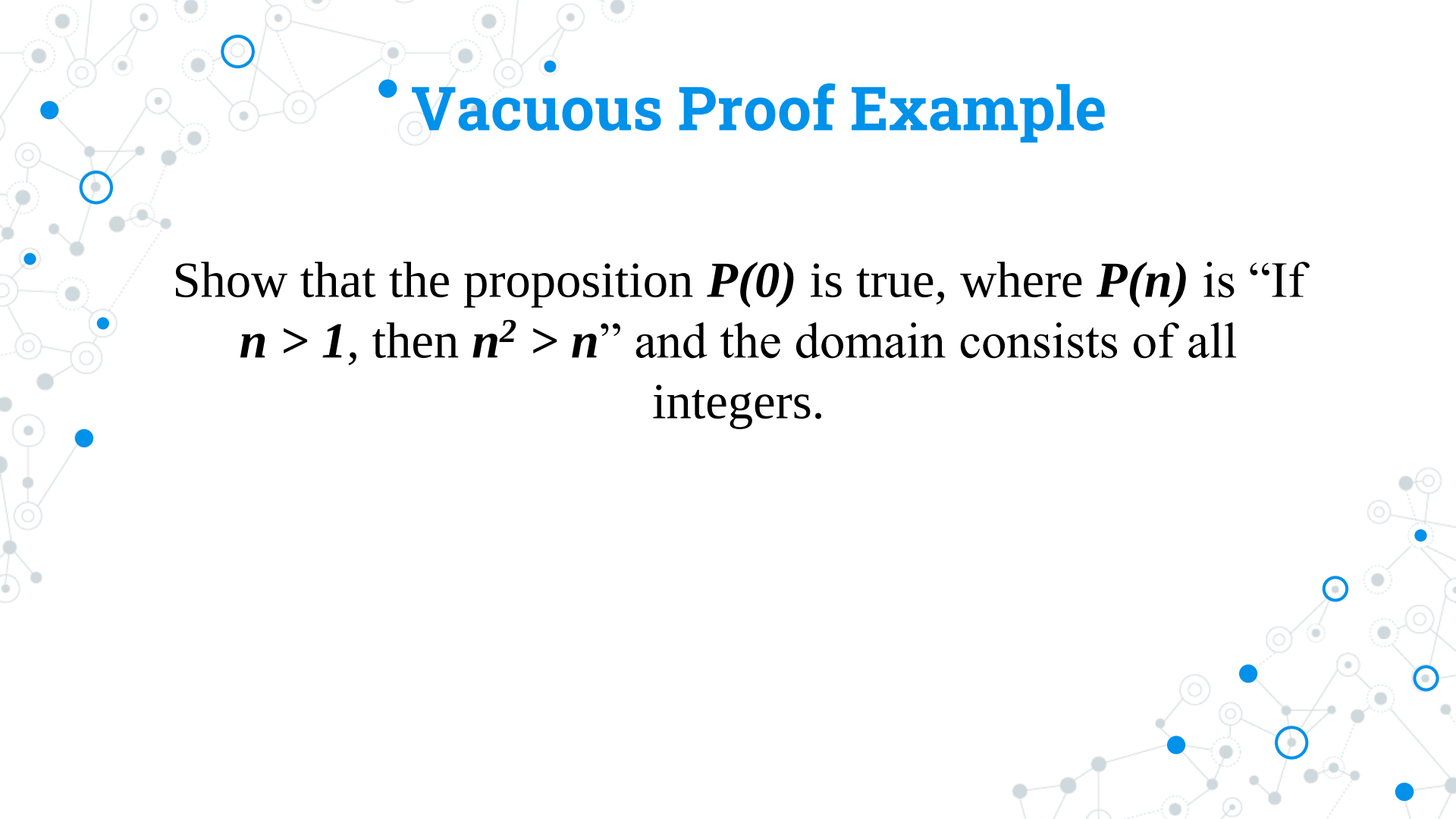
Case 2: $n \geq 1$, $1^2 \geq 1$, $\therefore n^2 \geq n$ is **true**

Case 3: $n \leq -1$, $(-1)^2 \geq -1$, $\therefore n^2 \geq n$ is **true**

Vacuous Proof

Showing that p is false in $p \rightarrow q$, means the statement is true.





• Vacuous Proof Example

Show that the proposition $P(0)$ is true, where $P(n)$ is “If $n > 1$, then $n^2 > n$ ” and the domain consists of all integers.

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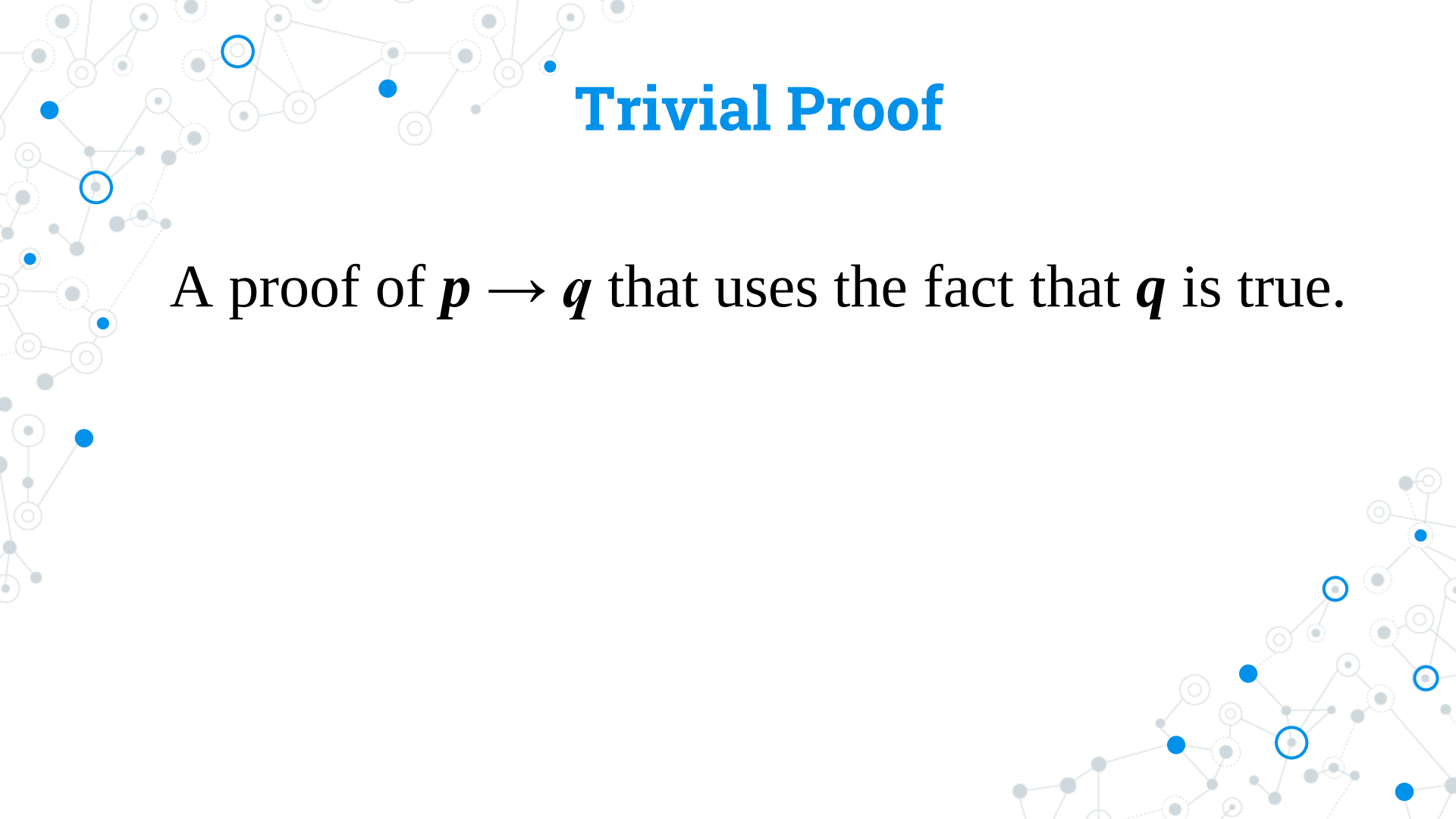
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Hypotheses is false, therefore the statement is true.



Trivial Proof

A proof of $p \rightarrow q$ that uses the fact that q is true.

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p	q	$p \rightarrow q$
T	T	T
T	F	F
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Let $P(n)$ be “If a and b are positive integers with $a \geq b$, then $a^n \geq b^n$,” where the domain consists of all nonnegative integers. Show that $P(0)$ is true.



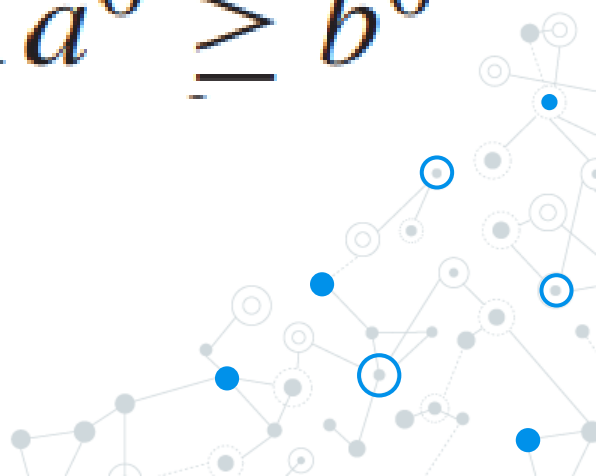


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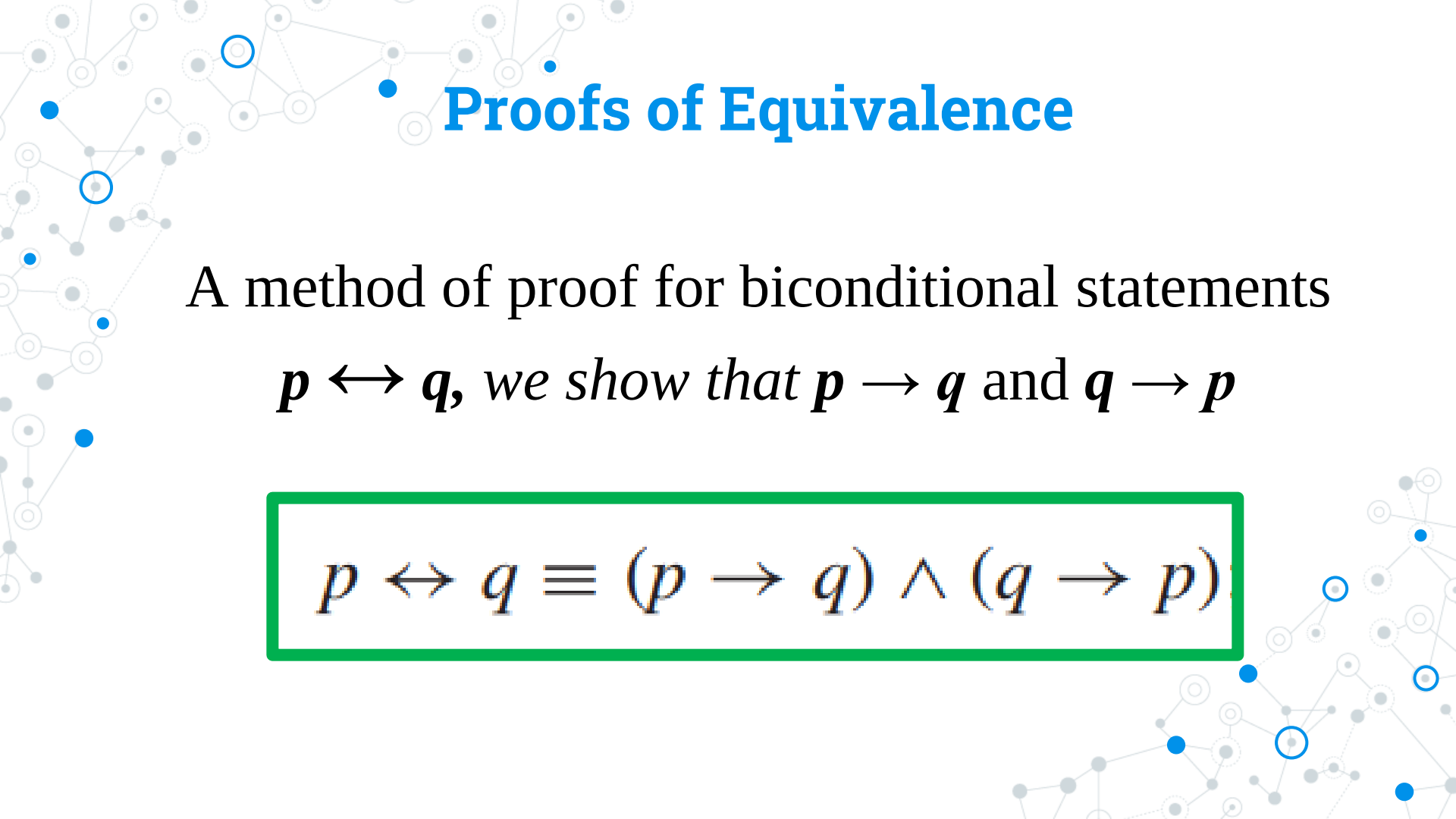


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$P(0)$ is “If $a \geq b$, then $a^0 \geq b^0$ ”

Always true



Proofs of Equivalence

A method of proof for biconditional statements

$p \leftrightarrow q$, we show that $p \rightarrow q$ and $q \rightarrow p$

$$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

Proofs of Equivalence Example

Prove that for every integer n , n is odd if and only if $n-1$ is even.

if n is odd then $n - 1$ is even

if $n - 1$ is even then n is odd.

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This formula is the definition of an even integer.

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If $n-1$ is even, then $n-1=2k$, $k \in \mathbb{Z}$

$$n=2k+1$$

This formula is the definition of an odd integer.

Proofs of Equivalence Example

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Existence Proof

$$\exists x P(x)$$

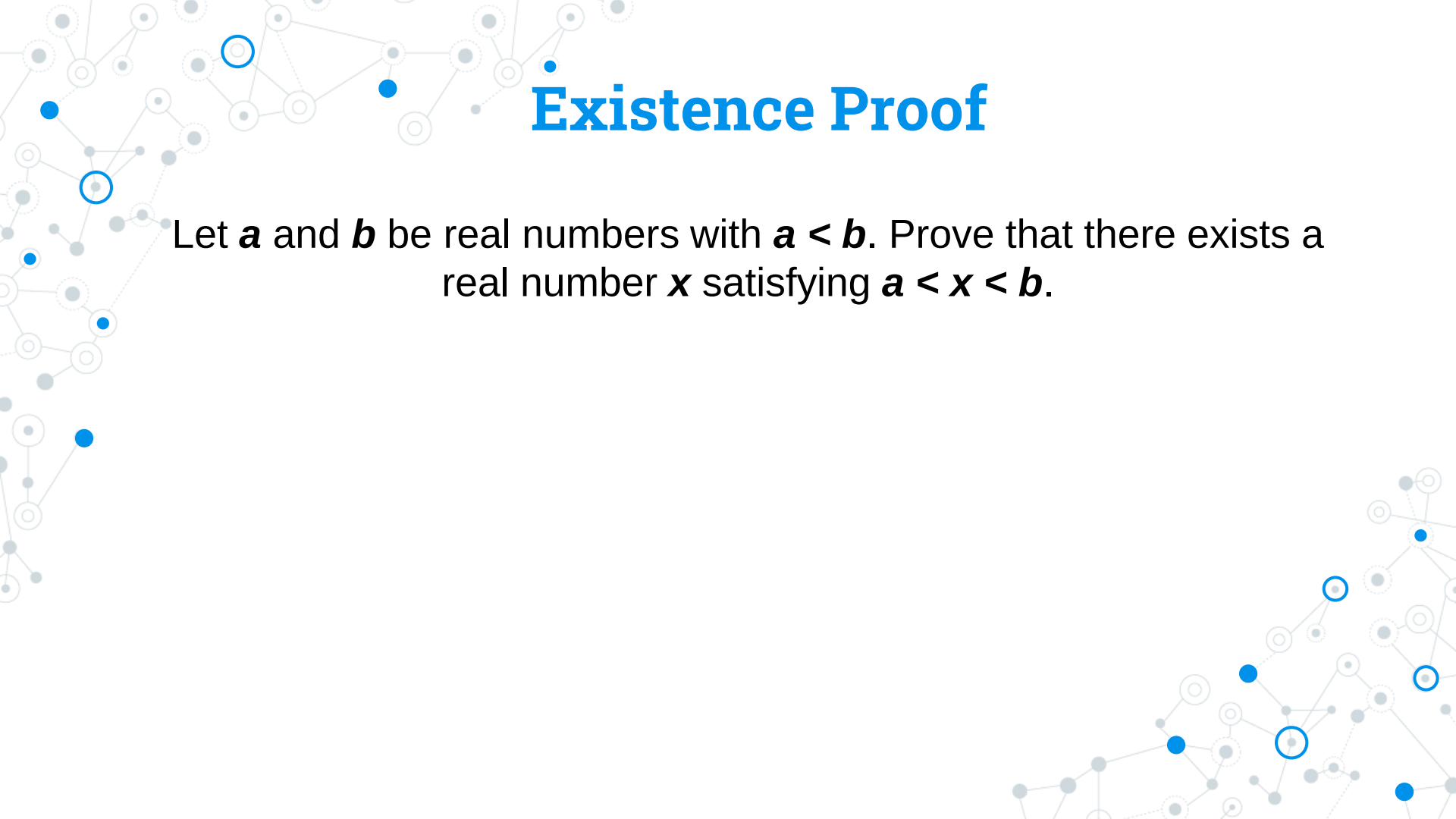



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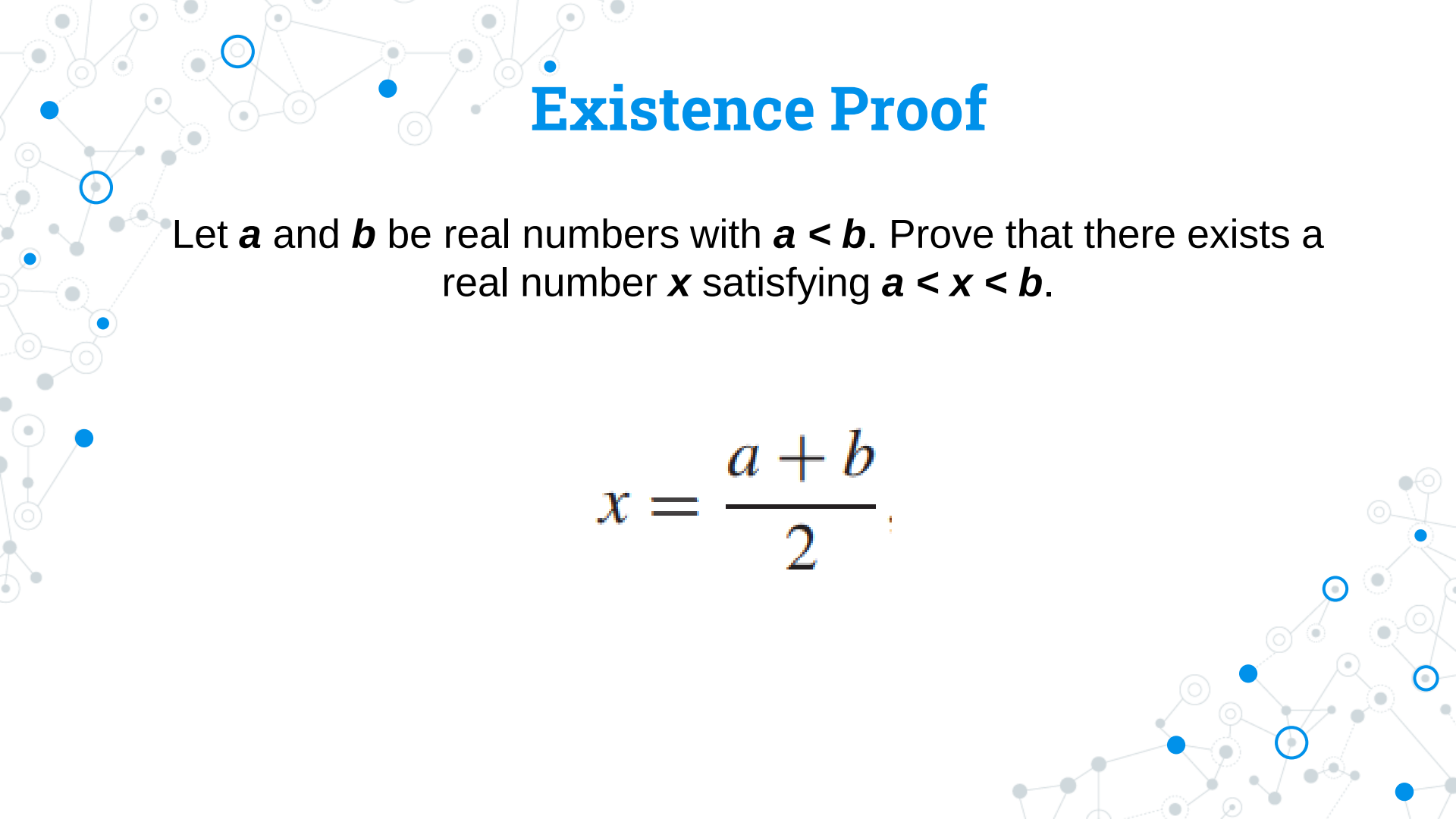
There exists at least one value of x for which the predicate $P(x)$ is true.





Existence Proof

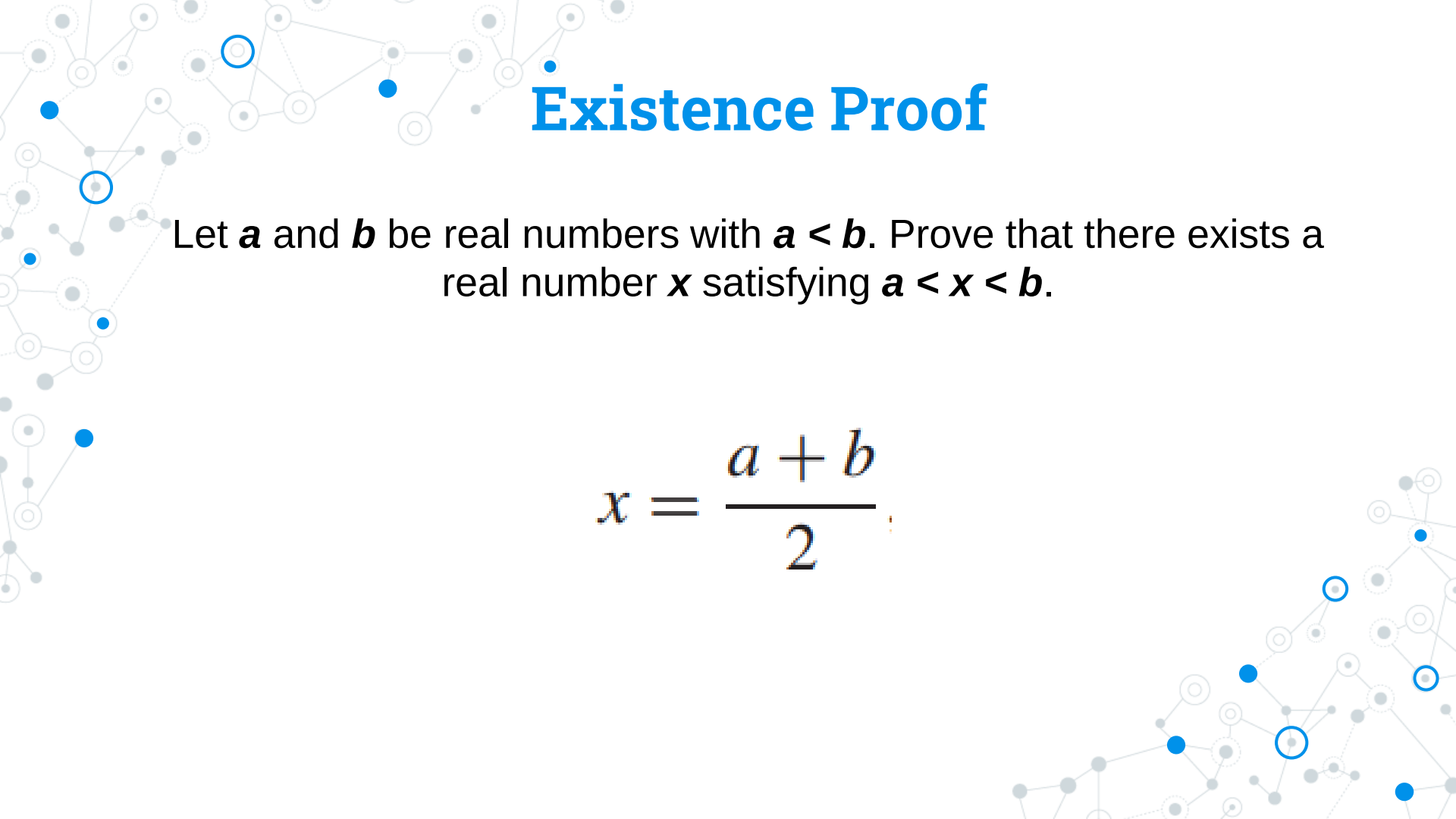
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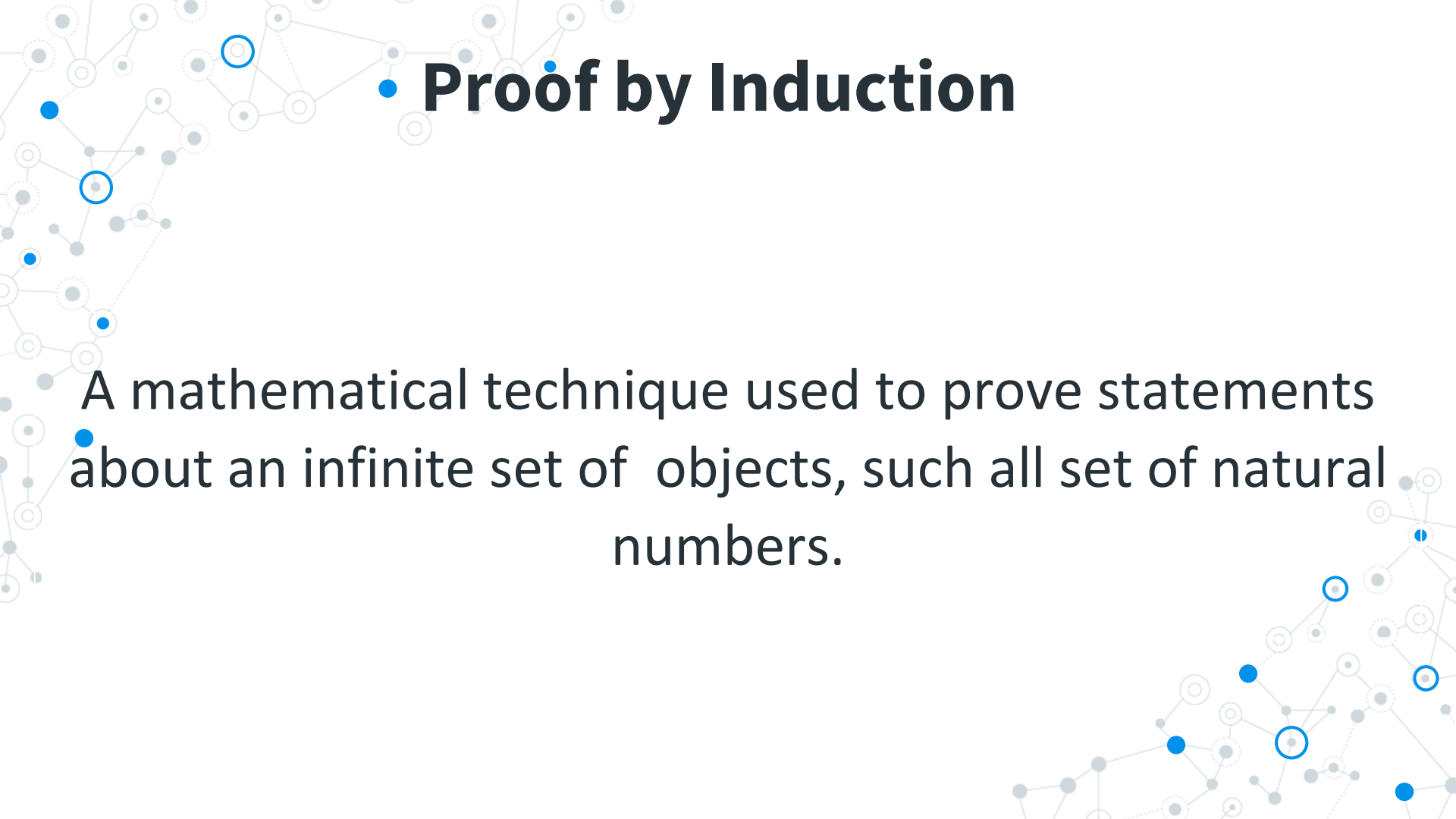
$$x = \frac{a + b}{2} ;$$



Existence Proof

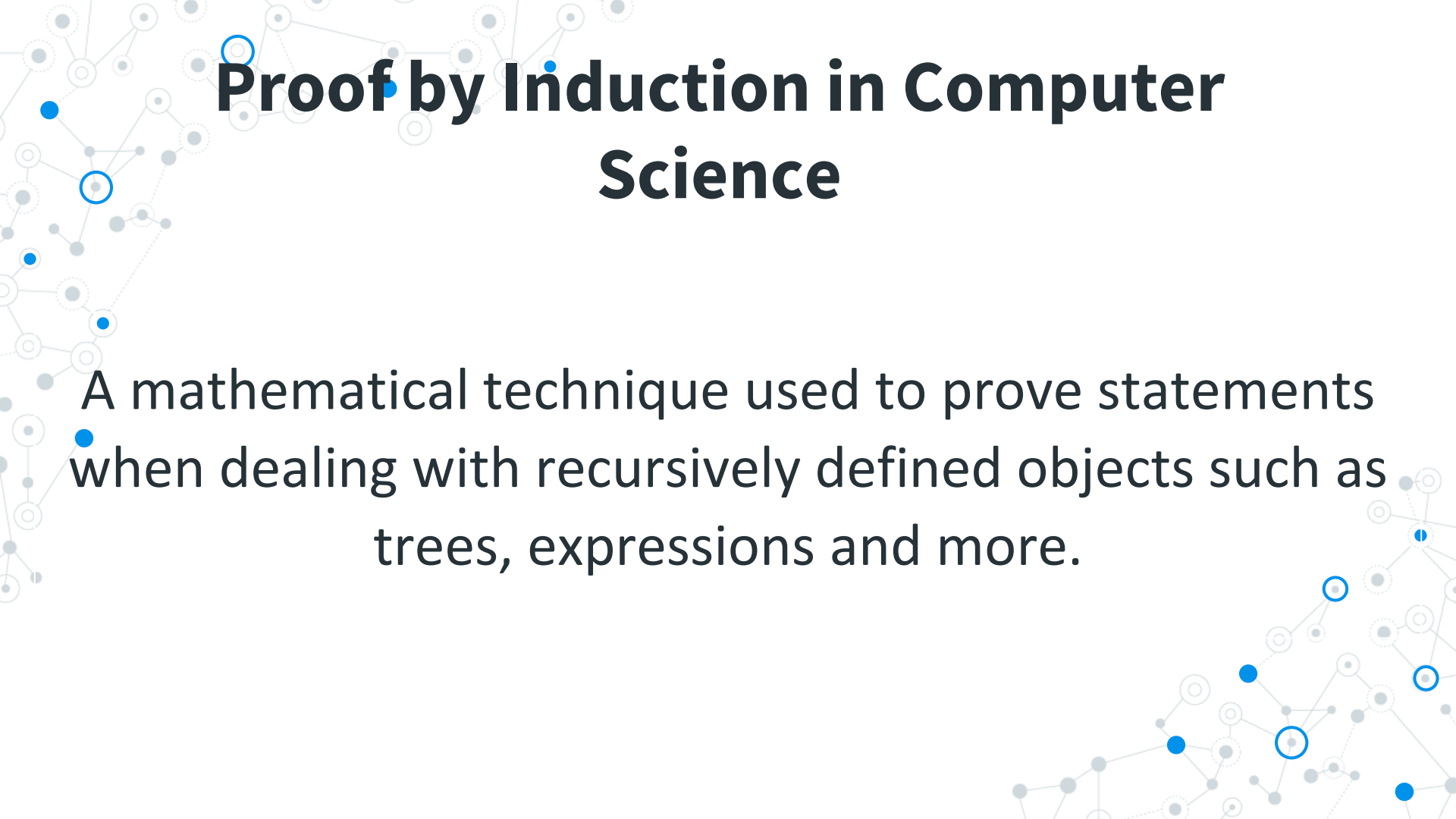
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• **Proof by Induction**

A mathematical technique used to prove statements about an infinite set of objects, such as all set of natural numbers.



Proof by Induction in Computer Science

A mathematical technique used to prove statements when dealing with recursively defined objects such as trees, expressions and more.

Proof by Induction Example

$S(n)$ = sum of all positive integers including n .

$$S(1) = 1$$

$$S(2) = 1 + 2 = 3$$

$$S(3) = 1 + 2 + 3 = 6$$

$$S(4) = 1 + 2 + 3 + 4 = 10$$

$$S(5) = 1 + 2 + 3 + 4 + 5 = 15$$

- **Proof by Induction**

Prove $S(n)$ is sum of all positive integers including n ,

$$S(n) = \frac{n(n+1)}{2}$$

• **Proof by Induction**

• Prove $S(n)$ is sum of all positive integers including n ,

* **Base case** $\rightarrow$ prove using $n=1$

* **Induction step** $\rightarrow$ assume $S(k)$ is **true**, where k is any positive integer.

$$S(n) = \frac{n(n+1)}{2}$$

then $S(k+1)$ must be true

- **Proof by Induction**

• Prove that the sum of all positive integers n is given by :

$$S(n) = \frac{n(n+1)}{2}$$

- **Proof by Induction**

$S(n)$ = sum of all positive integers including n .

$$S(n) = \frac{n(n+1)}{2}$$

Base Case:

$$S(1) = \frac{1(1+1)}{2} = \frac{1(2)}{2} = 1$$

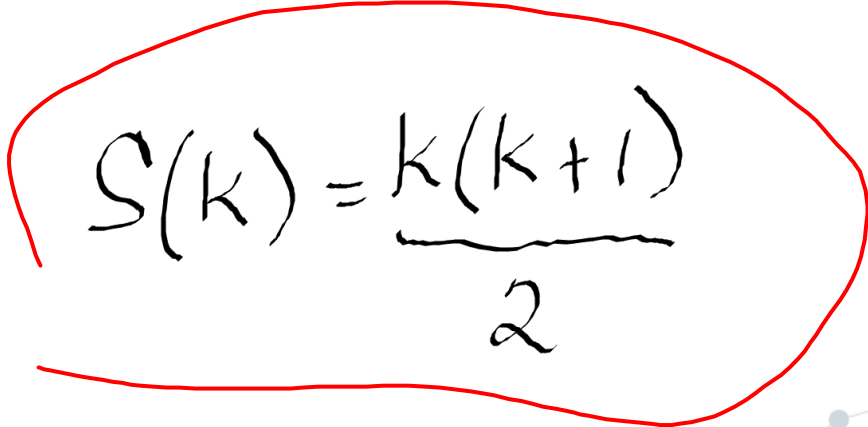
- **Proof by Induction**

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- **Induction Case:**

Assume:


$$S(k) = \frac{k(k+1)}{2}$$

• Induction Case:

• Prove:

$$S(n) = \frac{n(n+1)}{2}$$

Assume:

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$$= \frac{k(k+1)}{2} + (k+1)$$

Prove:

$$S(n) = \frac{n(n+1)}{2}$$

Induction Case:

Assume:

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$$\begin{aligned} S(k+1) &= 1 + 2 + 3 + 4 + \dots + k + (k+1) \\ &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \end{aligned}$$

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